

Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

University of Pisa, a.y. 2025-2026

Alessandro Gori*

Thesis for the Master's degree in Physics

Abstract

[To be continued. . .]

- Appendix on s -wave SC ghost simulations on HM
- Set ComputerModern font in all CairoMakie figures

Contents

1	Introduction to Hartree-Fock methods	3
1.1	Variational principle and gaussian states	3
1.2	The HF variational method	4
1.2.1	Properties of the target quadratic hamiltonian	5
1.2.2	Self-consistent variational solution	6
1.2.3	Operative hamiltonian approximation	9
1.3	The iHF algorithm	9
1.3.1	Fixed density simulations and the role of μ	11
1.3.2	The convergence scheme	12
2	Phase transitions and SSB in EHM	15
2.1	The EHM symmetries	15
2.1.1	Lattice symmetries	15
2.1.2	Spin and charge symmetries	16
2.1.3	Particle-hole symmetry at half-filling	20
2.2	The HF method and symmetry breaking	22
2.2.1	Local repulsion	23
2.2.2	Non-local attraction	25
2.3	The EHM in reciprocal space	27
2.4	Square lattice spatial symmetry channels	30
3	The normal phase	35
3.1	The normal phase and its symmetries	35
3.2	Application of HF methods to the normal phase	36
3.2.1	Hartree effects: chemical potential renormalisation	36
3.2.2	Fock effects: bare bands renormalisation	38
3.3	The HF hamiltonian and its properties	41
3.3.1	Nematicity: d -wave bands and anisotropic hopping	41
3.3.2	The entropy density of the SC ground-state	43
3.3.3	The free energy density of the SC ground-state	44

*a.gori23@studenti.unipi.it / nepero27178 @ GitHub

3.4	Analysis of the self-consistency equations	45
3.5	Discussion of numeric results	48
3.5.1	Mott localisation effect	49
3.5.2	Solution free energy and entropy densities	50
3.5.3	Convergence problems and mixing fine-tuning	51
4	Antiferromagnetic instability	53
4.1	The AF phase and its symmetries	53
4.1.1	Symmetries of the commensurate AF Ansatz	54
4.1.2	The AF Ansatz in reciprocal space	55
4.2	Local Hartree effects: antiferromagnetism in the HM	57
4.2.1	Local Hartree channel in the AF phase: analytic derivation	57
4.2.2	Local Hartree contraction energy	58
4.2.3	Antiferromagnetism in the HM	59
4.2.4	Features of the AF solution	61
4.3	Extension to the EHM	64
4.4	Non-local Hartree effects: chemical potential renormalisation	65
4.4.1	Non-local Hartree channel in the AF phase: analytic derivation	66
4.4.2	Non-local Hartree contraction energy	67
4.5	Fock effects: gap and bands renormalisation	68
4.5.1	Fock channel in the AF phase: analytic derivation	68
4.5.2	Fock contraction energy	71
4.6	Full MFT hamiltonian and gap complexification	72
4.6.1	The relative phase in the zero temperature half-filled ground state	73
4.6.2	AF phase flavours and spin degeneracy	74
4.7	AF free energy density	74
4.7.1	Energetic constraints on the HFPs	76
4.7.2	Free energy at half-filling	77
4.8	Discussion of numeric results	77
4.8.1	Magnetization boost	78
4.8.2	Bands shrinking	79
4.8.3	Solution free energy and entropy densities	80
4.8.4	Fixed-point interpretation of HF solution	81
A	Antiferromagnetism in the HM	83
A.1	The AF ground-state for the conventional HM	83
A.2	Explicit AF ground state at half-filling and $\beta \rightarrow \infty$	84
A.3	Instability of the commensurate AF phase	85
A.4	iHF results	86
A.5	Analytic expression of self-consistent magnetisation	87
	Bibliography	91

List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
$\Delta^{(\gamma)}$	γ -wave component of the gap function
DoF	Degree of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
iHF	Iterative Hartree-Fock
LRT	Linear Response Theory
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
PH	Particle-hole
PHS	Particle-hole symmetry
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Introduction

This thesis project is about my favorite ice cream flavor. [To be continued...]

Chapter 1

Introduction to Hartree-Fock methods

In this chapter we give a general introduction to the Hartree-Fock (HF) variational method and its numerical implementation. In Quantum Mechanics, variational methods allow to approximate the ground state of a many-body hamiltonian by minimising the energy computed for a specific substate of physical states. In particular, in the HF method we approximate the real ground-state with the best possible many-body state that is an eigenstate of a non-interacting fermionic hamiltonian.

Despite its simplicity, the method is very powerful for describing various types of quantum many-body phenomena and, in particular, spontaneous symmetry breaking (SSB). The description of HF method in this chapter is largely based on [4, 32].

1.1 Variational principle and gaussian states

Let $\hat{H}$ be a generic hamiltonian acting on the Hilbert space $\mathcal{H}$. For any many-body state $|\Psi\rangle \in \mathcal{H}$, the energy is given by the expectation value (EV)

$$E[\Psi] = \langle \Psi | \hat{H} | \Psi \rangle$$

Let $|\Psi_0\rangle$ be the ground state of $\hat{H}$, and $E_0 \equiv E[\Psi_0]$. Then

$$\forall |\Psi\rangle \in \mathcal{H} \quad : \quad E[\Psi] \geq E_0$$

Let $\mathcal{S} \subset \mathcal{H}$ be a subspace of states, ad let $|\Psi\rangle$ be a state inside that locally minimises energy,

$$|\Psi\rangle \in \mathcal{S} \quad \frac{\delta E[\Psi]}{\delta |\Psi\rangle} = 0$$

If the true ground state belongs to $\mathcal{S}$, the found local minimum coincides with $|\Psi_0\rangle$; otherwise, approximates it. In Fig. 1.1 we give a pictorial representation of this idea: the state $|\Psi\rangle \in \mathcal{S}$ is an approximation, found by constrained minimisation, of the true ground-state.

In HF methods, we identify $\mathcal{S}$ with the set of so-called “gaussian states”. These are all the ground-states of quadratic hamiltonians, namely the hamiltonians that are bilinear in the fermionic creations operators (see App. ?? for details). The restriction to gaussian states significantly simplifies the computation of expectation values. Indeed, thanks to Wick’s theorem, the expectation value of quartic operators exactly decomposes as

$$\langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\gamma \hat{c}_\delta \rangle = \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\delta \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\gamma \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle \langle \hat{c}_\gamma^\dagger \hat{c}_\delta \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \rangle \langle \hat{c}_\gamma \hat{c}_\delta \rangle}_{\text{Cooper}} \quad (1.1)$$

The subscripts indicate the standard names to the different types of contractions, that we adopt throughout the text.

Operatively, one can think of the HF method as an approximation of the the hamiltonian with a non-interacting hamiltonian

$$\hat{H} \stackrel{\text{HF}}{\simeq} \hat{H}_{\text{HF}} = \sum_{\alpha} E_{\alpha} \hat{\gamma}_{\alpha}^{\dagger} \hat{\gamma}_{\alpha}$$

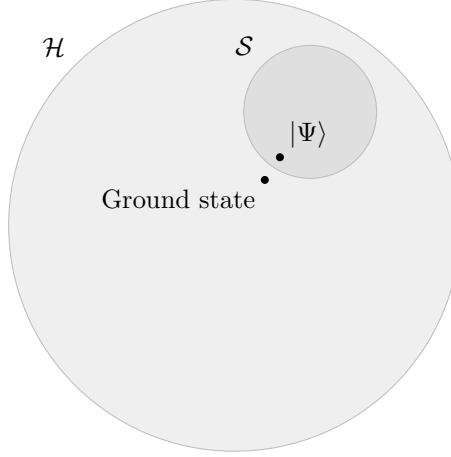


Figure 1.1 | Pictorial representation of the search for the best approximation $|\Psi\rangle$ to the true ground-state inside the subspace $\mathcal{S} \subset \mathcal{H}$.

where $\hat{\gamma}_\alpha$ are fermionic operators determined by the minimisation procedure

$$\{\hat{\gamma}_\alpha, \hat{\gamma}_\beta^\dagger\} = \delta_{\alpha\beta}$$

and are connected to the conventional electronic operators $\hat{c}_\alpha$ through an unitary map. See App. ?? for details.

Let us stress that the operatorial approximation $\hat{H}_{\text{HF}} \simeq^{\text{HF}}$ is a shorthand notation that must be intended as follows: $\hat{H}_{\text{HF}}$ is that non-interacting operator whose ground-state is the best approximation to the ground-state of $\hat{H}$. This does not mean that in general the two operators are related.

1.2 The HF variational method

We now discuss the implementation of the variational HF method for the generic quartic hamiltonian:

$$\hat{H} = \sum_{\alpha,\beta} A_{\alpha\beta} \hat{c}_\alpha^\dagger \hat{c}_\beta + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\gamma \hat{c}_\delta \quad (1.2)$$

where

$$A_{\alpha\beta} = A_{\beta\alpha}^* \quad B_{\alpha\beta\gamma\delta} = B_{\delta\gamma\beta\alpha}^* \quad (1.3)$$

to ensure hermiticity. Moreover, the following equation must hold:

$$B_{\alpha\beta\gamma\delta} = B_{\beta\alpha\delta\gamma} \quad (1.4)$$

due to the property $\hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\gamma \hat{c}_\delta = \hat{c}_\beta^\dagger \hat{c}_\alpha^\dagger \hat{c}_\delta \hat{c}_\gamma$, which follows from Fermi anticommutation rules.

Our aim is to approximate the above hamiltonian with a non-interacting hamiltonian. The most general quadratic hamiltonian is given by:

$$\hat{H} \simeq \hat{H}_{\text{HF}} \equiv E_0 + \sum_{\alpha,\beta} k_{\alpha\beta} \hat{\gamma}_\alpha^\dagger \hat{\gamma}_\beta \quad (1.5)$$

As is specified in Sec. ??, γ and c operators must be linked by an unitary map. To find the best approximation to the original hamiltonian means to identify:

- the offset E_0 ;
- the fermionic operators $\hat{\gamma}_\alpha, \hat{\gamma}_\alpha^\dagger$;
- the $k_{\alpha\beta}$ coefficients.

Note that, in terms of algebraic solution, E_0 is irrelevant. To determine it correctly is only relevant in order to get the correct approximation of the actual ground-state energy. Let us discuss briefly the features of $\hat{H}_{\text{HF}}$ and then propose a solution.

1.2.1 Properties of the target quadratic hamiltonian

In this section we enumerate some useful properties of $\hat{H}_{\text{HF}}$. We will use also the following notation:

$$\hat{a}_\alpha^- \equiv \hat{c}_\alpha \quad \hat{a}_\alpha^+ \equiv \hat{c}_\alpha^\dagger$$

Due to linearity of the unitary mapping, the operator $\hat{H}_{\text{HF}}$ of Eq. (1.5) can also be expressed as the most general quadratic hamiltonians for electronic operators:

$$\hat{H}_{\text{HF}} = E_0 + \sum_{\alpha, \beta} \sum_{p, q} h_{\alpha\beta}^{pq} \hat{a}_\alpha^p \hat{a}_\beta^q \quad (1.6)$$

By changing basis from $\hat{\gamma}$ to $\hat{c}$ operators, we moved all our ignorance into the new $h_{\alpha\beta}^{pq}$ matrix elements. These satisfy the relations:

$$h_{\alpha\beta}^{++} = \left(h_{\beta\alpha}^{--}\right)^* \quad (1.7)$$

$$h_{\beta\alpha}^{+-} = -h_{\alpha\beta}^{-+} \quad (1.8)$$

Because of these properties, we actually need to determine just half of the fields, since the other half will be uniquely determined.

Proof 1.1 (Symmetry properties of the tensor $h_{\alpha\beta}^{pq}$).

Let us start by proving Eq. (1.7). The HF hamiltonian can be represented through Nambu spinors as

$$\begin{aligned} \hat{H}_{\text{HF}} &= E_0 + \hat{\Phi}^\dagger \mathbf{h} \hat{\Phi} \\ &= E_0 + \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^\dagger \end{bmatrix}^\dagger \begin{bmatrix} h^{+-} & h^{++} \\ h^{--} & h^{-+} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^\dagger \end{bmatrix} \end{aligned}$$

Our hamiltonian must be hermitian. We conclude that:

$$\hat{H}_{\text{HF}}^\dagger = \hat{H}_{\text{HF}} \implies h^{++} = [h^{--}]^\dagger$$

which reads, element-wise:

$$h_{\alpha\beta}^{++} = \left(h_{\beta\alpha}^{--}\right)^*$$

This proves Eq. (1.7).

Eq. (1.8) is proven as follows. We can set, without loss of generality,

$$\text{Tr } h^{+-} + \text{Tr } h^{-+} = 0$$

This passage is guaranteed by the fact that if $\mathbf{h}$ is not traceless, we can redefine it by subtracting its trace and including the latter in the energy shift E_0 . Then

$$\begin{aligned} \hat{H}_{\text{HF}} &= E_0 + \sum_{\alpha, \beta} \left[h_{\alpha\beta}^{+-} \hat{c}_\alpha^\dagger \hat{c}_\beta + h_{\alpha\beta}^{-+} \hat{c}_\alpha \hat{c}_\beta^\dagger \right] + (\text{same sign terms}) \\ &= E_0 + \sum_{\alpha, \beta} \left[h_{\alpha\beta}^{+-} \left(\delta_{\alpha\beta} - \hat{c}_\beta \hat{c}_\alpha^\dagger \right) + h_{\alpha\beta}^{-+} \left(\delta_{\alpha\beta} - \hat{c}_\beta^\dagger \hat{c}_\alpha \right) \right] + (\text{same sign terms}) \\ &= E_0 - \sum_{\alpha, \beta} \left[h_{\beta\alpha}^{+-} \hat{c}_\alpha \hat{c}_\beta^\dagger + h_{\beta\alpha}^{-+} \hat{c}_\alpha^\dagger \hat{c}_\beta \right] + (\text{same sign terms}) \end{aligned}$$

In second row we got rid of the traces sum and exchanged indices $\alpha \leftrightarrow \beta$. Confronting last and first row we conclude:

$$h_{\beta\alpha}^{+-} = -h_{\alpha\beta}^{-+}$$

which proves Eq. (1.8).

These properties were all we needed to get to the main argument of the method.

1.2.2 Self-consistent variational solution

This section gives the operational method to obtain the variational solution. We follow and extend the argument proposed by Fabrizio [32]. Let:

$$\Delta_{\alpha\beta}^{pq} \equiv \langle \hat{a}_{\alpha}^p \hat{a}_{\beta}^q \rangle$$

thus, explicitly

$$\Delta_{\alpha\beta}^{+-} = \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta} \rangle \quad \Delta_{\alpha\beta}^{-+} = \langle \hat{c}_{\alpha} \hat{c}_{\beta}^{\dagger} \rangle \quad \Delta_{\alpha\beta}^{++} = \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \rangle \quad \Delta_{\alpha\beta}^{--} = \langle \hat{c}_{\alpha} \hat{c}_{\beta} \rangle \quad (1.9)$$

The first two are “normal densities”; the second two are “anomalous densities”. The reason for such names is the fact that these expectation values are non-zero if total charge conservation is broken. Conjugation of the last two gives a symmetry relation similar to Eq. (1.7):

$$\left(\Delta_{\alpha\beta}^{++} \right)^{\dagger} = \Delta_{\beta\alpha}^{--} \quad (1.10)$$

Here $\Delta_{\alpha\beta}^{pq}$ are functions of the fields $h_{\alpha\beta}^{pq}$. These quantities satisfy the equation:

$$\sum_{\alpha,\beta} \sum_{p,q} h_{\alpha\beta}^{pq} \frac{\partial \Delta_{\alpha\beta}^{pq}}{\partial h_{\mu\nu}^{rs}} = 0 \quad (1.11)$$

Let us prove this.

Proof 1.2 (Proof of Eq. (1.11)).

The HF ground-state energy is given by:

$$\begin{aligned} E_{\text{HF}} &= \langle \hat{H}_{\text{HF}} \rangle \\ &= E_0 + \sum_{\alpha,\beta} \sum_{p,q} h_{\alpha\beta}^{pq} \Delta_{\alpha\beta}^{pq} \end{aligned}$$

where the expectation value is taken over the (up to now unknown) HF ground-state. Derive it:

$$\begin{aligned} \frac{\partial E_{\text{HF}}}{\partial h_{\mu\nu}^{rs}} &= \frac{\partial}{\partial h_{\mu\nu}^{rs}} \sum_{\alpha,\beta} \sum_{p,q} h_{\alpha\beta}^{pq} \Delta_{\alpha\beta}^{pq} \\ &= \Delta_{\mu\nu}^{rs} + \sum_{\alpha,\beta} \sum_{p,q} h_{\alpha\beta}^{pq} \frac{\partial \Delta_{\alpha\beta}^{pq}}{\partial h_{\mu\nu}^{rs}} \end{aligned} \quad (1.12)$$

Note that by construction E_{HF} is greater than the actual ground-state energy, since our approximated wavefunction is not the actual ground-state wavefunction. From Hellman-Feynman theorem [4], we know that for a parametric hamiltonian it holds:

$$\begin{aligned} \frac{\partial E_{\text{HF}}}{\partial h_{\mu\nu}^{rs}} &= \langle \Psi | \frac{\partial \hat{H}_{\text{HF}}}{\partial h_{\mu\nu}^{rs}} | \Psi \rangle \\ &= \langle \Psi | \frac{\partial}{\partial h_{\mu\nu}^{rs}} \sum_{\alpha,\beta} \sum_{p,q} h_{\alpha\beta}^{pq} \hat{a}_{\alpha}^p \hat{a}_{\beta}^q | \Psi \rangle \\ &= \langle \hat{a}_{\mu}^r \hat{a}_{\nu}^s \rangle \\ &= \Delta_{\mu\nu}^{rs} \end{aligned} \quad (1.13)$$

Eqns.(1.12) and (1.13), combined, imply:

$$\sum_{\alpha,\beta} \sum_{p,q} h_{\alpha\beta}^{pq} \frac{\partial \Delta_{\alpha\beta}^{pq}}{\partial h_{\mu\nu}^{rs}} = 0$$

which ends the proof.

We now use Eq. (1.11) to minimise the variational energy $E = \langle \hat{H} \rangle$. Performing Wick's contractions, the variational energy becomes

$$\begin{aligned}
E &= \langle \hat{H} \rangle \\
&= \sum_{\alpha, \beta} A_{\alpha\beta} \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta} \rangle + \frac{1}{2} \sum_{\alpha, \beta, \gamma, \delta} B_{\alpha\beta\gamma\delta} \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \hat{c}_{\gamma} \hat{c}_{\delta} \rangle \\
&= \sum_{\alpha, \beta} A_{\alpha\beta} \langle \hat{a}_{\alpha}^{+} \hat{a}_{\beta}^{-} \rangle + \frac{1}{2} \sum_{\alpha, \beta, \gamma, \delta} B_{\alpha\beta\gamma\delta} \left(\langle \hat{a}_{\alpha}^{+} \hat{a}_{\beta}^{+} \rangle \langle \hat{a}_{\gamma}^{-} \hat{a}_{\delta}^{-} \rangle - \langle \hat{a}_{\alpha}^{+} \hat{a}_{\gamma}^{-} \rangle \langle \hat{a}_{\beta}^{+} \hat{a}_{\delta}^{-} \rangle + \langle \hat{a}_{\alpha}^{+} \hat{a}_{\delta}^{-} \rangle \langle \hat{a}_{\beta}^{+} \hat{a}_{\gamma}^{-} \rangle \right) \\
&= \sum_{\alpha, \beta} A_{\alpha\beta} \Delta_{\alpha\beta}^{+-} + \frac{1}{2} \sum_{\alpha, \beta, \gamma, \delta} B_{\alpha\beta\gamma\delta} \left(\Delta_{\alpha\beta}^{++} \Delta_{\gamma\delta}^{--} - \Delta_{\alpha\gamma}^{+-} \Delta_{\beta\delta}^{+-} + \Delta_{\alpha\delta}^{+-} \Delta_{\beta\gamma}^{+-} \right) \\
&= \sum_{\alpha, \beta} A_{\alpha\beta} \Delta_{\alpha\beta}^{+-} + \frac{1}{2} \sum_{\alpha, \beta, \gamma, \delta} (B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}) \Delta_{\alpha\beta}^{+-} \Delta_{\gamma\delta}^{+-} + \frac{1}{2} \sum_{\alpha, \beta, \gamma, \delta} B_{\alpha\beta\gamma\delta} \Delta_{\alpha\beta}^{++} \Delta_{\gamma\delta}^{--} \quad (1.14)
\end{aligned}$$

where in the second line, we inserted the relations (1.9). Our argument is completely contained in the following statement: the HF approximation to the real ground state it obtained as the ground state of a quadratic hamiltonian whose fields $h_{\alpha\beta}^{pq}$ satisfy the following relations:

$$h_{\alpha\beta}^{+-} = \frac{1}{2} \left[A_{\alpha\beta} + \sum_{\gamma, \delta} (B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}) \Delta_{\gamma\delta}^{+-} \right] \quad (1.15)$$

$$h_{\alpha\beta}^{++} = \sum_{\gamma, \delta} \frac{B_{\alpha\beta\gamma\delta} \Delta_{\gamma\delta}^{--}}{2} \quad (1.16)$$

These equations, operationally, give the instructions to build the best possible fermionic non-interacting hamiltonian to reproduce our original hamiltonian. These equations are proven as follows:

Proof 1.3 (Proof of Eqns. (1.15), (1.16)).

Consider E as expressed in Eq. (1.14). Upon taking the derivative, we obtain

$$\begin{aligned}
\frac{\partial E}{\partial h_{\mu\nu}^{rs}} &= \sum_{\alpha, \beta} A_{\alpha\beta} \frac{\partial \Delta_{\alpha\beta}^{+-}}{\partial h_{\mu\nu}^{rs}} + \frac{1}{2} \sum_{\alpha, \beta, \gamma, \delta} (B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}) \left(\frac{\partial \Delta_{\alpha\beta}^{+-}}{\partial h_{\mu\nu}^{rs}} \Delta_{\gamma\delta}^{+-} + \Delta_{\alpha\beta}^{+-} \frac{\partial \Delta_{\gamma\delta}^{+-}}{\partial h_{\mu\nu}^{rs}} \right) \\
&\quad + \frac{1}{2} \sum_{\alpha, \beta, \gamma, \delta} B_{\alpha\beta\gamma\delta} \left(\frac{\partial \Delta_{\alpha\beta}^{++}}{\partial h_{\mu\nu}^{rs}} \Delta_{\gamma\delta}^{--} + \Delta_{\alpha\beta}^{++} \frac{\partial \Delta_{\gamma\delta}^{--}}{\partial h_{\mu\nu}^{rs}} \right)
\end{aligned}$$

which, by using Eq. (1.4), becomes:

$$\begin{aligned}
\frac{\partial E}{\partial h_{\mu\nu}^{rs}} &= \sum_{\alpha, \beta} \left[A_{\alpha\beta} + \sum_{\gamma, \delta} (B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}) \Delta_{\gamma\delta}^{+-} \right] \frac{\partial \Delta_{\alpha\beta}^{+-}}{\partial h_{\mu\nu}^{rs}} \\
&\quad + \sum_{\alpha, \beta, \gamma, \delta} \frac{B_{\alpha\beta\gamma\delta} \Delta_{\gamma\delta}^{--}}{2} \frac{\partial \Delta_{\alpha\beta}^{++}}{\partial h_{\mu\nu}^{rs}} + \sum_{\alpha, \beta, \gamma, \delta} \frac{B_{\alpha\beta\gamma\delta} \Delta_{\alpha\beta}^{++}}{2} \frac{\partial \Delta_{\gamma\delta}^{--}}{\partial h_{\mu\nu}^{rs}}
\end{aligned}$$

Let us define

$$C_{\alpha\beta} \equiv \frac{1}{2} \left[A_{\alpha\beta} + \sum_{\gamma, \delta} (B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}) \Delta_{\gamma\delta}^{+-} \right] \quad D_{\alpha\beta} \equiv \sum_{\gamma, \delta} \frac{B_{\alpha\beta\gamma\delta} \Delta_{\gamma\delta}^{--}}{2} \quad (1.17)$$

From Fermi anticommutation rules, we have

$$\Delta_{\alpha\beta}^{+-} = \delta_{\alpha\beta} - \Delta_{\beta\alpha}^{-+} \implies \frac{\partial \Delta_{\alpha\beta}^{+-}}{\partial h_{\mu\nu}^{rs}} = -\frac{\partial \Delta_{\beta\alpha}^{-+}}{\partial h_{\mu\nu}^{rs}}$$

which implies

$$\begin{aligned} \sum_{\alpha,\beta} 2C_{\alpha\beta} \frac{\partial \Delta_{\alpha\beta}^{+-}}{\partial h_{\mu\nu}^{rs}} &= \sum_{\alpha,\beta} \left(C_{\alpha\beta} \frac{\partial \Delta_{\alpha\beta}^{+-}}{\partial h_{\mu\nu}^{rs}} + C_{\alpha\beta} \frac{\partial \Delta_{\alpha\beta}^{+-}}{\partial h_{\mu\nu}^{rs}} \right) \\ &= \sum_{\alpha,\beta} \left(C_{\alpha\beta} \frac{\partial \Delta_{\alpha\beta}^{+-}}{\partial h_{\mu\nu}^{rs}} - C_{\beta\alpha} \frac{\partial \Delta_{\beta\alpha}^{-+}}{\partial h_{\mu\nu}^{rs}} \right) \end{aligned}$$

Moreover, using Eqns. (1.3), (1.10),

$$\sum_{\alpha,\beta} \frac{B_{\alpha\beta\gamma\delta} \Delta_{\alpha\beta}^{++}}{2} = \sum_{\alpha,\beta} \left(\frac{B_{\delta\gamma\beta\alpha} \Delta_{\beta\alpha}^{--}}{2} \right)^*$$

and, recognising $D_{\gamma\delta}$ from Eq. (1.17), we conclude

$$\sum_{\alpha,\beta,\gamma,\delta} \frac{B_{\alpha\beta\gamma\delta} \Delta_{\alpha\beta}^{++}}{2} \frac{\partial \Delta_{\gamma\delta}^{--}}{\partial h_{\mu\nu}^{rs}} = \sum_{\gamma,\delta} D_{\delta\gamma}^* \frac{\partial \Delta_{\gamma\delta}^{--}}{\partial h_{\mu\nu}^{rs}}$$

Recollect everything and impose the solution to be a stationary point of energy,

$$\frac{\partial E}{\partial h_{\mu\nu}^{rs}} \stackrel{!}{=} 0$$

The notation “ $\stackrel{!}{=}$ ” indicates, throughout the thesis, that we are imposing some condition. In the end we have:

$$\frac{\partial E}{\partial h_{\mu\nu}^{rs}} = \sum_{\alpha,\beta} \left(C_{\alpha\beta} \frac{\partial \Delta_{\alpha\beta}^{+-}}{\partial h_{\mu\nu}^{rs}} - C_{\beta\alpha} \frac{\partial \Delta_{\alpha\beta}^{-+}}{\partial h_{\mu\nu}^{rs}} + D_{\alpha\beta} \frac{\partial \Delta_{\alpha\beta}^{++}}{\partial h_{\mu\nu}^{rs}} + D_{\beta\alpha}^* \frac{\partial \Delta_{\alpha\beta}^{--}}{\partial h_{\mu\nu}^{rs}} \right) \stackrel{!}{=} 0 \quad (1.18)$$

Comparing this equation with Eq. (1.11). It must hold:

$$h_{\alpha\beta}^{+-} = C_{\alpha\beta} \quad h_{\alpha\beta}^{++} = D_{\alpha\beta}$$

Recalling their explicit form from Eq. (1.17), we prove Eqns. (1.15), (1.16).

We notice that the right-hand side of both Eqns. (1.15), (1.16) depends on the fields themselves, as $\Delta_{\alpha\beta}^{pq}$ are defined as expectation values of two-points fermionic operators over the ground-state $|\Psi\rangle$. For a given set of fields $h_{\alpha\beta}^{pq}$, the latter is determined (up to some unitary transformation) as the ground-state of a fermionic non-interacting model. In other words, being $\mathbf{h}$ the tensor containing all fields,

$$\Delta_{\alpha\beta}^{pq}[\mathbf{h}] = \langle \Psi[\mathbf{h}] | \hat{a}_\alpha^p \hat{a}_\beta^q | \Psi[\mathbf{h}] \rangle \quad (1.19)$$

The three Eqns. (1.15), (1.16) and (1.19) define the solution of the HF variational method. Specifically, we need to determine $\Delta_{\alpha\beta}^{pq}$ such that, for given fields, Eq. (1.19) is respected.

The list of parameters $\{\Delta_{\alpha\beta}^{pq}\}$ to be determined self-consistently may be not that short: by looking at Eq. (1.19), the indices α, β, p, q are spanning a large Hilbert space. However, this equation can be manipulated and transformed, reducing the set of parameters to be determined self-consistently to a very manageable number. For each phase we define these last as “Hartree-Fock parameters” (HFPs): a set of objects self-consistently determined by a set equivalent, but transformed, version of Eq. (1.19).

The great advantage of this scheme is that, with an appropriate choice of the Hilbert space basis (for example, moving to reciprocal space for a translationally-invariant model), we can implement easily model symmetries.

1.2.3 Operative hamiltonian approximation

In this section we give an alternative expression for Eqns. (1.15), (1.16), by using an equivalent (but more straightforward) operatorial approximation. For a generic two-body quartic operator, the HF approximation is given by the decomposition

$$\begin{aligned}\hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\beta \hat{c}_\alpha &\stackrel{\text{HF}}{\simeq} \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \langle \hat{c}_\beta \hat{c}_\alpha \rangle + \langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \rangle \hat{c}_\beta \hat{c}_\alpha - \langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \rangle \langle \hat{c}_\beta \hat{c}_\alpha \rangle & (\text{Hartree}) \\ &\quad - \hat{c}_\alpha^\dagger \hat{c}_\beta \langle \hat{c}_\beta^\dagger \hat{c}_\alpha \rangle - \langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle \hat{c}_\beta^\dagger \hat{c}_\alpha + \langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\alpha \rangle & (\text{Fock}) \\ &\quad + \hat{c}_\alpha^\dagger \hat{c}_\alpha \langle \hat{c}_\beta^\dagger \hat{c}_\beta \rangle + \langle \hat{c}_\alpha^\dagger \hat{c}_\alpha \rangle \hat{c}_\beta^\dagger \hat{c}_\beta - \langle \hat{c}_\alpha^\dagger \hat{c}_\alpha \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\beta \rangle & (\text{Cooper})\end{aligned}$$

The proof of this statement is given in App. ??, where we also argue the interpretation of HF method as a mean-field theory (MFT). Let us summarise this section through a practical rule to decompose the quartic part of the local repulsion $\hat{H}_U$ and the non-local attraction $\hat{H}_V$ in the EHM, by applying Eq. (??).

Local repulsion. Applying the HF method, the local repulsion $\hat{H}_U$ is approximated by the quadratic operator:

$$\begin{aligned}\hat{H}_U &= U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} & (1.20) \\ &\stackrel{\text{HF}}{\simeq} U \sum_i \left(\hat{c}_{i\uparrow}^\dagger \hat{c}_{i\uparrow} \langle \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\downarrow} \rangle + \langle \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\uparrow} \rangle \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\downarrow} - \langle \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\uparrow} \rangle \langle \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\downarrow} \rangle \right. & (\text{Hartree}) \\ &\quad - \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\downarrow} \langle \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\uparrow} \rangle - \langle \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\downarrow} \rangle \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\uparrow} + \langle \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\downarrow} \rangle \langle \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\uparrow} \rangle & (\text{Fock}) \\ &\quad \left. + \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\downarrow}^\dagger \langle \hat{c}_{i\downarrow} \hat{c}_{i\uparrow} \rangle + \langle \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\downarrow}^\dagger \rangle \hat{c}_{i\downarrow} \hat{c}_{i\uparrow} - \langle \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\downarrow}^\dagger \rangle \langle \hat{c}_{i\downarrow} \hat{c}_{i\uparrow} \rangle \right) & (\text{Cooper})\end{aligned}$$

Non-local attraction. Applying the HF method, the local repulsion $\hat{H}_V$ is approximated by the quadratic operator:

$$\begin{aligned}\hat{H}_V &= -V \sum_{\langle ij \rangle} \sum_{\sigma, \sigma'} \hat{n}_{i\sigma} \hat{n}_{j\sigma'} & (1.21) \\ &\stackrel{\text{HF}}{\simeq} -V \sum_{\langle ij \rangle} \sum_{\sigma, \sigma'} \left(\hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} \langle \hat{c}_{j\sigma'}^\dagger \hat{c}_{j\sigma'} \rangle + \langle \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} \rangle \hat{c}_{j\sigma'}^\dagger \hat{c}_{j\sigma'} - \langle \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} \rangle \langle \hat{c}_{j\sigma'}^\dagger \hat{c}_{j\sigma'} \rangle \right. & (\text{Hartree}) \\ &\quad - \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma'} \langle \hat{c}_{j\sigma'}^\dagger \hat{c}_{i\sigma} \rangle - \langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma'} \rangle \hat{c}_{j\sigma'}^\dagger \hat{c}_{i\sigma} + \langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma'} \rangle \langle \hat{c}_{j\sigma'}^\dagger \hat{c}_{i\sigma} \rangle & (\text{Fock}) \\ &\quad \left. + \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma'}^\dagger \langle \hat{c}_{j\sigma'} \hat{c}_{i\sigma} \rangle + \langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma'}^\dagger \rangle \hat{c}_{j\sigma'} \hat{c}_{i\sigma} - \langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma'}^\dagger \rangle \langle \hat{c}_{j\sigma'} \hat{c}_{i\sigma} \rangle \right) & (\text{Cooper})\end{aligned}$$

We will use these decomposition referring to each sector in dedicated chapters.

1.3 The iHF algorithm

In this section we describe the algorithm we used to compute Eq. (1.19). We refer to this technique as “iterative Hartree Fock” (iHF). Let $\mathbf{v}$ be the vector containing all HFPs. Suppose we have N HFPs v_i

$$\mathbf{v} \in \mathbb{C}^N \quad \mathbf{v} \equiv \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix}$$

In order to get a physical solution, each HFP must satisfy a self-consistency equation similar to Eq. (1.19). Each equation has the structure:

$$v_i = (\text{Some non-linear function of } \mathbf{v})$$

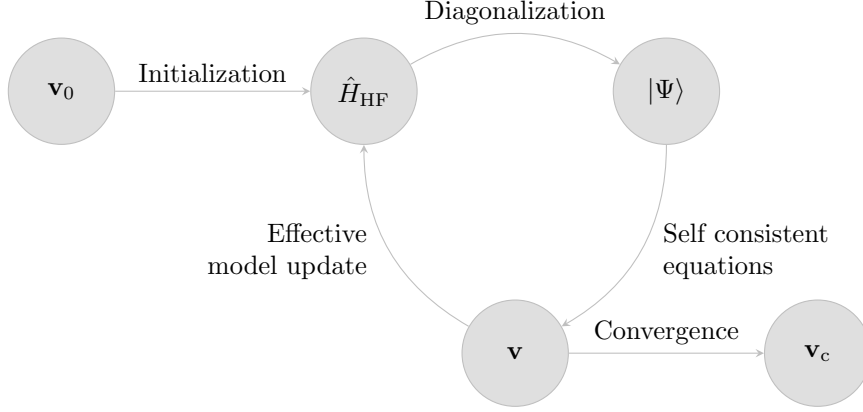


Figure 1.2 | Schematisation of the self-consistent Hartree Fock method described in Sec. 1.3. The effective model is solved, the solution is used to estimate self-consistently the Hartree Fock vector $\mathbf{v}$, which is then used to update iteratively the model itself. The notation $\mathbf{v}_0$ indicates the initialiser HFPs vector, $\mathbf{v}_c$ the converged HFPs vector.

Let A be the non-linear operator collecting all these functions. The problem of finding a self-consistent solution can be reformulated as follows: $\mathbf{v}$ represents a physical solution if it is a fixed point of the map A ,

$$A: \mathbb{C}^N \rightarrow \mathbb{C}^N \quad A(\mathbf{v}) = \mathbf{v} \quad \implies \quad \mathbf{v} \text{ is a solution.} \quad (1.22)$$

We implement the search for a self-consistent solution as follows: first, we obtain an analytic expression for A . Then we choose an initialiser $\mathbf{v}_0$ and apply A to it, obtaining a new HFPs vector. Iterating this procedure, we “follow” the non-linear map A until we converge. Specifically:

1. We start with a generic quartic hamiltonian:

$$\hat{H} = \sum_{\alpha\beta} A_{\alpha\beta} \hat{c}_\alpha^\dagger \hat{c}_\beta + \sum_{\alpha\beta\gamma\delta} B_{\alpha\beta\gamma\delta} \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\gamma \hat{c}_\delta$$

We choose the model parameters and set the electronic density target;

2. Following the discussion of Sec. 1.2, we approximate analytically $\hat{H}$ with a quadratic hamiltonian

$$\hat{H} \simeq \hat{H}_{\text{HF}} \equiv \sum_{\alpha\beta} \left[h_{\alpha\beta}^{+-} \hat{c}_\alpha^\dagger \hat{c}_\beta - h_{\beta\alpha}^{+-} \hat{c}_\alpha \hat{c}_\beta^\dagger + h_{\alpha\beta}^{++} \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger + \left(h_{\beta\alpha}^{++} \right)^* \hat{c}_\alpha \hat{c}_\beta \right]$$

with the fields given by Eqns. (1.15), (1.16);

3. We impose selection rules on the fields $h_{\alpha\beta}^{+-}$, $h_{\alpha\beta}^{++}$, eventually eliminating those not respecting the symmetry sector we are targeting. For example, if we choose not to break total charge conservation, set $\forall \alpha, \beta: h_{\alpha\beta}^{++} = 0$;
4. We choose a starting value for the HFPs vector $\mathbf{v}_0$;
5. We set $\mathbf{v} = \mathbf{v}_0$ and enter the iterative scheme of Fig. 1.2:
 - a) Get a numeric approximation of the hamiltonian for the current value of $\mathbf{v}$, $\hat{H}_{\text{HF}}[\mathbf{v}]$;
 - b) Determine the value $\mu[\mathbf{v}]$ giving the desired density;
 - c) Diagonalise the effective hamiltonian $\hat{H}_{\text{HF}}[\mathbf{v}] - \mu[\mathbf{v}] \hat{N}$, and obtain the approximated ground-state $|\Psi[\mathbf{v}]\rangle$;
 - d) Compute $A(\mathbf{v})$ using self-consistency equations;
 - e) If the algorithm converged, exit; otherwise, update the value of $\mathbf{v}$ repeat from point a).

In the following subsections we describe technical details concerning the chemical potential fixing and the convergence criteria used for the practical implementation of the above iterative scheme.

1.3.1 Fixed density simulations and the role of μ

In this thesis all derivations and simulations are carried out at fixed target density. This is done by fixing the chemical potential μ during iterations of the algorithm. This must be done iteratively: changing at each step the HFPs vector, we are actively modifying the approximate hamiltonian. Chemical potential must be determined accordingly to maintain the required density.

For a given choice of HFPs vector and chemical potential, the ground state $|\Psi[\mathbf{v}, \mu]\rangle$ is uniquely determined; then we compute density as

$$n(\mathbf{v}, \mu) \equiv \frac{1}{\mathcal{V}} \sum_{\alpha} \langle \Psi[\mathbf{v}, \mu] | \hat{c}_{\alpha}^{\dagger} \hat{c}_{\alpha} | \Psi[\mathbf{v}, \mu] \rangle$$

being $\mathcal{V}$ the volume. Suppose we fix density to a value $n_0 \in [0, 1]$. At step i , naming the current HFPs vector $\mathbf{v}_i$, μ_i (the chemical potential at current step) is determined as

$$\mu_i \equiv \min_{\mu} |n(\mathbf{v}_i, \mu) - n_0| \quad (1.23)$$

The density n must be a monotonically increasing function of μ : then we can translate the above equation, which would inspire a minimisation procedure, to a much simpler node-search rule:

$$\mu_i = \mu \mid_{\delta n(\mathbf{v}_i, \mu)=0} \quad \text{being} \quad \delta n(\mathbf{v}_i, \mu) \equiv n(\mathbf{v}_i, \mu) - n_0$$

In order to where the function changes sign (which is, the position of μ_i) we use a basic bisection algorithm:

1. Set an arbitrary tolerance $\Delta n > 0$ and a shifting parameter $\Delta\mu > 0$;
2. Choose starting lower and upper bounds $\mu_L < \mu_U$;
3. Compute $\delta n(\mathbf{v}_i, \mu_L)$. Evaluate:
 - if $|\delta n(\mathbf{v}_i, \mu_L)| \leq \Delta n$, set $\mu_i = \mu_L$ and halt;
 - if $\delta n(\mathbf{v}_i, \mu_L) < -\Delta n$ do nothing;
 - if $\delta n(\mathbf{v}_i, \mu_L) > \Delta n$, shift the lower bound down, $\mu_L \rightarrow \mu_L - \Delta\mu$ and repeat from step 3;
4. Compute $\delta n(\mathbf{v}_i, \mu_U)$. Evaluate:
 - if $|\delta n(\mathbf{v}_i, \mu_U)| \leq \Delta n$, set $\mu_i = \mu_U$ and halt;
 - if $\delta n(\mathbf{v}_i, \mu_U) > \Delta n$ do nothing;
 - if $\delta n(\mathbf{v}_i, \mu_U) < -\Delta n$, shift the upper bound up, $\mu_U \rightarrow \mu_U + \Delta\mu$ and repeat from step 4;
5. At this point, the the lower bound underestimates density while the upper bound overestimates. Start iterations:
 - a) Compute the midpoint,
$$\mu_M = \frac{\mu_L + \mu_U}{2}$$
 - b) Evaluate:
 - If $|\delta n(\mathbf{v}_i, \mu_M)| \leq \Delta n$, set $\mu_i = \mu_M$ and halt;
 - If $\delta n(\mathbf{v}_i, \mu_M) > \Delta n$, set $\mu_U = \mu_M$ and repeat from point a);
 - If $\delta n(\mathbf{v}_i, \mu_M) < -\Delta n$, set $\mu_L = \mu_M$ and repeat from point a);

The bottleneck of the procedure is our ability to compute expectation values. This depends on how much we are able to simplify analytically the computations for each phase.

As will turn out in the derivation, chemical potential will be renormalised by interaction. This means that applying self-consistent HF methods to the EHM will produce an effective hamiltonian containing a shift to the chemical potential,

$$\mu \rightarrow \mu + F[\mathbf{v}] \equiv \tilde{\mu}[\mathbf{v}]$$

being F some function of the HFPs. The chemical potential μ is an example of Renormalised Model Parameter (RMP).

Note that at step i of the iHF algorithm, our bisection procedure determines $\tilde{\mu}_i$, not μ_i . This consideration is relevant when dealing with ground state energy estimation. Let us abstract from the EHM and iHF, and consider a generic hamiltonian $\hat{H}$. To find the ground state $|\Psi\rangle$ means precisely to solve

$$\frac{\delta}{\delta \langle \Psi |} \langle \Psi | \hat{H} | \Psi \rangle = 0$$

Now, working with a fixed particles number N means we are performing a constrained minimisation (which is, taking the functional derivative) on the composite operator

$$\hat{H} - \mu(\hat{N} - N) \equiv \hat{K} + \mu N$$

being μ the Lagrange multiplier, $\hat{N}$ the number operator and $\hat{K} \equiv \hat{H} - \mu\hat{N}$. Finding the constrained ground state parametrised by a vector $\mathbf{v}$ now means to solve

$$\mathbf{v} \implies |\Psi[\mathbf{v}]\rangle \quad : \quad \begin{cases} \frac{\delta}{\delta \langle \Psi |} \langle \Psi | \hat{K} + \mu N | \Psi \rangle \Big|_{|\Psi\rangle=|\Psi[\mathbf{v}]\rangle} = 0 \\ \langle \Psi[\mathbf{v}] | \hat{N} | \Psi[\mathbf{v}] \rangle = N \end{cases}$$

In the first line μN is derived to zero; the constraint is imposed in the second line. Energy is given by:

$$\begin{aligned} \langle \hat{H} \rangle &= \langle \Psi[\mathbf{v}] | \hat{K} | \Psi[\mathbf{v}] \rangle + \mu N \\ &= \Omega[\mathbf{v}] + (\tilde{\mu}[\mathbf{v}] - F[\mathbf{v}])N \end{aligned}$$

being Ω the expectation value (EV) of $\hat{K}$. To correctly compute energy, we need to add $+\mu N$; since we determine $\tilde{\mu}[\mathbf{v}] \neq \mu$, everything needs to be expressed in terms of it. Using this relation, when the algorithm has converged, we can estimate ground-state energy.

1.3.2 The convergence scheme

Let us discuss briefly the convergence criterion of our algorithm and the iteration mechanism. The iHF scheme we design is a sequential loop that runs up to a logical halt condition. In order to specify such condition, we need to define our notion of “convergence”. Recall the definition of $A(\mathbf{v})$, the non-linear map in HFPs space whose component are given by a set of self-consistency equations: we claim to have reached convergence when $\mathbf{v}$ comes “close enough” to a fixed point of the map A , as in Eq. (1.22). Explicitly, we stop whenever the following logic condition is satisfied for each component of $\mathbf{v} = (v_1, v_2, \dots)$:

$$\forall v_j \quad : \quad |v_j - A_j(\mathbf{v})| \leq \Delta v_j$$

being $A(\mathbf{v})$ the map image of the HFP vector $\mathbf{v}$, and A_j its j -th component. The tolerance Δv_j is arbitrary. If just one component of $\mathbf{v}$ is mapped “too far” (with respect to Δv_j) the step is rejected and iterations continue. In this text we deal in with quantities of order 10^{-1} , so convergence values of order $10^{-3} \div 10^{-4}$ will be in general sufficient.

At step i , we check the distance of the two vectors $A(\mathbf{v}_i)$ and $\mathbf{v}_i$ and if possible, halt. If instead convergence is not reached, we need repeat the cycle with a new initialiser $\mathbf{v}_{i+1}$. A very simple choice would be to set $\mathbf{v}_{i+1} = \mathbf{v}_i$. We do something slightly different:

$$\mathbf{v}_{i+1} \equiv gA(\mathbf{v}_i) + (1 - g)\mathbf{v}_i \tag{1.24}$$

The “mixing parameter” g is a simple but useful knob to tune algorithmic evolution. Its usefulness comes from this observation: even if a fixed point exists, we don’t know how easy it is to actually get there following A . Setting $g = 1$, the new initialiser is just the image of the previous initialiser. By moving “slower” ($g < 1$) with an accurate optimisation of the mixing, it is possible to reach convergence in a controlled way.

Let us give some concrete justification about the relevance of controlling g . When it comes to HF iterative schemes of this kind, it can sometimes happen for the algorithm to remain stuck

in infinite loops. An example of this behaviour is given in Sec. 3.5.3. A typical situation is the following: during algorithm evolution we reach a couple of points $\mathbf{v}^{(a)}$, $\mathbf{v}^{(b)}$ such that

$$\dots \xrightarrow{A,g} \mathbf{v}^{(a)} \xrightarrow{A,g} \mathbf{v}^{(b)} \xrightarrow{A,g} \mathbf{v}^{(a)} \xrightarrow{A,g} \mathbf{v}^{(b)} \xrightarrow{A,g} \dots$$

With the notation “ $\xrightarrow{A,g}$ ” we intend one algorithmic step at mixing parameter g . We are stuck forever in this loop. Now, to predict if couples like $(\mathbf{v}^{(a)}, \mathbf{v}^{(b)})$ exist, how they depend on g , where they are located in HFPs space... is impossible at this level. Thus there is no immediate way to avoid these loops *a priori*. One workaround we found to work in most cases, however, was to just make g smaller. We gained empirical evidence that to “slow down” in HFPs space makes numeric instabilities less frequent. The trade-off, however, is that if we move too slow, runtime could get very large. Note that inaccurate setup of the algorithm could lead to resources-intensive runs also in “ordinary” situations, when we are not stuck in a loop.

To keep the number of iterations under control, we also defined an upper bound p to the total number of steps. If after p steps we algorithm failed to converge, it halts. In that case, $\mathbf{v}$ is saved as a “NaN” (Not a number) vector and we move to another simulation. Introducing this simple rule, runtime is controlled and we gain an easy method to identify the regions of parameter space requiring more care – just read from the raw data where NaNs have been saved and repeat simulations fine-tuning the parameters. This basically means to lower g and increase p . These considerations end this technical introductory chapter on iHF methods.

Chapter 2

Phase transitions and SSB in EHM

In this chapter we discuss the symmetries of the EHM, and describe the phases we will investigate in next chapter based on which symmetry they break. Then we connect this discussion with the HF method by deriving a set of selection rules on the HF approximation of the EHM hamiltonian for each phase.

This chapter serves as groundwork for the solution of the self-consistency problem posed in Eq. (1.19) for each phase. We start by enumerating and demonstrating the symmetries of the EHM.

2.1 The EHM symmetries

In this section we present the symmetries of the extended Hubbard model introduced in Eq. (??)

$$\hat{H}_{\text{EHM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma}}_{\hat{H}_t} + \underbrace{U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\hat{H}_U} - \underbrace{V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{n}_{i\sigma} \hat{n}_{j\sigma'}}_{\hat{H}_V}$$

The symmetries of the model can be divided in two groups: the ones related to the geometry of the square lattice, and those related to the algebraic properties of $\hat{H}_{\text{EHM}}$. We start with the lattice symmetries.

2.1.1 Lattice symmetries

We study the EHM on the planar square lattice, with one orbital per site. We indicate with L_x the number of sites in direction x , and L_y in direction y . The hamiltonian of Eq. (??) respects three fundamental lattice symmetries:

1. Discrete translations symmetry in both the x and y directions. We refer to the groups of one-site translations as $T_1^{(x)}$, $T_1^{(y)}$. It holds:

$$T_n^{(\ell)} \supset T_{kn}^{(\ell)} \quad k \in \mathbb{N} \quad (2.1)$$

A shift in direction ℓ by kn sites can be decomposed in k shift by n sites. Invariance by n -sites shifts, thus, implies invariance also for kn -sites shifts.

2. Discrete $\pi/2$ rotations symmetry. This symmetry is described by the four-fold cyclic point group C_4 , containing rotations by angle $\pi/2, \pi, 3\pi/2$ and identity. A similar property as for the translations holds for cyclic point groups,

$$C_n \supset C_{n/k} \quad k, n/k \in \mathbb{N} \quad (2.2)$$

Invariance for rotations of angle $2\pi/n$ implies invariance for rotations of angle $k \times 2\pi/n$.

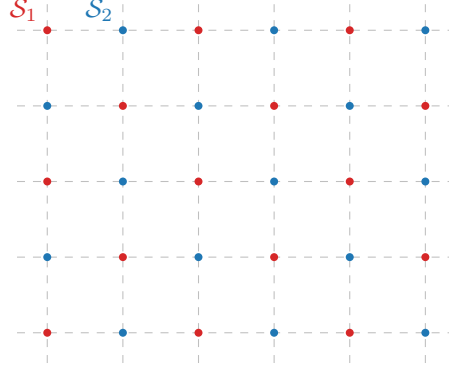


Figure 2.1 | Square lattice $\mathcal{L}$ bipartition in two sublattices $\mathcal{S}_1$ (red), $\mathcal{S}_2$ (blue).

3. Inversion symmetry. We define I_2 as the inversion group in two-dimensional space, containing the reflection operator and identity. This point may appear redundant, because

$$I_2 \equiv C_2 \quad (2.3)$$

In 2D space, to rotate by π around the origin and to mirror the position of a point are the same operation.

The planar square lattice is bipartite. Let us give a definition. Consider a graph $\mathcal{L}$ and the set of its vertices $V(\mathcal{L}) = \{v_i\}$. Let $E(\mathcal{L})$ be the set of graph edges,

$$E(\mathcal{L}) = \{(v, w) \mid v, w \in V(\mathcal{L}) \wedge v, w \text{ are connected}\}$$

The graph is said to be bipartite if two sub-graphs $\mathcal{S}_1, \mathcal{S}_2$ exist such that:

- The vertices of $\mathcal{L}$ split in the two sub-graphs,

$$V(\mathcal{L}) = V(\mathcal{S}_1) \cup V(\mathcal{S}_2)$$

- Vertices of one sub-graph are linked only to vertices of the other sub-graph. In other words

$$v, w \in \mathcal{S}_i \implies (v, w) \notin E(\mathcal{L})$$

The planar square lattice is bipartite: it can be divided in two square lattices, as represented in Fig. 2.1. Two sites are “connected” if linked by a non-zero hopping operator. Which is, if they are NNs. Note that one-site translations leave the hamiltonian invariant but exchange the two sublattices, while two-sites translations don’t.

2.1.2 Spin and charge symmetries

The conventional HM shows a global $U(2)$ symmetry, extended to $SU(2) \times SU(2)/\mathbb{Z}_2 = SO(4)$ symmetry at half-filling on a bipartite lattice [6], such as the square lattice. For the EHM, we limit to prove that we can identify a $U(2) \subset SO(4)$ global symmetry.

Gauge symmetry. The EHM is symmetric under the global transformation

$$\hat{c}_{i\sigma} \rightarrow e^{-i\eta} \hat{c}_{i\sigma} \quad \hat{c}_{i\sigma}^\dagger \rightarrow e^{i\eta} \hat{c}_{i\sigma}^\dagger \quad \eta \in \mathbb{R} \quad (2.4)$$

To prove this, it suffices to note that

$$\hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma'} \rightarrow e^{i\eta} \hat{c}_{i\sigma}^\dagger e^{-i\eta} \hat{c}_{j\sigma'} = \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma'} \quad (2.5)$$

Any product of fermionic operators with an equal number of annihilations and creations is gauge-symmetric. Hence the kinetic part $\hat{H}_t$ and the interactions $\hat{H}_U, \hat{H}_V$ are invariant.

We observe that the on-site density is gauge invariant,

$$\hat{n}_{i\sigma} \rightarrow \hat{n}_{i\sigma} \quad (2.6)$$

It follows that the total number of particles (charge)

$$\hat{N} = \sum_{i\sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma}$$

is a good quantum number.

Gauge symmetry is described by the transformation of Eq. (2.4), which is a pure phase shift by η . The gauge group is $U^c(1)$, with a superscript “c” indicating “charge”.

Global spin symmetry The EHM is described by three operators: the kinetic operator $\hat{H}_t$, independent of σ ; the local repulsion $\hat{H}_U$ between opposite-spin densities; the non-local attraction $\hat{H}_V$, independent of σ . If we rotate all spins by an equal amount, the hamiltonian does not change. An identical observation holds for the number operator, $\hat{N}$. Thus $\hat{H} - \mu\hat{N}$ must exhibit a global $SU(2)$ symmetry, related to the rotation in 3D space of all spins altogether.

Let us discuss this rigorously. We define the spinors $\hat{\Psi}_i$,

$$\hat{\Psi}_i \equiv \begin{bmatrix} \hat{c}_{i\uparrow} \\ \hat{c}_{i\downarrow} \end{bmatrix} \quad (2.7)$$

and the Pauli matrices

$$\tau^x \equiv \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} \quad \tau^y \equiv \begin{bmatrix} & -i \\ i & \end{bmatrix} \quad \tau^z \equiv \begin{bmatrix} 1 & \\ & -1 \end{bmatrix}$$

The local spin operator are defined as

$$\hat{S}_i^\alpha \equiv \frac{1}{2} \hat{\Psi}_i^\dagger \tau^\alpha \hat{\Psi}_i \quad [\hat{S}_j^\alpha, \hat{S}_j^\beta] = i \sum_\gamma \epsilon_{\alpha\beta\gamma} \hat{S}_j^\gamma \quad (2.8)$$

with $\epsilon_{\alpha\beta\gamma}$ the Levi-Civita tensor. The $\mathfrak{su}(2)$ spin algebra is actually respected:

Proof 2.1 (Spin operators respect the $\mathfrak{su}(2)$ algebra).

To prove that $\mathfrak{su}(2)$ algebra is respected, we must compute all possible commutators. This involves many very similar calculations. For the sake of conciseness, we sketch the proof for $\alpha = x, \beta = y$:

$$\begin{aligned} [\hat{S}_j^x, \hat{S}_j^y] &= \frac{1}{4} \left[\hat{\Psi}_j^\dagger \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} \hat{\Psi}_j, \hat{\Psi}_j^\dagger \begin{bmatrix} & -i \\ i & \end{bmatrix} \hat{\Psi}_j \right] \\ &= \frac{1}{4} \left[\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow} + \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow}, -i \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow} + i \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow} \right] \\ &= \frac{i}{4} \left[\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow}, \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow} \right] - \frac{i}{4} \left[\hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow}, \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow} \right] \\ &= \frac{i}{2} \left[\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow}, \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow} \right] \\ &= \frac{i}{2} \left(\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow} \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow} - \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow} \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow} \right) \\ &= \frac{i}{2} \left(-\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\downarrow} \hat{c}_{j\uparrow} + \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\uparrow} + \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\uparrow} \hat{c}_{j\downarrow} - \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\downarrow} \right) \\ &= \frac{i}{2} \left(\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\uparrow} - \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\downarrow} \right) \\ &= i \hat{S}_j^z \end{aligned}$$

We used Fermi anticommutation rules, $\{\hat{c}_{i\sigma}, \hat{c}_{j\sigma'}^\dagger\} = \delta_{ij} \delta_{\sigma\sigma'}$. Identical derivations can be carried

out for all other commutators, giving in the end the $\mathfrak{su}(2)$ spin algebra.

We define the on-site raising and lowering operators:

$$\hat{S}_j^+ \equiv \frac{\hat{S}_j^x + i\hat{S}_j^y}{2} \quad \hat{S}_j^- \equiv \frac{\hat{S}_j^x - i\hat{S}_j^y}{2}$$

and the global spin operators

$$\hat{S}^\alpha \equiv \sum_{i \in \mathcal{L}} \hat{S}_i^\alpha \quad [\hat{S}^\alpha, \hat{S}^\beta] = i \sum_{\gamma} \epsilon_{\alpha\beta\gamma} \hat{S}^\gamma$$

being $\mathcal{L}$ the lattice. Explicitly,

$$\hat{S}^z \equiv \frac{1}{2} \sum_{i \in \mathcal{L}} (\hat{n}_{i\uparrow} - \hat{n}_{i\downarrow}) \quad \hat{S}^+ \equiv \frac{1}{2} \sum_{i \in \mathcal{L}} \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\downarrow} \quad \hat{S}^- \equiv (\hat{S}^+)^\dagger$$

The EHM hamiltonian and the number operator commute with the global spin operator. For the kinetic part $\hat{H}_t$ and the local repulsion $\hat{H}_U$ this is a well known result [6, 30],

$$[\hat{H}_t + \hat{H}_U, \hat{\mathbf{S}}] = 0 \quad [\hat{N}, \hat{\mathbf{S}}] = 0$$

For the non-local attraction, since it depends solely on total densities on neighbouring sites, spin-rotational invariance must hold,

$$[\hat{H}_V, \hat{\mathbf{S}}] = 0 \tag{2.9}$$

Let us give a formal proof.

Proof 2.2 (The non-local attraction $\hat{H}_V$ is symmetric under global spin rotations.).

The proof is organised through a sequence of results:

1. The non-local attraction can be written as:

$$\begin{aligned} \hat{H}_V &= V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{n}_{i\sigma} \hat{n}_{j\sigma'} \\ &= V \sum_{\langle ij \rangle} \hat{\Psi}_i^\dagger \hat{\Psi}_i \hat{\Psi}_j^\dagger \hat{\Psi}_j \end{aligned} \tag{2.10}$$

having we used $\hat{\Psi}_i^\dagger \hat{\Psi}_i = \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\uparrow} + \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\downarrow}$.

2. Let now $\hat{R}(\boldsymbol{\theta})$ be any global spin rotation,

$$\hat{R}(\boldsymbol{\theta}) \equiv \exp(-i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}) \quad \text{where } \boldsymbol{\theta} = (\theta_x, \theta_y, \theta_z)$$

Spinors transform as:

$$\hat{R}(\boldsymbol{\theta}) \hat{\Psi}_i \hat{R}(\boldsymbol{\theta})^\dagger = \mathcal{R}(\boldsymbol{\theta}) \hat{\Psi}_i$$

with $\mathcal{R}(\boldsymbol{\theta}) \in \text{SU}(2)$ the unitary matrix associated to said rotation,

$$\mathcal{R}(\boldsymbol{\theta}) = \exp\left(-\frac{i}{2} \boldsymbol{\theta} \cdot \boldsymbol{\tau}\right) \quad \text{where } \boldsymbol{\tau} = (\tau_x, \tau_y, \tau_z) \tag{2.11}$$

Let us prove this. Let $\hat{A} = i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}$ e $\hat{B} = \hat{\Psi}_i$. The following identity, a consequence of Baker-Campbell-Housdorff formula, holds:

$$e^{\hat{A}} \hat{B} e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2!} [\hat{A}, [\hat{A}, \hat{B}]] + \frac{1}{3!} [\hat{A}, [\hat{A}, [\hat{A}, \hat{B}]]] + \dots$$

The first order of the expansion gives:

$$\begin{aligned}
[i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, \hat{\Psi}_i] &= i \sum_{\alpha} \theta_{\alpha} [\hat{S}^{\alpha}, \hat{\Psi}_i] \\
&= -\frac{i}{2} \sum_{\alpha} \theta_{\alpha} \tau^{\alpha} \hat{\Psi}_i \\
&= -\frac{i}{2} \boldsymbol{\theta} \cdot \boldsymbol{\tau} \hat{\Psi}_i
\end{aligned}$$

We omitted some purely algebraic passages. The second order:

$$\begin{aligned}
[i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, [i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, \hat{\Psi}_i]] &= [i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, -\frac{i}{2}(\boldsymbol{\theta} \cdot \boldsymbol{\tau}) \hat{\Psi}_i] \\
&= -\frac{i}{2} \boldsymbol{\theta} \cdot \boldsymbol{\tau} [i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, \hat{\Psi}_i] \\
&= \frac{1}{4} (\boldsymbol{\theta} \cdot \boldsymbol{\tau})^2 \hat{\Psi}_i
\end{aligned}$$

and so on. Collecting all terms,

$$\begin{aligned}
\hat{R}(\boldsymbol{\theta}) \hat{\Psi}_i \hat{R}(\boldsymbol{\theta})^{\dagger} &= \hat{\Psi}_i - \frac{i}{2} \boldsymbol{\theta} \cdot \boldsymbol{\tau} \hat{\Psi}_i + \frac{1}{2! \times 4} (\boldsymbol{\theta} \cdot \boldsymbol{\tau})^2 \hat{\Psi}_i + \dots \\
&= \sum_{k=1}^{\infty} \frac{1}{k!} \left(-\frac{i}{2} \boldsymbol{\theta} \cdot \boldsymbol{\tau} \right)^k \hat{\Psi}_i \\
&= \mathcal{R}(\boldsymbol{\theta}) \hat{\Psi}_i
\end{aligned}$$

In the last line, we identified the series as the Taylor expansion of the exponential, recognising the expression for $\mathcal{R}(\boldsymbol{\theta})$ given in Eq. (2.11).

3. Then, the group acts on the product $\hat{\Psi}_i^{\dagger} \hat{\Psi}_i$ as

$$\begin{aligned}
\hat{R} \hat{\Psi}_i^{\dagger} \hat{\Psi}_i \hat{R}^{\dagger} &= \hat{R} \hat{\Psi}_i^{\dagger} \hat{R}^{\dagger} \hat{R} \hat{\Psi}_i \hat{R}^{\dagger} \\
&= \hat{\Psi}_i^{\dagger} \mathcal{R}^{\dagger} \mathcal{R} \hat{\Psi}_i \\
&= \hat{\Psi}_i^{\dagger} \hat{\Psi}_i
\end{aligned}$$

being $\mathcal{R} \in \text{SU}(2)$. Recalling Eqns. (2.10), we conclude that $\hat{H}_V$ is $\text{SU}(2)$ symmetric – which means, it does not change under global spin rotations. This fact implies Eq. (2.9).

We observe that the above proof can be used to show that the on-site number operator is spin-invariant,

$$\hat{n}_{i\sigma} = \hat{\Psi}_i^{\dagger} \hat{\Psi}_i \implies [\hat{n}_{i\sigma}, \hat{\mathbf{S}}] = 0 \quad (2.12)$$

The EHM is symmetric under global spin rotations. Specifically, both $\hat{H}_U$ and $\hat{H}_V$ are symmetric also under local spin rotations, meaning that we could rotate the spin at each site by a different amount and get the same interaction. It is the kinetic operator that makes the EHM globally (and not locally) $\text{SU}(2)$ symmetric.

The $\text{U}^c(1)$ gauge symmetry and the $\text{SU}(2)$ spin rotations symmetry can be combined,

$$\text{U}^c(1) \otimes \text{SU}(2) = \text{U}(2)$$

We will refer to $\text{U}(2)$ symmetry to indicate full charge-spin conservation. Finally, note the group of 3D rotations contains the subgroup of 2D rotations around a given axis z ,

$$\text{U}^z(1) \subset \text{SU}(2)$$

where the z superscript serves to distinguish from the gauge group. The z axis does not need to be identified with the “real” z axis, perpendicular with the lattice plane. Any given direction works.

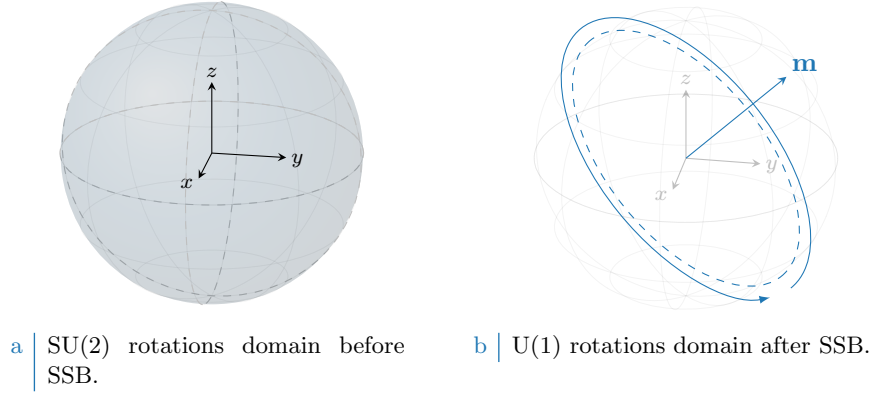


Figure 2.2 | Graphic representation of the concept of $SU(2) \rightarrow U^z(1)$ SSB. “Before” SSB, we have full symmetry under any possible rotation (solid color sphere, left panel). “After” SSB we have symmetry only for rotations around some vector $\mathbf{m}$ (dashed tilted circumference, right panel).

Inverting our logic, we can build the $SU(2)$ generators, starting from $\hat{S}^z$, after we have chosen the direction z . In terms of commutation relations, a given operator can respect $U^z(1)$ but not $SU(2)$: in that case, it will commute with $\hat{S}^z$, and not the other spin operators. A graphic rendition of this SSB is given in Fig. 2.2, representing for a given vector $\mathbf{m}$ the passage from full $SU(2)$ rotational symmetry around any direction to reduced $U^z(1)$ rotational symmetry around just one direction.

2.1.3 Particle-hole symmetry at half-filling

The half-filled EHM on the square lattice exhibits particle-hole symmetry (PHS). The presence of this additional symmetry is a consequence of the fact that the square lattice is bipartite. The generic unitary particle-hole (PH) transformation is given by

$$P : \hat{c}_{i\sigma} \rightarrow e^{i\eta_{i\sigma}} \hat{h}_{i\sigma}^\dagger \quad (2.13)$$

being in general η_i a phase dependent on the fermionic quantum state ($i\sigma$). Inserting the above transformation in the EHM, we get

$$P\{\hat{H}\} = -t \sum_{\langle ij \rangle} \sum_{\sigma} e^{i(\eta_{i\sigma} + \eta_{j\sigma})} \hat{h}_{i\sigma} \hat{h}_{j\sigma}^\dagger + U \sum_i \hat{h}_{i\uparrow} \hat{h}_{i\downarrow} \hat{h}_{i\downarrow}^\dagger \hat{h}_{i\uparrow}^\dagger - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma} \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^\dagger \hat{h}_{i\sigma}^\dagger$$

For the two interactions $\hat{H}_U, \hat{H}_V$ the phase factors cancelled out. For a specific choice of the η_i phases,

$$P : \begin{cases} \hat{c}_{i\sigma} \rightarrow \hat{h}_{i\sigma}^\dagger & \text{for } i \in \mathcal{S}_1 \\ \hat{c}_{i\sigma} \rightarrow -\hat{h}_{i\sigma}^\dagger & \text{for } i \in \mathcal{S}_2 \end{cases} \quad (2.14)$$

the above hamiltonian is mapped onto the EHM of (??),

$$P\{\hat{H}\} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma} + U \sum_{i \in \mathcal{L}} \hat{h}_{i\uparrow}^\dagger \hat{h}_{i\downarrow}^\dagger \hat{h}_{i\downarrow} \hat{h}_{i\uparrow} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^\dagger \hat{h}_{i\sigma}$$

The equality $P\{\hat{H}\} = \hat{H}$ expresses PHS, which holds as long as we work at half-filling.

Proof 2.3 (PHS at half filling, with η_i given by Eq. (2.14)).

This proof is an extension of the argument carried out for the conventional HM [6]. For the hole

operators, Fermi anticommutation rules hold regardlessly of the phase factor:

$$\begin{aligned}\{\hat{h}_{i\sigma}, \hat{h}_{j\sigma'}^\dagger\} &= e^{i(\eta_{i\sigma} - \eta_{j\sigma'})} \{\hat{c}_{i\sigma}^\dagger, \hat{c}_{j\sigma'}\} \\ &= e^{i(\eta_{i\sigma} - \eta_{j\sigma'})} \delta_{ij} \delta_{\sigma\sigma'} \\ &= \delta_{ij} \delta_{\sigma\sigma'}\end{aligned}$$

The on-site density operator is given by:

$$\hat{h}_{i\sigma} \hat{h}_{i\sigma}^\dagger = \hat{n}_{i\sigma}$$

being $\hat{n}_{i\sigma}$ the fermionic number operator on site i with spin σ . It follows, for quartic interactions

$$\begin{aligned}\hat{h}_{i\sigma} \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^\dagger \hat{h}_{i\sigma}^\dagger &= -\hat{h}_{i\sigma} \hat{h}_{j\sigma'} \hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma'}^\dagger \\ &= \hat{h}_{i\sigma} \hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^\dagger - \hat{h}_{i\sigma} \hat{h}_{j\sigma'}^\dagger \delta_{ij} \delta_{\sigma\sigma'} \\ &= -\hat{h}_{i\sigma}^\dagger \hat{h}_{i\sigma} \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^\dagger + \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^\dagger - \hat{h}_{i\sigma} \hat{h}_{j\sigma'}^\dagger \delta_{ij} \delta_{\sigma\sigma'} \\ &= \hat{h}_{i\sigma}^\dagger \hat{h}_{i\sigma} \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^\dagger - \hat{h}_{i\sigma}^\dagger \hat{h}_{i\sigma} + \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^\dagger - \hat{h}_{i\sigma} \hat{h}_{j\sigma'}^\dagger \delta_{ij} \delta_{\sigma\sigma'} \\ &= -\hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma'} \hat{h}_{i\sigma} \hat{h}_{j\sigma'}^\dagger - \hat{h}_{i\sigma}^\dagger \hat{h}_{i\sigma} + \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^\dagger - \left(\hat{h}_{i\sigma} \hat{h}_{j\sigma'}^\dagger - \hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma'} \right) \delta_{ij} \delta_{\sigma\sigma'} \\ &= \hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^\dagger \hat{h}_{i\sigma} + (\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) - 1 - \left(\hat{h}_{i\sigma} \hat{h}_{j\sigma'}^\dagger - \hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma'} \right) \delta_{ij} \delta_{\sigma\sigma'}\end{aligned}\quad (2.15)$$

The first term of the last passage is mirrored with respect to the starting term. In next paragraphs we derive the constraint of Eq. (2.14) and the requirement to work at half-filling.

Kinetic term. Consider the kinetic term,

$$\hat{H}_t = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma}$$

We apply the PH transform of Eq. (2.13)

$$P\{\hat{H}_t\} = -t \sum_{\langle ij \rangle} \sum_{\sigma} e^{i(\eta_{i\sigma} + \eta_{j\sigma})} \hat{h}_{i\sigma} \hat{h}_{j\sigma}^\dagger$$

Anti-commutating the two fermionic operators, and exchanging site labels $i \leftrightarrow j$, we get:

$$P\{\hat{H}_t\} = -t \sum_{\langle ij \rangle} \sum_{\sigma} e^{i(\eta_{i\sigma} + \eta_{j\sigma} + \pi)} \hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma}$$

To obtain $P\{\hat{H}_t\} \stackrel{!}{=} \hat{H}_t$, a necessary condition on the unitary operation P is the following,

$$\eta_{i\sigma} + \eta_{j\sigma} \stackrel{!}{=} (2k - 1)\pi \quad \text{with} \quad k \in \mathbb{Z} \quad (2.16)$$

valid for NN sites. This requirement on the phases η_i gives Eq. (2.14).

Local repulsion. Substitute Eq. (2.15) in $\hat{H}_U$,

$$\begin{aligned}P\{\hat{H}_U\} &= U \sum_{i \in \mathcal{L}} \hat{h}_{i\uparrow} \hat{h}_{i\downarrow} \hat{h}_{i\downarrow}^\dagger \hat{h}_{i\uparrow}^\dagger \\ &= U \sum_{i \in \mathcal{L}} \hat{h}_{i\uparrow}^\dagger \hat{h}_{i\downarrow}^\dagger \hat{h}_{i\downarrow} \hat{h}_{i\uparrow} + U \sum_{i \in \mathcal{L}} (\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow}) - U \sum_{i \in \mathcal{L}} (1) \\ &= U \sum_i \hat{h}_{i\uparrow}^\dagger \hat{h}_{i\downarrow}^\dagger \hat{h}_{i\downarrow} \hat{h}_{i\uparrow} + U(\hat{N} - L_x L_y)\end{aligned}$$

Symmetry	Operations
$T_1^{(x)} \otimes T_1^{(y)}$	One-site discrete translations separately in both directions
C_4	Four-fold point group (real $\pi/2$ angle rotations)
$U^c(1)$	Charge-conserving global phase shifts
$SU(2)$	Three-dimensional rotations in global spin space

Table 2.1 | Model symmetries and relative group operations. Note that $U^c(1)$ represents the gauge symmetry linked to charge conservation.

At half filling, $\langle \hat{N} \rangle = L_x L_y$. The unitary map of Eq. (2.13), independently of the condition of Eq. (2.16), leaves the hamiltonian identical at half-filling.

Non-local attraction. We proceed identically for $\hat{H}_V$,

$$\begin{aligned}
P\{\hat{H}_V\} &= -V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma} \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^\dagger \hat{h}_{i\sigma}^\dagger \\
&= -V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma'}^\dagger \hat{h}_{j\sigma'} \hat{h}_{i\sigma} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} (\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) + V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} (1) \\
&= -V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma'}^\dagger \hat{h}_{j\sigma'} \hat{h}_{i\sigma} - 2zV(\hat{N} - L_x L_y)
\end{aligned}$$

with $z = 4$ the lattice coordination number. In the last line, we used that on the square lattice

$$\begin{aligned}
\sum_{\langle ij \rangle} (1) &= \sum_{i \in \mathcal{L}/2} z \\
&= \frac{L_x L_y}{2} \times z
\end{aligned} \tag{2.17}$$

As before, the non-local attraction is particle-hole symmetric at half-filling. We conclude that the entire EHM exhibits PHS at half filling, if P is defined as in Eq. (2.14).

As for the conventional HM [6], this result can be generalised to any bipartite lattice. In turn, if a lattice is not bipartite, PHS is not present. To summarise, we report in Tab. 2.1 all identified symmetries.

2.2 The HF method and symmetry breaking

In Quantum Mechanics, a phase transition is described by the breaking of some natural symmetry of the hamiltonian. In this thesis we investigate three phases:

- Normal phase: no broken symmetry.
- Antiferromagnetic phase: simultaneous $SU(2) \xrightarrow{\text{SSB}} U^z(1)$ and $T_1^{(x)} \otimes T_1^{(y)} \xrightarrow{\text{SSB}} T_2^{(x)} \otimes T_2^{(y)}$. We study the so-called “commensurate antiferromagnetism”.
- Superconducting phase: simultaneous $U^c(1) \xrightarrow{\text{SSB}} 0$ and $C_4 \xrightarrow{\text{SSB}} C_2$. We investigate anisotropic superconductivity and breaking of planar rotational symmetry, observed in unconventional superconductors [24].

In the subsections that follow we discuss how to implement these SSBs in the context of the HF method. Indeed, we investigate each of these phases by approximating the EHM hamiltonian

$$\hat{H} \xrightarrow{\text{HF}} \hat{H}_{\text{HF}}$$

with an operator $\hat{H}_{\text{HF}}$ that respects the symmetries of the given phase. Hence, starting by the general decomposition of the two interactions $\hat{H}_U$ and $\hat{H}_V$, given respectively in Eqns. (1.20) and (1.21), we discuss the symmetry properties of each sector.

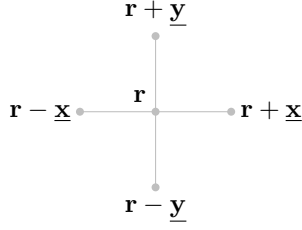


Figure 2.3 | Schematic representation of the four NNs of a given site i for a planar square lattice.

In next subsections, we switch from the site index $i \in \mathcal{L}$ to the vector notation $\mathbf{r} \in \mathbb{N}^2$, indicating the site 2D position on the lattice

$$i \rightarrow \mathbf{r} = (x, y)$$

with $x, y \in \mathbb{N}$. The lattice versors are:

$$\underline{\mathbf{x}} \equiv (1, 0) \quad \underline{\mathbf{y}} \equiv (0, 1)$$

as in Fig. 2.3.

2.2.1 Local repulsion

The local repulsion $\hat{H}_U$ is given by:

$$\begin{aligned} \hat{H}_U &= U \sum_{\mathbf{r} \in \mathcal{L}} \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}\downarrow} \\ &= U \sum_{\mathbf{r} \in \mathcal{L}} \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \end{aligned}$$

Applying Wick's theorem and using the denominations of Eq. (1.1) we have:

$$\langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \rangle = \underbrace{\langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\uparrow} \rangle \langle \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} \rangle \langle \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\uparrow} \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger \rangle \langle \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \rangle}_{\text{Cooper}} \quad (2.18)$$

where the equality holds exactly as long as we perform the averaging operation on the subspace of gaussian states. We now discuss the above EVs separately, obtaining selection rules for each phase.

Hartree term. The Hartree part of the above decomposition is a product of density EVs,

$$\langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle$$

From Eq. (2.6), (2.12) we know that single-site number operators are invariant under the entire group $U(2)$. Moreover, shifting or rotating the lattice leaves density operators unchanged. Thus no selection rule applies, and the above EVs are in principle non-zero.

Fock terms. For the Fock part, notice that

$$\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} = \hat{S}_{\mathbf{r}}^+$$

being $\hat{S}_{\mathbf{r}}^+$ the local raising operator. From $\mathfrak{su}(2)$ algebra we have:

$$[\hat{S}_{\mathbf{r}}^+, \hat{S}_{\mathbf{r}}^\alpha] \neq 0 \quad \alpha \in \{x, y, z\}$$

If the ground-state is invariant under global spin rotations, $\langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} \rangle$ must be zero. instead, gauge symmetry gives no selection rules, due to Eq. (2.5).

Term	Normal phase	Antiferromagnet	Superconductor
Hartree			
Fock	Prohibited	Prohibited	Prohibited
Cooper	Prohibited	Prohibited	

Table 2.2 Summary of selection rules for the Wick's decomposition of Eq. (2.18) of the local repulsion $\hat{H}_U$. An empty entry indicates no selection rule.

Cooper terms. The Cooper part can be non-zero only if charge is not conserved. In terms of the gauge symmetry, it suffices to apply the map of Eq. (2.4) to $\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger$

$$\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger \rightarrow e^{2i\eta} \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger$$

to conclude that this operator is in general not gauge invariant. It is instead SU(2)-symmetric.

Proof 2.4 (Local Cooper terms are SU(2)-symmetric).

We must compute the commutators of $\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger$ with $\hat{S}^z$, $\hat{S}^+$:

$$\begin{aligned} [\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{S}^z] &= \left[\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger, \frac{1}{2} \sum_{\mathbf{r}' \in \mathcal{L}} (\hat{n}_{\mathbf{r}'\uparrow} - \hat{n}_{\mathbf{r}'\downarrow}) \right] \\ &= \frac{1}{2} [\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}] \\ &= \frac{1}{2} [\hat{c}_{\mathbf{r}\uparrow}^\dagger, \hat{n}_{\mathbf{r}\uparrow}] \hat{c}_{\mathbf{r}\downarrow}^\dagger - \frac{1}{2} \hat{c}_{\mathbf{r}\uparrow}^\dagger [\hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{n}_{\mathbf{r}\downarrow}] \end{aligned}$$

Since:

$$\begin{aligned} [\hat{c}_{\mathbf{r}\sigma}^\dagger, \hat{n}_{\mathbf{r}\sigma}] &= \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} - \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}\sigma}^\dagger \\ &= -\hat{c}_{\mathbf{r}\sigma}^\dagger (1 - \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma}) \\ &= -\hat{c}_{\mathbf{r}\sigma}^\dagger \end{aligned} \tag{2.19}$$

we have:

$$[\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{S}^z] = -\frac{1}{2} \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger + \frac{1}{2} \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger = 0$$

Hence, $\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger$ is U^z(1)-symmetric.

Moreover, considering the raising operator:

$$\begin{aligned} [\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{S}^+] &= \left[\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger, \frac{1}{2} \sum_{\mathbf{r}' \in \mathcal{L}} \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} \right] \\ &= [\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}] \\ &= [\hat{c}_{\mathbf{r}\uparrow}^\dagger, \hat{c}_{\mathbf{r}\uparrow}^\dagger] \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} - (\hat{c}_{\mathbf{r}\uparrow}^\dagger)^2 [\hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{c}_{\mathbf{r}\downarrow}] \\ &= 0 \end{aligned}$$

where we used the facts that every operator commutes with itself, and a squared creation operator is null. These commutation relations, together, prove that Cooper EVs are legitimately non-zero for a SU(2)-preserving phase.

In Tab. 2.2 we collect the selection rules for the Hartree, Fock and Cooper terms of Eq. (2.18), based on the symmetry specifications given for each phase in above paragraphs. Note that Fock-like EVs are prohibited for all phases.

2.2.2 Non-local attraction

The non-local attraction $\hat{H}_V$ is given by:

$$\begin{aligned}\hat{H}_V &= -V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma, \sigma'} \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}'\sigma'} \\ &= V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma, \sigma'} \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}'\sigma'}^\dagger \hat{c}_{\mathbf{r}'\sigma'} \hat{c}_{\mathbf{r}\sigma}\end{aligned}\quad (2.20)$$

Let

$$\hat{H}_V^{\sigma\sigma'} \equiv -V \sum_{\langle \mathbf{r}, \mathbf{r}' \rangle} \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}'\sigma'}$$

The interaction decomposes in two “spin sectors”

$$\begin{aligned}\hat{H}_V &= \underbrace{\hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow}}_{\text{Same spin}} + \underbrace{\hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow}}_{\text{Opposite spin}} \\ &\equiv \hat{H}^{(\text{s.s.})} + \hat{H}^{(\text{o.s.})}\end{aligned}\quad (2.21)$$

where we further separated in s.s. and o.s. operators.

The operator $\hat{H}^{(\text{s.s.})}$ can be written as follows:

$$\begin{aligned}\hat{H}_V^{(\text{s.s.})} &\equiv \hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow} \\ &= -V \sum_{\mathbf{r} \in \mathcal{L}/2} \sum_{\delta \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}} \sum_{\sigma \in \{\uparrow, \downarrow\}} (\hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}+\delta\sigma} + \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}-\delta\sigma})\end{aligned}\quad (2.22)$$

where the sum over $\mathcal{L}/2$, in last line, means that we are summing over just one sublattice (it does not matter which one). Applying Wick’s theorem in the subspace of gaussian states, we get:

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r} \pm \delta\sigma} \rangle &= \langle \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r} \pm \delta\sigma}^\dagger \hat{c}_{\mathbf{r} \pm \delta\sigma} \hat{c}_{\mathbf{r}\sigma} \rangle \\ &= \underbrace{\langle \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} \rangle \langle \hat{c}_{\mathbf{r} \pm \delta\sigma}^\dagger \hat{c}_{\mathbf{r} \pm \delta\sigma} \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r} \pm \delta\sigma} \rangle \langle \hat{c}_{\mathbf{r} \pm \delta\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r} \pm \delta\sigma}^\dagger \rangle \langle \hat{c}_{\mathbf{r} \pm \delta\sigma} \hat{c}_{\mathbf{r}\sigma} \rangle}_{\text{Cooper}}\end{aligned}\quad (2.23)$$

Let us demonstrate Eq. (2.22).

Proof 2.5 (Proof of Eq. (2.22)).

Let $f(\mathbf{r}, \mathbf{r}')$ be a function of two variables on the square lattice. Summing over NNs, we get the identity

$$\sum_{\langle \mathbf{r}\mathbf{r}' \rangle} f(\mathbf{r}, \mathbf{r}') = \sum_{\mathbf{r}\mathbf{r}'} f(\mathbf{r}, \mathbf{r}') \left(\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}} \right)$$

where we used the notation of Fig. 2.3. The above sum is identically written as:

$$\sum_{\langle \mathbf{r}\mathbf{r}' \rangle} f(\mathbf{r}, \mathbf{r}') = \sum_{\mathbf{r} \in \mathcal{L}/2} \sum_{\delta \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}} [f(\mathbf{r}, \mathbf{r} + \delta) + f(\mathbf{r}, \mathbf{r} - \delta)] \quad (2.24)$$

The symbol “ $\mathcal{L}/2$ ” indicates one of the two sublattices $\mathcal{S}_1, \mathcal{S}_2$ represented in Fig. 2.1. Inserting Eq. (2.24) in Eq. (2.20), we obtain:

$$\hat{H}_V = -V \sum_{\mathbf{r} \in \mathcal{L}/2} \sum_{\delta \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}} \sum_{\sigma, \sigma'} (\hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}+\delta\sigma'} + \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}-\delta\sigma'}) \quad (2.25)$$

By taking the s.s. sector of the above equation, namely $\sigma = \sigma'$, we obtain Eq. (2.22).

The operator $\hat{H}^{(\text{o.s.})}$ can be written as follows:

$$\begin{aligned}\hat{H}_V^{(\text{o.s.})} &\equiv \hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow} \\ &= -V \sum_{\mathbf{r} \in \mathcal{L}} \sum_{\delta \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}} (\hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}+\delta\downarrow} + \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}-\delta\downarrow})\end{aligned}\quad (2.26)$$

From Wick's theorem:

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}\pm\delta\downarrow} \rangle &= \langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\pm\delta\downarrow}^\dagger \hat{c}_{\mathbf{r}\pm\delta\downarrow} \hat{c}_{\mathbf{r}\uparrow} \rangle \\ &= \underbrace{\langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\uparrow} \rangle \langle \hat{c}_{\mathbf{r}\pm\delta\downarrow}^\dagger \hat{c}_{\mathbf{r}\pm\delta\downarrow} \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\pm\delta\downarrow} \rangle \langle \hat{c}_{\mathbf{r}\pm\delta\downarrow}^\dagger \hat{c}_{\mathbf{r}\uparrow} \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\pm\delta\downarrow} \rangle \langle \hat{c}_{\mathbf{r}\pm\delta\downarrow}^\dagger \hat{c}_{\mathbf{r}\uparrow} \rangle}_{\text{Cooper}}\end{aligned}\quad (2.27)$$

Again, equality holds in the subspace of gaussian states. Let us prove Eq. (2.26).

Proof 2.6 (Proof of Eq. (2.26)).

Consider Eq. (2.25), restricting to $\sigma = \uparrow$, $\sigma' = \downarrow$ and setting $\mathcal{L}/2 = \mathcal{S}_1$. We obtain

$$\hat{H}_V^{\uparrow\downarrow} = -V \sum_{\mathbf{r} \in \mathcal{S}_1} \sum_{\delta \in \{\mathbf{x}, \mathbf{y}\}} (\hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}+\delta\downarrow} + \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}-\delta\downarrow})$$

Note that spin $\uparrow$ operators act on the sublattice $\mathcal{S}_1$; spin $\downarrow$ operators act on the sublattice $\mathcal{S}_2$. If instead, from Eq. (2.25), we restrict to $\sigma = \downarrow$, $\sigma' = \uparrow$ and set $\mathcal{L}/2 = \mathcal{S}_2$, we get:

$$\hat{H}_V^{\downarrow\uparrow} = -V \sum_{\mathbf{r} \in \mathcal{S}_2} \sum_{\delta \in \{\mathbf{x}, \mathbf{y}\}} (\hat{n}_{\mathbf{r}\downarrow} \hat{n}_{\mathbf{r}+\delta\uparrow} + \hat{n}_{\mathbf{r}\downarrow} \hat{n}_{\mathbf{r}-\delta\uparrow})$$

We observe that if $\mathbf{r} \in \mathcal{S}_2$, all its NNs belong to $\mathcal{S}_1$. Again, spin $\uparrow$ operators act on the sublattice $\mathcal{S}_1$; spin $\downarrow$ operators act on the sublattice $\mathcal{S}_2$.

Therefore, summing $\hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow}$, we obtain

$$\hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow} = -V \sum_{\mathbf{r} \in \mathcal{L}} \sum_{\delta \in \{\mathbf{x}, \mathbf{y}\}} (\hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}+\delta\downarrow} + \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}-\delta\downarrow})$$

which coincides with Eq. (2.26).

The separation of $\hat{H}_V$ in o.s. and s.s. sector is necessary in order to derive a set of selection rules for the Wick's contractions of Eqns. (2.23), (2.27). This is done in next paragraphs.

Hartree terms. As was for the local repulsion $\hat{H}_U$, the non-local Hartree terms of Eqns. (2.26), (2.22) are not subject to selection rules for any symmetry of Tab. 2.1. Thus they are allowed in both sectors.

Fock terms. The Fock EVs from the two sectors behave differently. Let us compute the following generic commutation relations:

$$\begin{aligned}\left[\hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}+\delta\sigma'}, \hat{S}^z \right] &= \left[\hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}+\delta\sigma'}, \frac{1}{2} \sum_{\mathbf{r}' \in \mathcal{L}} (\hat{n}_{\mathbf{r}'\uparrow} - \hat{n}_{\mathbf{r}'\downarrow}) \right] \\ &= \frac{(-1)^{\delta_{\sigma=\downarrow}}}{2} [\hat{c}_{\mathbf{r}\sigma}^\dagger, \hat{n}_{\mathbf{r}\sigma}] \hat{c}_{\mathbf{r}+\delta\sigma'} + \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2} \hat{c}_{\mathbf{r}\sigma}^\dagger [\hat{c}_{\mathbf{r}+\delta\sigma'}, \hat{n}_{\mathbf{r}+\delta\sigma'}] \\ &= \left(-\frac{(-1)^{\delta_{\sigma=\downarrow}}}{2} + \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2} \right) \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}+\delta\sigma'}\end{aligned}\quad (2.28)$$

where in last passage we used Eq. (2.19) and its conjugate version,

$$[\hat{c}_{\mathbf{r}+\delta\sigma'}, \hat{n}_{\mathbf{r}+\delta\sigma}] = \hat{c}_{\mathbf{r}+\delta\sigma'}$$

Now, if $\sigma = \sigma'$ the last passage of Eq. (2.28) is zero; otherwise is not. This proves that the o.s. Fock EVs are non-zero only for a phase that completely breaks SU(2); instead, in the s.s. sector we can have non-zero EVs for a phase that reduces SU(2) to U^z(1). [Commutation with $\hat{S}^+$]

Sector	Term	Normal phase	Antiferromagnet	Superconductor
o.s.	Hartree			
	Fock	Prohibited	Prohibited	Prohibited
	Cooper	Prohibited	Prohibited	
s.s.	Hartree			
	Fock			
	Cooper	Prohibited	Prohibited	Prohibited

Table 2.3 | Summary of selection rules for the Wick's decomposition of Eqns. (2.27), (2.23) of the non-local attraction $\hat{H}_V$. An empty entry indicates no selection rule.

Cooper terms. Cooper terms are not gauge invariant. The proof is analogous to the one given for $\hat{H}_U$. Instead, the o.s. and s.s. spin sectors behave differently under global spin rotations. With an identical procedure as for Eq. (2.28), we get:

$$\begin{aligned}
\left[\hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}+\delta\sigma'}^\dagger, \hat{S}^z \right] &= \left[\hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}+\delta\sigma'}^\dagger, \frac{1}{2} \sum_{\mathbf{r}' \in \mathcal{L}} (\hat{n}_{\mathbf{r}'\uparrow} - \hat{n}_{\mathbf{r}'\downarrow}) \right] \\
&= \frac{(-1)^{\delta_{\sigma=\downarrow}}}{2} [\hat{c}_{\mathbf{r}\sigma}^\dagger, \hat{n}_{\mathbf{r}\sigma}] \hat{c}_{\mathbf{r}+\delta\sigma'}^\dagger + \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2} \hat{c}_{\mathbf{r}\sigma}^\dagger [\hat{c}_{\mathbf{r}+\delta\sigma'}^\dagger, \hat{n}_{\mathbf{r}+\delta\sigma'}] \\
&= \left(-\frac{(-1)^{\delta_{\sigma=\downarrow}}}{2} - \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2} \right) \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}+\delta\sigma'}^\dagger
\end{aligned} \tag{2.29}$$

We have a mirrored situation with respect to the Fock terms. For the Cooper channel, the s.s. EVs are zero if $U^z(1)$ is unbroken, while the o.s. sector are not.

In Tab. 2.3 we summarise these considerations, listing a set of selection rules about the EVs of Eqns. (2.27), (2.23), dividing in the two sectors “o.s.” and “s.s.”. By identifying these selection rules, we set grounds for the detailed derivation of each chapter and heavily simplify application of HF model to the EHM. We now move to the last part of this chapter: reciprocal space description of the EHM.

2.3 The EHM in reciprocal space

The EHM is symmetric under discrete lattice translations. We will investigate phases that do not (completely) break translational invariance. Therefore, It is convenient to move to reciprocal space, by Fourier-transforming the hamiltonian. We will use the following convention:

$$\hat{c}_{\mathbf{r}\sigma} \stackrel{\mathcal{F}}{=} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k} \in \text{BZ}} e^{-i\mathbf{k} \cdot \mathbf{r}} \hat{c}_{\mathbf{k}\sigma} \tag{2.30}$$

Throughout the text, $\stackrel{\mathcal{F}}{=}$ will indicate a passage where a Fourier transform is used. “BZ” is shorthand notation for the Brillouin Zone.

It is useful to recall that the following equation for Kronecker's delta holds:

$$\frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} e^{i\mathbf{k} \cdot \mathbf{r}} = \delta_{\mathbf{k}=\mathbf{0}} \tag{2.31}$$

On one sublattice:

$$\frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}/2} e^{i\mathbf{k} \cdot \mathbf{r}} = \frac{1}{2} \delta_{\mathbf{k}=\mathbf{0}} \tag{2.32}$$

The notation $\mathcal{L}/2$ is shorthand to indicate $\mathcal{S}_1$ or $\mathcal{S}_2$.

Kinetic part. The kinetic operators transforms as follows:

$$\hat{H}_t \stackrel{\mathcal{F}}{=} \sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \tag{2.33}$$

where we defined the bare bands:

$$\epsilon_{\mathbf{k}} \equiv -2t (\cos k_x + \cos k_y) \quad (2.34)$$

Let us prove this first transform.

Proof 2.7 ($\mathcal{F}$ -transform of $\hat{H}_t$).

Consider the kinetic operator $\hat{H}_t$:

$$\hat{H}_t = -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{c}_{\mathbf{r}\sigma}^{\dagger} \hat{c}_{\mathbf{r}'\sigma} + \text{h.c.}]$$

Applying the convention of Eq. (2.30), we get:

$$\begin{aligned} \hat{H}_t &\stackrel{\mathcal{F}}{=} -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \left(\frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k} \in \text{BZ}} e^{i\mathbf{k} \cdot \mathbf{r}} \hat{c}_{\mathbf{k}\sigma}^{\dagger} \right) \left(\frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}' \in \text{BZ}} e^{-i\mathbf{k}' \cdot \mathbf{r}'} \hat{c}_{\mathbf{k}'\sigma} \right) + \text{h.c.} \\ &= -2t \sum_{\mathbf{k}, \mathbf{k}'} \sum_{\sigma} \hat{c}_{\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{k}'\sigma} \times \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}/2} e^{i(\mathbf{k}-\mathbf{k}') \cdot \mathbf{r}} \times \sum_{\delta \in \{\underline{x}, \underline{y}\}} e^{i\mathbf{k}' \cdot \delta} \end{aligned}$$

We used Eq. (2.24) and recognised that the h.c. term gives an identical contribution. We recognise in the central term the relation of Eq. (2.32). Hence,

$$\begin{aligned} \hat{H}_t &\stackrel{\mathcal{F}}{=} -2t \sum_{\mathbf{k}, \mathbf{k}'} \sum_{\sigma} \hat{c}_{\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{k}\sigma} \left[\frac{e^{ik_x} + e^{-ik_x}}{2} + \frac{e^{ik_y} + e^{-ik_y}}{2} \right] \\ &= \sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \hat{c}_{\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{k}\sigma} \end{aligned}$$

which proves Eq. (2.33).

Local repulsion. The local repulsion transforms as follows:

$$\hat{H}_U \stackrel{\mathcal{F}}{=} \frac{U}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \quad (2.35)$$

where sums over the variables $\mathbf{K}, \mathbf{k}, \mathbf{k}'$ must be intended as over the BZ.

Proof 2.8 ($\mathcal{F}$ -transform of $\hat{H}_U$).

Using the convention of Eq. (2.30), we get for $\hat{H}_U$:

$$\begin{aligned} \hat{H}_U &= U \sum_{\mathbf{r} \in \mathcal{L}} \hat{c}_{\mathbf{r}\uparrow}^{\dagger} \hat{c}_{\mathbf{r}\downarrow}^{\dagger} \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \\ &\stackrel{\mathcal{F}}{=} U \sum_{\mathbf{r} \in \mathcal{L}} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} e^{i\mathbf{k}_1 \cdot \mathbf{r}} \hat{c}_{\mathbf{k}_1\uparrow}^{\dagger} e^{i\mathbf{k}_2 \cdot \mathbf{r}} \hat{c}_{\mathbf{k}_2\downarrow}^{\dagger} e^{-i\mathbf{k}_3 \cdot \mathbf{r}} \hat{c}_{\mathbf{k}_3\downarrow} e^{-i\mathbf{k}_4 \cdot \mathbf{r}} \hat{c}_{\mathbf{k}_4\uparrow} \\ &= \frac{U}{L_x L_y} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \hat{c}_{\mathbf{k}_1\uparrow}^{\dagger} \hat{c}_{\mathbf{k}_2\downarrow}^{\dagger} \hat{c}_{\mathbf{k}_3\downarrow} \hat{c}_{\mathbf{k}_4\uparrow} \times \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)] \cdot \mathbf{r}} \\ &= \frac{U}{L_x L_y} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \hat{c}_{\mathbf{k}_1\uparrow}^{\dagger} \hat{c}_{\mathbf{k}_2\downarrow}^{\dagger} \hat{c}_{\mathbf{k}_3\downarrow} \hat{c}_{\mathbf{k}_4\uparrow} \delta_{\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4} \end{aligned}$$

where we used Eq. (2.31). Let $\mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}_3 + \mathbf{k}_4 = \mathbf{K}$, and define $\mathbf{k}, \mathbf{k}'$ such that

$$\mathbf{k}_1 \equiv \mathbf{K} + \mathbf{k} \quad \mathbf{k}_2 \equiv \mathbf{K} - \mathbf{k} \quad \mathbf{k}_3 \equiv \mathbf{K} - \mathbf{k}' \quad \mathbf{k}_4 \equiv \mathbf{K} + \mathbf{k}'$$

Substituting, we get the expression of Eq. (2.35).

Non-local attraction. In o.s. sector, the non-local attraction transforms as:

$$\hat{H}_V^{(o.s.)} \stackrel{\mathcal{F}}{=} -\frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \quad (2.36)$$

In the s.s. sector:

$$\begin{aligned} \hat{H}_V^{(s.s.)} \stackrel{\mathcal{F}}{=} & -\frac{V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \\ & \times \left[\hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\uparrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} + \hat{c}_{\mathbf{K}+\mathbf{k}\downarrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\downarrow} \right] \end{aligned} \quad (2.37)$$

Sums over the variables $\mathbf{K}, \mathbf{k}, \mathbf{k}'$ must be intended as over the BZ.

Proof 2.9 ($\mathcal{F}$ -transform of $\hat{H}_V$).

To prove Eqns. (2.36), (2.37), it is useful to start by defining:

$$h_{\mathbf{r}\sigma\sigma'} \equiv -V \sum_{\delta \in \{\underline{x}, \underline{y}\}} (\hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}+\delta\sigma'} + \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}-\delta\sigma'})$$

with:

$$\hat{H}_V = \sum_{\mathbf{r} \in \mathcal{L}/2} \sum_{\sigma, \sigma'} h_{\mathbf{r}\sigma\sigma'}$$

The product of densities on neighbouring sites transforms as:

$$\begin{aligned} \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r} \pm \delta\sigma'} &= \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r} \pm \delta\sigma'}^\dagger \hat{c}_{\mathbf{r} \pm \delta\sigma'} \hat{c}_{\mathbf{r}\sigma} \\ &\stackrel{\mathcal{F}}{=} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} e^{i[(\mathbf{k}_1 + \mathbf{k}_2) - (\mathbf{k}_3 + \mathbf{k}_4)] \cdot \mathbf{r}} e^{\pm i(\mathbf{k}_2 - \mathbf{k}_3) \cdot \delta} \hat{c}_{\mathbf{k}_1\sigma}^\dagger \hat{c}_{\mathbf{k}_2\sigma'}^\dagger \hat{c}_{\mathbf{k}_3\sigma'} \hat{c}_{\mathbf{k}_4\sigma} \end{aligned}$$

Then:

$$\begin{aligned} h_{\mathbf{r}\sigma\sigma'} &\stackrel{\mathcal{F}}{=} -\frac{V}{(L_x L_y)^2} \sum_{\delta \in \{\underline{x}, \underline{y}\}} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} e^{i[(\mathbf{k}_1 + \mathbf{k}_2) - (\mathbf{k}_3 + \mathbf{k}_4)] \cdot \mathbf{r}} \\ &\quad \times \left(e^{i(\mathbf{k}_2 - \mathbf{k}_3) \cdot \delta} + e^{-i(\mathbf{k}_2 - \mathbf{k}_3) \cdot \delta} \right) \hat{c}_{\mathbf{k}_1\sigma}^\dagger \hat{c}_{\mathbf{k}_2\sigma'}^\dagger \hat{c}_{\mathbf{k}_3\sigma'} \hat{c}_{\mathbf{k}_4\sigma} \\ &= -\frac{2V}{(L_x L_y)^2} \sum_{\delta \in \{\underline{x}, \underline{y}\}} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} e^{i[(\mathbf{k}_1 + \mathbf{k}_2) - (\mathbf{k}_3 + \mathbf{k}_4)] \cdot \mathbf{r}} \cos[(\mathbf{k}_2 - \mathbf{k}_3) \cdot \delta] \hat{c}_{\mathbf{k}_1\sigma}^\dagger \hat{c}_{\mathbf{k}_2\sigma'}^\dagger \hat{c}_{\mathbf{k}_3\sigma'} \hat{c}_{\mathbf{k}_4\sigma} \end{aligned}$$

The full non-local interaction is given by summing over all sites of $\mathcal{L}/2$ (which is, one sublattice). Applying Eq. (2.32) gives back momentum conservation

$$\frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}/2} e^{i[(\mathbf{k}_1 + \mathbf{k}_2) - (\mathbf{k}_3 + \mathbf{k}_4)] \cdot \mathbf{r}} = \frac{1}{2} \delta_{\mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}_3 + \mathbf{k}_4}$$

where we used Eq. (2.32). Define $\mathbf{k}, \mathbf{k}'$ and $\mathbf{K}$ as we did for the local repulsion, and

$$\delta \mathbf{k} \equiv \mathbf{k} - \mathbf{k}'$$

Hence:

$$\begin{aligned}
\hat{H}_V &= \sum_{\mathbf{r} \in \mathcal{L}/2} \sum_{\sigma, \sigma'} h_{\mathbf{r}\sigma\sigma'} \\
&\stackrel{\mathcal{F}}{=} -\frac{V}{L_x L_y} \sum_{\sigma\sigma'} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} \cos(\delta\mathbf{k} \cdot \boldsymbol{\delta}) \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\sigma'}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \\
&= -\frac{V}{L_x L_y} \sum_{\sigma\sigma'} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\sigma'}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \quad (2.38)
\end{aligned}$$

Finally, we can separate in same spin and opposite spin channels. This gives:

$$\begin{aligned}
\hat{H}_V &\stackrel{\mathcal{F}}{=} \overbrace{-\frac{V}{L_x L_y} \sum_{\sigma} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}^{\text{s.s.}} \\
&\quad - \underbrace{\frac{V}{L_x L_y} \sum_{\sigma} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\bar{\sigma}}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\bar{\sigma}} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}_{\text{o.s.}}
\end{aligned}$$

The s.s. part confirms Eq. (2.37). The o.s. part can be simplified: since

$$\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\bar{\sigma}}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\bar{\sigma}} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} = \hat{c}_{\mathbf{K}-\mathbf{k}\bar{\sigma}}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\bar{\sigma}}$$

due to fermionic anti-commutation rules, and since the composite transform

$$\mathbf{k} \rightarrow -\mathbf{k} \quad \mathbf{k}' \rightarrow -\mathbf{k}'$$

leaves the form factor $\cos(\delta k_x) + \cos(\delta k_y)$ invariant, we reduce the spin sum to two times the $\sigma = \uparrow$ part. This confirms Eq. (2.36)

In reciprocal space, following the general decomposition presented in Eq. (??) for a quartic operator $\hat{c}_{\mathbf{k}_1}^\dagger \hat{c}_{\mathbf{k}_2}^\dagger \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}$, we distinguish contractions as follows:

- Hartree contractions:

$$\overbrace{\hat{c}_{\mathbf{k}_1}^\dagger \hat{c}_{\mathbf{k}_2}^\dagger \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}} \quad \hat{c}_{\mathbf{k}_1}^\dagger \overbrace{\hat{c}_{\mathbf{k}_2}^\dagger \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}}$$

- Fock contractions:

$$\overbrace{\hat{c}_{\mathbf{k}_1}^\dagger \hat{c}_{\mathbf{k}_2}^\dagger \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}} \quad \hat{c}_{\mathbf{k}_1}^\dagger \overbrace{\hat{c}_{\mathbf{k}_2}^\dagger \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}}$$

- Cooper contractions:

$$\overbrace{\hat{c}_{\mathbf{k}_1}^\dagger \hat{c}_{\mathbf{k}_2}^\dagger \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}} \quad \hat{c}_{\mathbf{k}_1}^\dagger \hat{c}_{\mathbf{k}_2}^\dagger \overbrace{\hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}}$$

with appropriate sign factors.

2.4 Square lattice spatial symmetry channels

In this section we give two mathematical identities that are going to be used extensively. We define the set of functions $\varphi_{\mathbf{r}\mathbf{r}'}^{(\gamma)}$, with $\gamma \in \{s, s^*, p_x, p_y, d\}$, in Tab. 2.4; we plot them in Fig. 2.4. We will often refer to them as “harmonics”. By construction, these functions obey an orthonormality condition:

$$\frac{1}{L_x L_y} \sum_{\mathbf{r}\mathbf{r}'} \varphi_{\mathbf{r}\mathbf{r}'}^{(\gamma)} \varphi_{\mathbf{r}\mathbf{r}'}^{(\gamma')*} = \delta_{\gamma\gamma'} \quad (2.39)$$

which is proven by direct computation.

Channel	Form factor	Graph
s -wave	$\varphi_{\mathbf{r}\mathbf{r}'}^{(s)} = \delta_{\mathbf{r}\mathbf{r}'}$	Fig. 2.4a
s^* -wave	$\varphi_{\mathbf{r}\mathbf{r}'}^{(s^*)} = \frac{1}{2} [\delta_{\mathbf{r}'=\mathbf{r}+\underline{x}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{x}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{y}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{y}}]$	Fig. 2.4b
p_x -wave	$\varphi_{\mathbf{r}\mathbf{r}'}^{(p_x)} = \frac{1}{\sqrt{2}} [\delta_{\mathbf{r}'=\mathbf{r}+\underline{x}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{x}}]$	Fig. 2.4c
p_y -wave	$\varphi_{\mathbf{r}\mathbf{r}'}^{(p_y)} = \frac{1}{\sqrt{2}} [\delta_{\mathbf{r}'=\mathbf{r}+\underline{y}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{y}}]$	Fig. 2.4d
$d_{x^2-y^2}$ -wave	$\varphi_{\mathbf{r}\mathbf{r}'}^{(d)} = \frac{1}{2} [\delta_{\mathbf{r}'=\mathbf{r}+\underline{x}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{x}} - \delta_{\mathbf{r}'=\mathbf{r}+\underline{y}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{y}}]$	Fig. 2.4e

Table 2.4 | The harmonics $\varphi_{\mathbf{r}\mathbf{r}'}^{(\gamma)}$. The notation of Fig. 2.3 is used. In the last column, we indicate a graphic representation given in the subfigures Fig. 2.4.

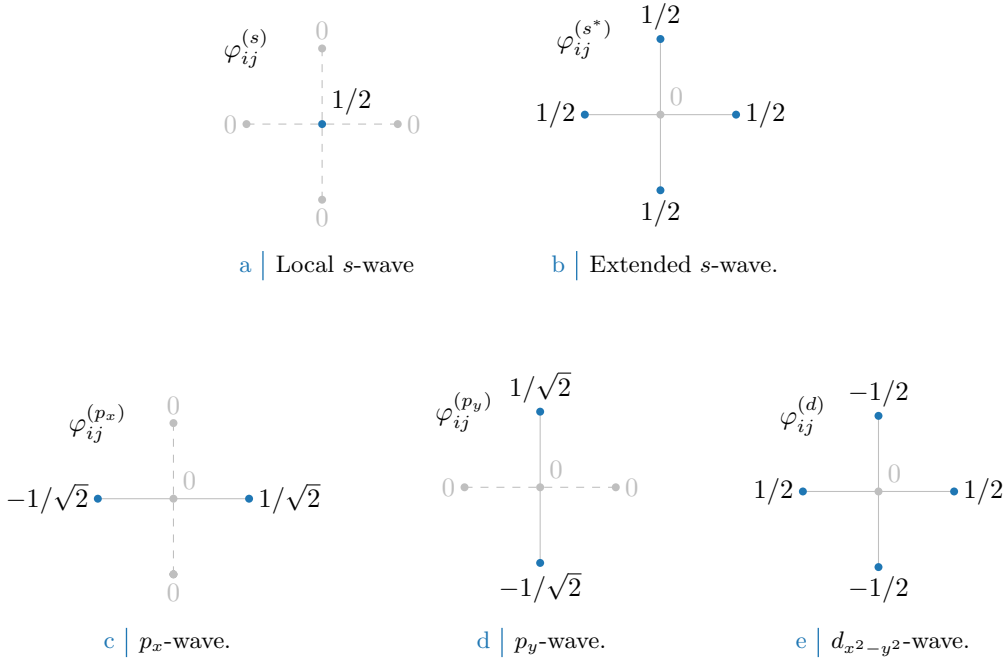


Figure 2.4 | The harmonics listed in Tab. 2.4. In figures five sites are represented: the hub site i and its four NNs. Solid lines connect the hub sites to a rim site where the harmonic is non-zero. Dashed lines connect to a rim site where the harmonic is zero.

Decomposition in symmetry channels. Consider a generic function of two variables $f(\mathbf{r}, \mathbf{r}')$. Suppose that, for a given centre $\mathbf{r}$, the function is non-zero only on the site itself ($\mathbf{r}' = \mathbf{r}$) and its NNs $\mathbf{r}' = \mathbf{r} \pm \boldsymbol{\delta}$, with $\boldsymbol{\delta} \in \{\underline{x}, \underline{y}\}$. In other words,

$$f(\mathbf{r}, \mathbf{r}') \neq 0 \iff |\mathbf{r} - \mathbf{r}'| = 0, 1 \quad (2.40)$$

in units of the lattice step. Using the harmonics $\varphi_{\mathbf{r}\mathbf{r}'}^{(\gamma)}$, we get the following decomposition:

$$f(\mathbf{r}, \mathbf{r}') = \sum_{\gamma} f^{(\gamma)} \varphi_{\mathbf{r}\mathbf{r}'}^{(\gamma)} \quad (2.41)$$

where we defined:

$$f^{(\gamma)} = \frac{1}{L_x L_y} \sum_{\mathbf{r}, \mathbf{r}'} f(\mathbf{r}, \mathbf{r}') \varphi_{\mathbf{r}\mathbf{r}'}^{(\gamma)} \quad (2.42)$$

Eq. (2.41) will be referred to as “decomposition in symmetry channels”. The sum over γ is extended to the five symmetries of Tab. 2.4.

Proof 2.10 (Decomposition in symmetry channels of Eq. (2.41)).

Consider a function $f(\mathbf{r}, \mathbf{r}')$ defined as in Eq. (2.40) and the harmonics of Tab. 2.4. By direct computation, the following identity holds

$$\begin{aligned} f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{x}}) &= \sum_{\mathbf{r}' \in \mathcal{L}} f(\mathbf{r}, \mathbf{r}') \delta_{\mathbf{r}' = \mathbf{r} + \underline{\mathbf{x}}} \\ &= \sum_{\mathbf{r}' \in \mathcal{L}} f(\mathbf{r}, \mathbf{r}') \left(\varphi_{\mathbf{r}\mathbf{r}'}^{(s^*)} + \varphi_{\mathbf{r}\mathbf{r}'}^{(d)} + \sqrt{2} \varphi_{\mathbf{r}\mathbf{r}'}^{(p_x)} \right) \end{aligned}$$

Similarly, also the following relations hold

$$\begin{aligned} f(\mathbf{r}, \mathbf{r}) &= \sum_{\mathbf{r}' \in \mathcal{L}} f(\mathbf{r}, \mathbf{r}') \varphi_{\mathbf{r}\mathbf{r}'}^{(s)} \\ f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{x}}) &= \sum_{\mathbf{r}' \in \mathcal{L}} f(\mathbf{r}, \mathbf{r}') \left(\varphi_{\mathbf{r}\mathbf{r}'}^{(s^*)} + \varphi_{\mathbf{r}\mathbf{r}'}^{(d)} - \sqrt{2} \varphi_{\mathbf{r}\mathbf{r}'}^{(p_x)} \right) \\ f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{y}}) &= \sum_{\mathbf{r}' \in \mathcal{L}} f(\mathbf{r}, \mathbf{r}') \left(\varphi_{\mathbf{r}\mathbf{r}'}^{(s^*)} - \varphi_{\mathbf{r}\mathbf{r}'}^{(d)} + \sqrt{2} \varphi_{\mathbf{r}\mathbf{r}'}^{(p_y)} \right) \\ f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{y}}) &= \sum_{\mathbf{r}' \in \mathcal{L}} f(\mathbf{r}, \mathbf{r}') \left(\varphi_{\mathbf{r}\mathbf{r}'}^{(s^*)} - \varphi_{\mathbf{r}\mathbf{r}'}^{(d)} - \sqrt{2} \varphi_{\mathbf{r}\mathbf{r}'}^{(p_y)} \right) \end{aligned}$$

Hence, by defining

$$\begin{aligned} f^{(s)} &\equiv f(\mathbf{r}, \mathbf{r}) \\ f^{(s^*)} &\equiv f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{x}}) + f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{x}}) + f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{y}}) + f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{y}}) \\ f^{(p_x)} &\equiv f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{x}}) - f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{x}}) \\ f^{(p_y)} &\equiv f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{y}}) - f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{y}}) \\ f^{(d)} &\equiv f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{x}}) + f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{x}}) - f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{y}}) - f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{y}}) \end{aligned}$$

we get the decomposition:

$$f(\mathbf{r}, \mathbf{r}') = \sum_{\gamma} f^{(\gamma)} \varphi_{\mathbf{r}\mathbf{r}'}^{(\gamma)}$$

The function f depends only on the difference of its two spatial variables. In other words, all the $f^{(\gamma)}$ components we defined above are independent of $\mathbf{r}$. Hence we can redefine them as in Eq. (2.42), averaging over $\mathbf{r}$. The decomposition obtained above still holds. With this redefinition of the $f^{(\gamma)}$ components, we recognise Eq. (2.41).

We decomposed the original function in its harmonics. Moving to reciprocal space, we get the $\mathcal{F}$ -transformed functions of Tab. 2.5, also satisfying the orthonormality condition:

$$\frac{1}{L_x L_y} \sum_{\mathbf{k}} \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma')*} = \delta_{\gamma\gamma'}$$

The functions of Tab. 2.5 are plotted in the BZ in Fig. 2.5.

A useful identity. We conclude this chapter by proving a useful mathematical identity:

$$\cos(\delta k_x) + \cos(\delta k_y) = \frac{1}{2} \sum_{\gamma \neq s} \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}'}^{(\gamma)*} \quad (2.43)$$

$$= \frac{1}{2} \sum_{\gamma \neq s} \varphi_{\mathbf{k}}^{(\gamma)*} \varphi_{\mathbf{k}'}^{(\gamma)} \quad (2.44)$$

Structure	Structure factor	Graph
s -wave	$\varphi_{\mathbf{k}}^{(s)} = 1$	Fig. 2.4a
Extended s -wave	$\varphi_{\mathbf{k}}^{(s^*)} = \cos k_x + \cos k_y$	Fig. 2.4b
p_x -wave	$\varphi_{\mathbf{k}}^{(p_x)} = i\sqrt{2} \sin k_x$	Fig. 2.4c
p_y -wave	$\varphi_{\mathbf{k}}^{(p_y)} = i\sqrt{2} \sin k_y$	Fig. 2.4d
$d_{x^2-y^2}$ -wave	$\varphi_{\mathbf{k}}^{(d)} = \cos k_x - \cos k_y$	Fig. 2.4e

Table 2.5 | $\mathcal{F}$ -transforms of the harmonics of Tab. 2.4. These functions are plotted in Fig. 2.5.

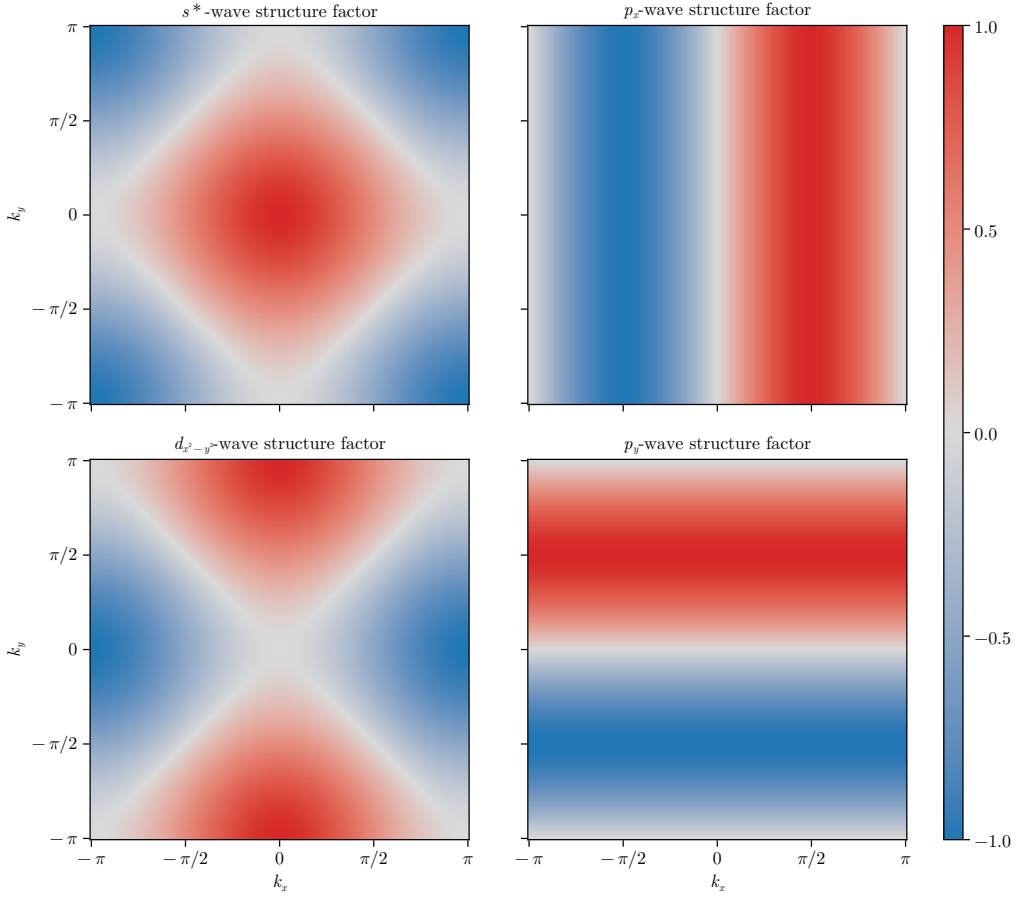


Figure 2.5 | Representation of the various structure factors listed in Tab. 2.5 on the BZ. For the s^* and $d_{x^2-y^2}$ -wave factors, the real part is plotted; for the p_ℓ factors, the imaginary part is plotted.

Note that in both rows above the sum is intended for $\gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\}$, excluding s .

Proof 2.11 (Proof of Eq. (2.43)).

Let $k_\ell \in \mathbb{R}$, with $\ell \in \{x, y\}$. We define:

$$c_\ell \equiv \cos k_\ell \quad s_\ell \equiv \sin k_\ell \quad c'_\ell \equiv \cos k'_\ell \quad s'_\ell \equiv \sin k'_\ell$$

Let $\delta k_\ell \equiv k_\ell - k'_\ell$. Using basic trigonometric identities we get:

$$\cos(\delta k_x) + \cos(\delta k_y) = \underbrace{(c_x c'_x + c_y c'_y)}_{\text{Symmetric}} + \underbrace{(s_x s'_x + s_y s'_y)}_{\text{Anti-symmetric}} \quad (2.45)$$

All four terms are invariant under simultaneous inversion of both momenta, $(\mathbf{k}, \mathbf{k}') \rightarrow (-\mathbf{k}, -\mathbf{k}')$. Instead, if we invert just one momentum of the pair, the first two are symmetric for both arguments $\mathbf{k}, \mathbf{k}'$; the second two are anti-symmetric. Decoupling the symmetric part, we obtain

$$c_x c'_x + c_y c'_y = \frac{1}{2}(c_x + c_y)(c'_x + c'_y) + \frac{1}{2}(c_x - c_y)(c'_x - c'_y)$$

which finally gives:

$$\begin{aligned} \cos(\delta k_x) + \cos(\delta k_y) &= \frac{1}{2}(c_x + c_y)(c'_x + c'_y) && (s^* \text{-wave}) \\ &+ s_x s'_x && (p_x \text{-wave}) \\ &+ s_y s'_y && (p_y \text{-wave}) \\ &+ \frac{1}{2}(c_x - c_y)(c'_x - c'_y) && (d_{x^2-y^2} \text{-wave}) \end{aligned}$$

which is written in compact form as in Eq. (2.43).

Chapter 3

The normal phase

In this chapter we apply the HF methods sketched in Sec. 1.2 to the “normal phase”. We are studying the model in a state where no symmetry of those enumerated in Tab. 2.1 is broken. The application of HF methods in this context will lead to a self-consistent renormalisation of the hopping parameter t , which will produce a Mott localisation effect.

This chapter is divided in three parts. First, we discuss the symmetries of the normal phase, following the general discussion of Chap. 2. Second, we apply HF methods to derive the self-consistent approximated hamiltonian and impose some theoretical constraints on the solution. Lastly, we present numeric results.

3.1 The normal phase and its symmetries

In the normal phase no symmetry is broken. Thus, the HF wavefunction approximating the ground state must exhibit the full $U(2) \otimes C_4 \otimes T_1^{(x)} \otimes T_1^{(y)}$ symmetry discussed in Sec. 2.1. Translational invariance and spin rotational symmetry imply that the site occupation is independent of site and spin indices,

$$\langle \hat{n}_{i\sigma} \rangle = n \quad (3.1)$$

with $n \in [0, 1]$.

We define the single-state occupation operator in reciprocal space:

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma}$$

and its EV

$$u_{\mathbf{k}\sigma} \equiv \langle \hat{n}_{\mathbf{k}\sigma} \rangle \quad u_{\mathbf{k}\sigma} = u_{\mathbf{k}\sigma}^* \quad (3.2)$$

The relation on the right follows from the fact that $\hat{n}_{\mathbf{k}\sigma}$ is hermitian by construction. Note that $\hat{n}_{\mathbf{k}\sigma}$ is not the Fourier-transform of $\hat{n}_{i\sigma}$. Translational invariance implies the following result:

$$\langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma'} \rangle = \delta_{\mathbf{k}=\mathbf{k}'} \delta_{\sigma=\sigma'} \langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \rangle \quad (3.3)$$

Proof 3.1 (Proof of Eq. (3.3)).

Consider the generic EV

$$\langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma'} \rangle$$

for two momenta $\mathbf{k}, \mathbf{k}'$. Moving to real space via an $\mathcal{F}$ -transform, we get

$$\langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma'} \rangle \stackrel{\mathcal{F}}{=} \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} \sum_{\mathbf{r}' \in \mathcal{L}} e^{-i(\mathbf{k} \cdot \mathbf{r} - \mathbf{k}' \cdot \mathbf{r}')} \langle \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}'\sigma'} \rangle$$

where we used the convention of Eq. (2.30). Let now $\Delta \mathbf{r} \equiv \mathbf{r}_2 - \mathbf{r}_1$. Translational invariance implies that

$$g(\Delta \mathbf{r}) = \langle \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}+\Delta \mathbf{r}\sigma'} \rangle$$

Operator	Sector	RWCs	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
	s.s.	Hartree Fock	μ shift by nzV Bands renormalisation and resymmetrization

Table 3.1 | Relevant Wick's contractions and net effect in the normal phase.

is independent of $\mathbf{r}$. Thus we have:

$$\begin{aligned} \langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma'}^\dagger \rangle &\stackrel{\mathcal{F}}{=} \sum_{\Delta\mathbf{r} \in \mathcal{L}} e^{-i\mathbf{k}' \cdot \Delta\mathbf{r}} g(\Delta\mathbf{r}) \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} e^{-i(\mathbf{k}-\mathbf{k}') \cdot \mathbf{r}} \\ &= \sum_{\Delta\mathbf{r} \in \mathcal{L}} e^{-i\mathbf{k}' \cdot \Delta\mathbf{r}} g(\Delta\mathbf{r}) \delta_{\mathbf{k}=\mathbf{k}'} \end{aligned}$$

where we used the identity of Eq. (2.31). Then the above EV vanishes for $\mathbf{k} \neq \mathbf{k}'$. This proves Eq. (3.3).

We anticipate here that we will actually allow for $C_4 \xrightarrow{\text{SSB}} C_2$, namely, breaking of planar rotational symmetry. As will result from our discussion, the normal phase does not break C_4 even if we allow it to. Nevertheless we carry out this general derivation because breaking of planar rotational symmetry will be observed in Chap. ?? in the context of superconductivity, and the computations carried out for the normal phase will be re-used.

Applying the HF methods to the interactions $\hat{H}_U$ and $\hat{H}_V$, from Tabs. 2.2 and 2.3, we know that in the normal phase the allowed contractions are:

$\hat{H}_U$	Hartree contractions
$\hat{H}_V$ (o.s. sector)	Hartree contractions
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions

As we will see in detail and is summarised in Tab. 3.1, Hartree contractions give a chemical potential shift, while the Fock contraction renormalises the bare bands.

In the normal phase, spin is a pure physical degeneracy, being the $\uparrow$ and $\downarrow$ sectors separated; moreover, $\hat{H}$ is $\mathbb{Z}_2$ invariant with respect to spin inversion, $\sigma \leftrightarrow \bar{\sigma}$. This implies that all physical quantities, as is for the normal phase, should not depend on the spin index. We will perform all computations leaving σ explicit and using this consideration at the end.

3.2 Application of HF methods to the normal phase

The quartic part of $\hat{H}$ is given by the the local repulsion $\hat{H}_U$ and the non-local attraction $\hat{H}_V$. Applying HF methods, we get the decomposition of Eqns. (1.20), (1.21). In the next section we analyse the Hartree channel, which contains all Hartree-like terms. We will see that $\hat{H}_U$ and $\hat{H}_V$ are approximated, in this channel, by a multiple of the number operator. In the section that follows, we deal with Fock terms, and show how $\hat{H}_V$ is approximated, in that channel, by an operator that is formally equivalent to the kinetic operator $\hat{H}_t$.

3.2.1 Hartree effects: chemical potential renormalisation

Consider the Hartree channel of Eqns. (1.20), (1.21),

$$\begin{aligned} \hat{H}_U &\stackrel{\text{HF}}{\simeq} U \sum_i \left(\hat{c}_{i\uparrow}^\dagger \hat{c}_{i\uparrow} \langle \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\downarrow} \rangle + \langle \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\uparrow} \rangle \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\downarrow} - \langle \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\uparrow} \rangle \langle \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\downarrow} \rangle \right) + (\text{all the rest}) \\ \hat{H}_V &\stackrel{\text{HF}}{\simeq} -V \sum_{\langle ij \rangle} \sum_{\sigma, \sigma'} \left(\hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} \langle \hat{c}_{j\sigma'}^\dagger \hat{c}_{j\sigma'} \rangle + \langle \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} \rangle \hat{c}_{j\sigma'}^\dagger \hat{c}_{j\sigma'} - \langle \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} \rangle \langle \hat{c}_{j\sigma'}^\dagger \hat{c}_{j\sigma'} \rangle \right) + (\text{all the rest}) \end{aligned}$$

Let us deal with those separately.

Local repulsion. In the normal phase, the local repulsion $\hat{H}_U$ decomposes in Hartree channel as specified in Eq. (1.20)

$$\hat{H}_U \stackrel{\text{HF}}{\simeq} U \sum_i (\langle \hat{n}_{i\uparrow} \rangle \hat{n}_{i\downarrow} + \hat{n}_{i\uparrow} \langle \hat{n}_{i\downarrow} \rangle - \langle \hat{n}_{i\downarrow} \rangle \langle \hat{n}_{i\uparrow} \rangle)$$

For a translationally invariant phase such as the normal phase, it holds

$$\langle \hat{n}_{i\sigma} \rangle = n \quad (3.4)$$

which gives

$$\begin{aligned} \hat{H}_U &\stackrel{\text{HF}}{\simeq} nU \sum_i (\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow}) - U \sum_i n^2 \\ &= nU \times \hat{N} - E_{\text{H}/U}^{(\text{N})} \end{aligned} \quad (3.5)$$

being $\hat{N}$ the total number operator and $E_{\text{H}/U}^{(\text{N})}$ the “local Hartree contraction energy” (also known as double counting term),

$$E_{\text{H}/U}^{(\text{N})} = L_x L_y \times n^2 U \quad (3.6)$$

This energy contributes negatively to the ground-state energy, for $U > 0$. The chemical potential is corrected by

$$\mu \rightarrow \mu - nU \quad (3.7)$$

The non-local attraction further corrects μ .

Non-local attraction. The non-local attraction is divided in two sectors, as in Eq. (2.21). From Eq. (1.21), its Hartree-like terms are:

$$\begin{aligned} \hat{H}_V &\stackrel{\text{HF}}{\simeq} \overbrace{-V \sum_{\langle ij \rangle} \sum_{\sigma} (\langle \hat{n}_{i\sigma} \rangle \hat{n}_{j\sigma} + \hat{n}_{i\sigma} \langle \hat{n}_{j\sigma} \rangle - \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\sigma} \rangle)}^{\text{s.s.}} \\ &\quad \underbrace{-V \sum_{\langle ij \rangle} \sum_{\sigma} (\langle \hat{n}_{i\sigma} \rangle \hat{n}_{j\bar{\sigma}} + \hat{n}_{i\sigma} \langle \hat{n}_{j\bar{\sigma}} \rangle - \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\bar{\sigma}} \rangle) + (\text{all the rest})}_{\text{o.s.}} \end{aligned}$$

where “all the rest” collects all non-Hartree terms. From Eq. (3.4), we get

$$\hat{H}_V \stackrel{\text{HF}}{\simeq} \underbrace{-nV \sum_{\langle ij \rangle} \sum_{\sigma} (\hat{n}_{j\sigma} + \hat{n}_{i\sigma})}_{\text{s.s.}} \underbrace{-nV \sum_{\langle ij \rangle} \sum_{\sigma} (\hat{n}_{j\bar{\sigma}} + \hat{n}_{i\sigma})}_{\text{o.s.}} - E_{\text{H}/V}^{(\text{N})} + (\text{all the rest}) \quad (3.8)$$

with $E_{\text{H}/V}^{(\text{N})}$ the normal state shift to energy brought by $\hat{H}_V$:

$$\begin{aligned} E_{\text{H}/V}^{(\text{N})} &= -V \sum_{\langle ij \rangle} \sum_{\sigma} (\langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\sigma} \rangle + \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\bar{\sigma}} \rangle) \\ &= -2V \sum_{\langle ij \rangle} \sum_{\sigma} n^2 \\ &= -L_x L_y \times 2z n^2 V \end{aligned} \quad (3.9)$$

Note the similarity with Eq. (3.6).

Getting back to Eq. (3.8), both sums give a number operator. Note that we sum over NNs: each site is connected to $z = 4$ sites. Including this combinatorial factor, we have

$$\hat{H}_V \simeq -2nzV \times \hat{N} - E_{\text{H}/V}^{(\text{N})} + (\text{all the rest}) \quad (3.10)$$

This accounts for the final Hartree shift of μ ,

$$\mu \rightarrow \mu + 2znV \quad (3.11)$$

Collecting everything, we get the HF approximation to the two interactions in the Hartree channel:

$$\hat{H}_U + \hat{H}_V \stackrel{\text{HF}}{\simeq} -n(2zV - U)\hat{N}$$

Considering the net shift to μ due to both interactions by combining Eqns. (3.7), (3.11), we get the final RMP anticipated in Tab. 3.1,

$$\tilde{\mu} \equiv \mu + n(2zV - U) \quad (3.12)$$

This result will remain valid throughout all phases discussed in this text.

3.2.2 Fock effects: bare bands renormalisation

We move to Fock terms. From Wick's decomposition of $\hat{H}_V$, the unique allowed Fock term comes from the same-spin sector. This selection rules is implied by the SU(2) symmetry of the normal phase and was stated in Tab. 2.3. Then, from Eq. (1.21):

$$\hat{H}_V \stackrel{\text{HF}}{\simeq} (\text{Hartree channel}) + V \sum_{\langle ij \rangle} \sum_{\sigma} \left(\langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \rangle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} + \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \langle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} \rangle - \langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \rangle \langle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} \rangle \right) \quad (3.13)$$

Note the plus sign in front of the displayed term: it comes from the minus sign in Fock term of Eq. (1.1).

Consider the Fourier Transform $\mathcal{F}$ given in Eq. (2.37), which gives

$$\begin{aligned} \hat{H}_V \stackrel{\text{HF}}{\simeq} (\text{Hartree channel}) \\ + \frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} \sum_{\sigma} (\cos(\delta k_x) + \cos(\delta k_y)) \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} - E_{\text{F/V}}^{(N)} \end{aligned} \quad (3.14)$$

Here, the 2 prefactor comes from the fact that the first two operators in square brackets in Eq. (3.13) are equal in reciprocal space. The double counting energy shift is given by

$$E_{\text{F/V}}^{(N)} \equiv \frac{V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} \sum_{\sigma} (\cos(\delta k_x) + \cos(\delta k_y)) \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \langle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \rangle \quad (3.15)$$

Recall the definition of $u_{\mathbf{k}\sigma}$ given in Eq. (3.2). Using the functions of Tab. 2.5, we define its harmonics:

$$u_{\sigma}^{(\gamma)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \rangle \quad (3.16)$$

The non-local attraction $\hat{H}_V$ is approximated, in Fock channel, by:

$$\hat{H}_V \stackrel{\text{HF}}{\simeq} (\text{Hartree channel}) + V \sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \left(u_{\sigma}^{(s^*)} \varphi_{\mathbf{k}}^{(s^*)*} + u_{\sigma}^{(d)} \varphi_{\mathbf{k}}^{(d)*} \right) \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} - E_{\text{F/V}}^{(N)} \quad (3.17)$$

We provide a proof.

Proof 3.2 (Proof of Eq. (3.17)).

Consider Eq. (3.14). Applying Eq. (3.3), it becomes:

$$\begin{aligned} \hat{H}_V \stackrel{\text{HF}}{\simeq} (\text{Hartree channel}) \\ + \frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} (\cos 2k_x + \cos 2k_y) \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} - E_{\text{F/V}}^{(N)} \end{aligned} \quad (3.18)$$

Let now $\mathbf{q} \equiv \mathbf{K} + \mathbf{k}$, $\mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$. A representation of this variables change is given in Fig. 3.1.

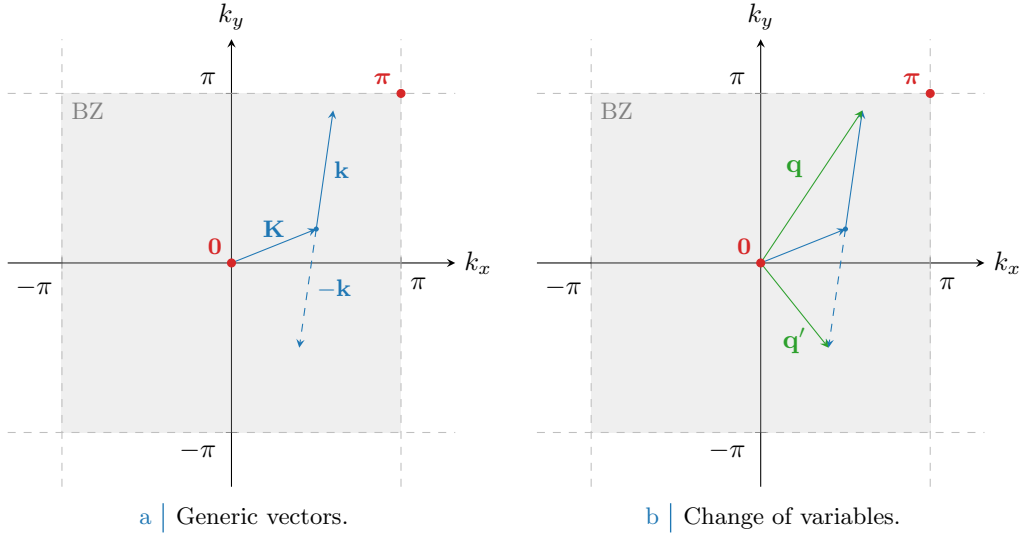


Figure 3.1 Representation of the variable change employed to pass from Eq. (3.18) to (3.19). In Fig. 3.1a generic vectors are considered, cycling over all values of $\mathbf{K}, \mathbf{k} \in \text{BZ}$. In Fig. 3.1b is depicted the variables change to the new vectors $\mathbf{q}, \mathbf{q}'$.

Being (for $\ell = x, y$)

$$\begin{aligned}\delta q_\ell &\equiv q_\ell - q'_\ell \\ &= (K_\ell + k_\ell) - (K_\ell - k_\ell) \\ &= 2k_\ell\end{aligned}$$

we reduce Eq. (3.14) to

$$\begin{aligned}\hat{H}_V &\stackrel{\text{HF}}{\simeq} (\text{Hartree channel}) \\ &+ \frac{2V}{L_x L_y} \sum_{\mathbf{q}, \mathbf{q}'} \sum_{\sigma} [\cos(\delta q_x) + \cos(\delta q_y)] \underbrace{\langle \hat{c}_{\mathbf{q}\sigma}^\dagger \hat{c}_{\mathbf{q}\sigma} \rangle}_{u_{\mathbf{q}\sigma}} \hat{c}_{\mathbf{q}'\sigma}^\dagger \hat{c}_{\mathbf{q}'\sigma} - E_{\text{F}/V}^{(N)}\end{aligned}\quad (3.19)$$

where we recognised $u_{\mathbf{q}\sigma}$, as was defined in Eq. (3.2). Using Eq. (2.43), we obtain

$$\begin{aligned}\hat{H}_V &\stackrel{\text{HF}}{\simeq} (\text{Hartree channel}) \\ &+ \sum_{\gamma, \sigma} \left(\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} u_{\mathbf{q}\sigma} \right) \times V \sum_{\mathbf{q}' \in \text{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} \hat{c}_{\mathbf{q}'\sigma}^\dagger \hat{c}_{\mathbf{q}'\sigma} - E_{\text{F}/V}^{(N)}\end{aligned}\quad (3.20)$$

where we used the functions defined in Tab. 2.5. We recognise the harmonics $u_{\sigma}^{(\gamma)}$ from Eq. (3.16),

$$\hat{H}_V \stackrel{\text{HF}}{\simeq} (\text{Hartree channel}) + V \sum_{\mathbf{q}' \in \text{BZ}} \sum_{\gamma, \sigma} u_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} \hat{c}_{\mathbf{q}'\sigma}^\dagger \hat{c}_{\mathbf{q}'\sigma} - E_{\text{F}/V}^{(N)}\quad (3.21)$$

which is almost Eq. (3.17).

To complete the proof, we recognise that in the normal phase inversion symmetry is not broken, thus

$$u_{\mathbf{q}\sigma} = u_{-\mathbf{q}\sigma}$$

Symmetry	Self-consistency equation	Graph
s^* -wave	$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$	Fig. 2.4b
$d_{x^2-y^2}$ -wave	$u^{(d)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$	Fig. 2.4e

Table 3.2 | Symmetry resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$.

This follows from the fact that in the normal phase the EV of the operator $\hat{n}_{\mathbf{k}\sigma}$ must be inversion-symmetric. It follows that antisymmetric harmonics are zero,

$$u_{\sigma}^{(p_{\ell})} = 0 \quad \text{for } \ell \in \{x, y\}$$

Using this observation in Eq. (3.21), we obtain Eq. (3.17).

Composing Eq. (3.17) with Eqns. (3.5), (3.10) we obtain the quadratic HF approximation to the EHM, for the normal phase. The chemical potential $\tilde{\mu}$ is fixed by density and determined during computations. To find the ground-state, we must determine self-consistently the two HFPs $u_{\sigma}^{(s^*)}$, $u_{\sigma}^{(d)}$ through the self-consistency Eq. (3.16). The symmetry-resolved equations are reported in Tab. 3.2.

The operator of Eq. (3.17) has the same algebraic structure of the kinetic operator $\hat{H}_t$ in reciprocal space, given in Eq. (2.33). Hence, we can collapse the two into one effective kinetic operator. Let the renormalised bare bands be

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u_{\sigma}^{(s^*)} V(\cos k_x + \cos k_y) + u_{\sigma}^{(d)} V(\cos k_x - \cos k_y) \quad (3.22)$$

In Fock channel, we have:

$$\hat{H}_t + \hat{H}_V^{\text{HF}} \simeq (\text{Hartree channel}) + \underbrace{\sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \tilde{\epsilon}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{k}\sigma}}_{\hat{H}_{\tilde{t}}} - E_{\text{F}/V}^{(\text{N})} \quad (3.23)$$

where we defined the effective kinetic operator, $\hat{H}_{\tilde{t}}$.

The renormalised bands of Eq. (3.22) are a sum of s^* and d -wave components. Instead, bare bands as defined Eq. (2.34) are purely s^* -symmetric functions,

$$\epsilon_{\mathbf{k}\sigma} = -2t(\cos k_x + \cos k_y) = -4t\varphi_{\mathbf{k}}^{(s^*)} \quad (3.24)$$

We define:

$$\tilde{t}_{\sigma}^{(s^*)} \equiv t - \frac{u_{\sigma}^{(s^*)} V}{2} \quad \tilde{t}_{\sigma}^{(d)} \equiv -\frac{u_{\sigma}^{(d)} V}{2} \quad (3.25)$$

Hence, Eq. (3.22) can be written as follows:

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv -4\tilde{t}_{\sigma}^{(s^*)} \varphi_{\mathbf{k}}^{(s^*)} - 4\tilde{t}_{\sigma}^{(d)} \varphi_{\mathbf{k}}^{(d)} \quad (3.26)$$

Note the formal similarity with Eq. (3.24). Bands renormalisation is actually a renormalisation of the hopping parameter t ,

$$t \text{ (} s^*\text{-wave)} \rightarrow \tilde{t}_{\sigma}^{(s^*)} \text{ (} s^*\text{-wave)} \text{ and } \tilde{t}_{\sigma}^{(d)} \text{ (} d\text{-wave)}$$

Therefore, t is an RMP. In Sec. 3.3.1 we will show how all of this is translated in real space.

We conclude this section by proving that the Fock contraction energy is given by:

$$E_{\text{F}/V}^{(\text{N})} = L_x L_y \times \frac{V}{2} \sum_{\sigma} \left(|u_{\sigma}^{(s^*)}|^2 + |u_{\sigma}^{(d)}|^2 \right) \quad (3.27)$$

Later, using spin degeneracy, we will omit the sum on σ and the $1/2$ prefactor.

Proof 3.3 (Non-local Fock contraction energy.).

Recall Eq. (3.15). Using Eq. (3.3), it reduces to:

$$E_{F/V}^{(N)} = \frac{V}{L_x L_y} \sum_{\mathbf{q}, \mathbf{q}'} \sum_{\sigma} [\cos(\delta q_x) + \cos(\delta q_y)] \langle \hat{c}_{\mathbf{q}\sigma}^\dagger \hat{c}_{\mathbf{q}\sigma} \rangle \langle \hat{c}_{\mathbf{q}'\sigma}^\dagger \hat{c}_{\mathbf{q}'\sigma} \rangle$$

Inserting Eq. (3.2), we get:

$$\begin{aligned} E_{F/V}^{(N)} &= L_x L_y \times \frac{V}{2} \sum_{\gamma, \sigma} \left(\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} u_{\mathbf{q}\sigma} \right) \left(\frac{1}{L_x L_y} \sum_{\mathbf{q}' \in \text{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} u_{\mathbf{q}'\sigma}^* \right) \\ &= L_x L_y \times \frac{V}{2} \sum_{\gamma, \sigma} |u_{\sigma}^{(\gamma)}|^2 \end{aligned}$$

which, using the fact that $u_{\sigma}^{(p\ell)} = 0$, proves Eq. (3.27).

In next section we collect all we found so far and discuss the properties of the HF hamiltonian.

3.3 The HF hamiltonian and its properties

In the normal phase, the HF approximation to the EHM hamiltonian is given by the following operator:

$$\hat{H} - \mu \hat{N} \stackrel{\text{HF}}{\simeq} \sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \tilde{\epsilon}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} - \tilde{\mu} \hat{N} - \left(E_{H/U}^{(N)} + E_{H/V}^{(N)} + E_{F/V}^{(N)} \right) \quad (3.28)$$

Apart from energy shifts, this hamiltonian describes a perfectly degenerate Fermi gas whose bare bands are $\tilde{\epsilon}_{\mathbf{k}\sigma}$ as defined in Eq. (3.22), and chemical potential $\tilde{\mu}$ as defined in Eq. (3.12). The latter is completely determined by the model parameters; the former must be determined self-consistently. The ground state of the approximated hamiltonian obeys Fermi-Dirac statistics,

$$\langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma'} \rangle = \delta_{\mathbf{k}=\mathbf{k}'} \delta_{\sigma=\sigma'} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \quad (3.29)$$

being f the Fermi-Dirac distribution,

$$f(\epsilon; \beta, \mu) \equiv \frac{1}{e^{\beta(\epsilon-\mu)} + 1}$$

Eq. (3.29) is the explicit version of Eq. (3.3).

In next subsection we discuss in detail the renormalisation of the hopping parameter in real space, with a focus on the meaning of the d -wave component. In the subsection that follow, we derive the free energy and entropy densities for the ground-state

3.3.1 Nematicity: d -wave bands and anisotropic hopping

In Eq. (3.23) we defined $\hat{H}_{\tilde{t}}$ in reciprocal space. Using Eq. (3.13) we get

$$\begin{aligned} \hat{H}_{\tilde{t}} &= \hat{H}_t + V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \rangle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} + \text{h.c.} \right] \\ &= - \sum_{i,j} \sum_{\sigma} \left[\tilde{t}_{ij\sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \text{h.c.} \right] \end{aligned}$$

Note that in second line we are not summing over NNs, but on two site indices $i, j \in \mathcal{L}$. We defined the effective hopping amplitude

$$\tilde{t}_{ij\sigma} \equiv \frac{1}{2} \left(t - V \langle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} \rangle \right) \quad (\text{for } i, j \text{ NNs})$$

The 1/2 prefactor was included to avoid double counting.

Due to translational invariance, the renormalised hopping $\tilde{t}_{ij\sigma}$ depends only on the difference $i - j$. Therefore we can use the decomposition of Eq. (2.41),

$$\tilde{t}_{ij\sigma} = \sum_{\gamma} \tilde{t}_{\sigma}^{(\gamma)} \varphi_{ij}^{(\gamma)}$$

The harmonics, given by Eq. (2.42), are identified with $\tilde{t}_{\sigma}^{(s^*)}$, $\tilde{t}_{\sigma}^{(d)}$ already introduced in Eq. (3.25). From the same argument we know that $\tilde{t}_{\sigma}^{(p_x)} = \tilde{t}_{\sigma}^{(p_y)} = 0$. Then, the renormalised hopping is explicitly given by

$$\begin{aligned} \tilde{t}_{\mathbf{r}\mathbf{r}'\sigma} = \frac{\tilde{t}_{\sigma}^{(s^*)}}{2} & \left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}} \right] \\ & + \frac{\tilde{t}_{\sigma}^{(d)}}{2} \left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} - \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}} \right] \end{aligned}$$

Let us define:

$$\tilde{t}_{\sigma}^{(x)} \equiv \frac{\tilde{t}_{\sigma}^{(s^*)} + \tilde{t}_{\sigma}^{(d)}}{2} \quad (3.30)$$

$$\tilde{t}_{\sigma}^{(y)} \equiv \frac{\tilde{t}_{\sigma}^{(s^*)} - \tilde{t}_{\sigma}^{(d)}}{2} \quad (3.31)$$

With this definition, we can rewrite $\hat{H}_{\tilde{t}}$ as:

$$\hat{H}_{\tilde{t}} = - \sum_{\mathbf{r} \in \mathcal{L}} \sum_{\sigma} \left[\tilde{t}_{\sigma}^{(x)} (\hat{c}_{\mathbf{r}\sigma}^{\dagger} \hat{c}_{\mathbf{r}+\underline{\mathbf{x}}\sigma} + \hat{c}_{\mathbf{r}\sigma}^{\dagger} \hat{c}_{\mathbf{r}-\underline{\mathbf{x}}\sigma}) + \tilde{t}_{\sigma}^{(y)} (\hat{c}_{\mathbf{r}\sigma}^{\dagger} \hat{c}_{\mathbf{r}+\underline{\mathbf{y}}\sigma} + \hat{c}_{\mathbf{r}\sigma}^{\dagger} \hat{c}_{\mathbf{r}-\underline{\mathbf{y}}\sigma}) + \text{h.c.} \right]$$

This is what we intend for “nematic effects”: the breaking of planar rotational symmetry, with the consequence that the kinetic operators in x and y directions differ. Breaking of planar rotational symmetry and uniform density are compatible.

Proof 3.4 (A d -wave hopping is compatible with uniform density).

Consider the on-site occupation operator $\hat{n}_{\mathbf{r}\sigma} = \hat{c}_{\mathbf{r}\sigma}^{\dagger} \hat{c}_{\mathbf{r}\sigma}$. Applying a Fourier transform as defined in Eq. (2.30), we obtain

$$\hat{c}_{\mathbf{r}\sigma}^{\dagger} \hat{c}_{\mathbf{r}\sigma} \stackrel{\mathcal{F}}{=} \frac{1}{L_x L_y} \sum_{\mathbf{k}, \mathbf{k}'} e^{i(\mathbf{k}-\mathbf{k}') \cdot \mathbf{r}} \hat{c}_{\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{k}'\sigma}$$

We compute the expectation value of the above operator and apply Eq. (3.29),

$$\begin{aligned} \langle \hat{n}_{\mathbf{r}\sigma} \rangle &= \frac{1}{L_x L_y} \sum_{\mathbf{k}, \mathbf{k}'} e^{i(\mathbf{k}-\mathbf{k}') \cdot \mathbf{r}} \langle \hat{c}_{\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{k}'\sigma} \rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \end{aligned}$$

The right-hand side of last line is independent of $\mathbf{r}$, therefore $\langle \hat{n}_{\mathbf{r}\sigma} \rangle$ is.

Note that we never used the actual topology of the renormalised bands $\tilde{\epsilon}_{\mathbf{k}}$, but only the fact that the ground-state obeys the Fermi-Dirac distribution. A perfectly degenerate Fermi gas is always translational invariant and thus cannot exhibit non-uniform symmetry.

A schematics of anisotropic hopping is given in Fig. 3.2: for each bond linking two sites, the hopping parameter can be a positive or negative combination of the two components, as the link direction is vertical or horizontal.

From now on, we will omit the spin index σ from both the renormalised bands and the renormalised hopping,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \rightarrow \tilde{\epsilon}_{\mathbf{k}} \quad \forall \alpha \quad : \quad \tilde{t}_{\sigma}^{(\alpha)} \rightarrow \tilde{t}^{(\alpha)}$$

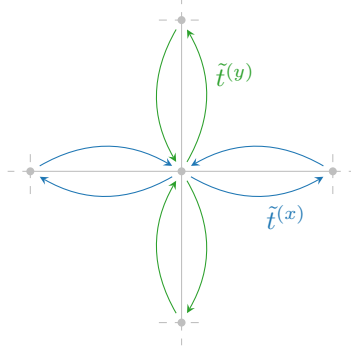


Figure 3.2 | Graphic rendition of anisotropic hopping, with a s^* -wave component (symmetric by $\pi/2$ rotations) and a d -wave component (antisymmetric).

The normal phase does not break spin rotational invariance. Therefore the renormalised bands must be independent of the spin index. Inverting Eq. (3.25), we can express $\tilde{t}^{(x)}$ and $\tilde{t}^{(y)}$ in terms of the HFPs $u^{(s^*)}$ and $u^{(d)}$

$$\begin{aligned}\tilde{t}^{(x)} &\equiv t - \frac{u^{(s^*)} + u^{(d)}}{2}V \\ \tilde{t}^{(y)} &\equiv t - \frac{u^{(s^*)} - u^{(d)}}{2}V\end{aligned}$$

Inverting these relations we obtain:

$$u^{(s^*)} = \frac{(t - \tilde{t}^{(x)}) + (t - \tilde{t}^{(y)})}{V} \quad (3.32)$$

$$u^{(d)} = \frac{\tilde{t}^{(x)} - \tilde{t}^{(y)}}{V} \quad (3.33)$$

It follows that if $u^{(d)} > 0$, the hopping in direction x is larger than in direction y is damped. If $u^{(d)} < 0$, roles are inverted.

3.3.2 The entropy density of the SC ground-state

In this section we compute the density of entropy of the ground state in the normal phase. Entropy density is given by

$$s^{(N)} = \frac{S^{(N)}}{L_x L_y}$$

being $S^{(N)}$ the total entropy.

The HF hamiltonian of Eq. (3.28) describes a non-interacting Fermi gas. The entropy density of a free Fermi gas in its ground state [4] is

$$s^{(N)} = -\frac{2k_B}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} [(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \ln f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})] \quad (3.34)$$

The 2 prefactor was included to account for spin degeneracy. The above equation can be written in another way:

$$s^{(N)} = -\frac{2k_B}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} [\ln(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) - \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu})f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})] \quad (3.35)$$

Proof 3.5 (Proof of Eq. (3.35)).

The following identity

$$\begin{aligned}\ln\left(1 - \frac{1}{e^x + 1}\right) &= \ln\left(\frac{e^x}{e^x + 1}\right) \\ &= x + \ln\left(\frac{1}{e^x + 1}\right)\end{aligned}\quad (3.36)$$

Substituting $x = \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu})$ and using Eq. (3.36) in Eq. (3.34) we obtain

$$\begin{aligned}s^{(N)} &= -\frac{2k_B}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \left[\ln(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \ln f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right. \\ &\quad \left. - \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \ln f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right]\end{aligned}$$

Second and fourth terms cancel out. The remaining parts give Eq. (3.35).

3.3.3 The free energy density of the SC ground-state

In this section we compute the free energy density of the ground state, for the normal phase. The density of free energy is given by

$$f^{(N)} \equiv \frac{F^{(N)}}{L_x L_y} \quad (3.37)$$

being $F^{(N)}$ the total free energy,

$$F^{(N)} = E_{\text{HF}}^{(N)} - \frac{S^{(N)}}{k_B \beta} \quad (3.38)$$

Here $E_{\text{HF}}^{(N)}$ is the ground-state energy for the HF hamiltonian. In the normal phase, we compute $f^{(N)}$ using the following formula:

$$f^{(N)} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) - V \left[|u^{(s^*)}|^2 + |u^{(d)}|^2 \right] + \tilde{\mu} n \quad (3.39)$$

Proof 3.6 (Free energy density for the normal phase).

Using Eqns. (3.37), (3.38), we have

$$f^{(N)} = \frac{E_{\text{HF}}^{(N)}}{L_x L_y} - \frac{s^{(N)}}{k_B \beta}$$

with $s^{(N)}$ the entropy density computed in Eq. (3.35). Then, $f^{(N)}$ is obtained by summing the following terms:

1. The free particles ground state energy density,

$$e_g^{(N)} = \frac{2}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

The prefactor 2 accounts for spin degeneracy.

2. The contraction energy density,

$$\begin{aligned}e_c^{(N)} &\equiv \frac{E_c^{(N)}}{L_x L_y} \\ &= -n^2 U + V \left[2zn^2 - |u^{(s^*)}|^2 - |u^{(d)}|^2 \right]\end{aligned}$$

where in first line we included the all contraction energies from Eqns. (3.6), (3.9) and (3.27) in $E_c^{(N)}$,

$$E_c^{(N)} \equiv - \left[E_{H/U}^{(N)} + E_{H/V}^{(N)} + E_{F/V}^{(N)} \right]$$

3. The entropic contribution to free energy

$$\begin{aligned} e_s^{(N)} &\equiv - \frac{s^{(N)}}{k_B \beta} \\ &= \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) - \frac{2}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \end{aligned}$$

We used Eq. (3.35).

4. The chemical potential correction

$$e_n^{(N)} = \mu n$$

We explained at the end of Sec. 1.3.1 why this correction is needed. Note here that we are using the “physical” value of μ , not its RMP counterpart $\tilde{\mu}$.

Thus, composing everything

$$\begin{aligned} f^{(N)} &\equiv e_g^{(N)} + e_c^{(N)} + e_s^{(N)} + e_n^{(N)} \\ &= \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) - V \left[|u^{(s^*)}|^2 + |u^{(d)}|^2 \right] + [\mu + n(2zV - U)] n \end{aligned}$$

Using Eq. (3.12), we recognise

$$\mu + n(2zV - U) = \tilde{\mu}$$

Hence Eq. (3.39) is proven.

3.4 Analysis of the self-consistency equations

In this section we collect some results that follow from a formal analysis of the self-consistency problem for the normal phase. This phase is described by the HFPs $u^{(s^*)}$ and $u^{(d)}$. It is possible to prove that:

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} = 0 \quad (3.40)$$

for any choice of $U, V > 0$ and doping δ . The proof follows from the detailed analysis of the self-consistency equations of Tab. 3.2, and is reported in App. ???. Eq. (3.40) implies that in the normal phase there is no nematic effect, and the renormalised hopping parameter $\tilde{t}_{ij}$ is isotropic with an amplitude smaller than t ,

$$0 < \tilde{t}^{(s^*)} < t$$

Here we used Eq. (3.25). Then, bands renormalisation

$$\epsilon_{\mathbf{k}} \rightarrow \tilde{\epsilon}_{\mathbf{k}}$$

is actually a flattening. We denote this effect as “bands shrinking”. The flattening of electronic bands is the signature of a Mott transition.

Let us define the Magnetic Brillouin Zone (MBZ):

$$\text{MBZ} \equiv \left\{ \mathbf{k} \in \text{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \right\} \quad (3.41)$$

The MBZ coincides with the set of points (k_x, k_y) that satisfy the relation

$$|k_x| + |k_y| \leq \pi$$

We draw the MBZ in Fig. 3.3. The renormalised bands satisfy

$$\tilde{\epsilon}_{\mathbf{k}} = -4\tilde{t}^{(s^*)} \varphi_{\mathbf{k}}^{(s^*)}$$

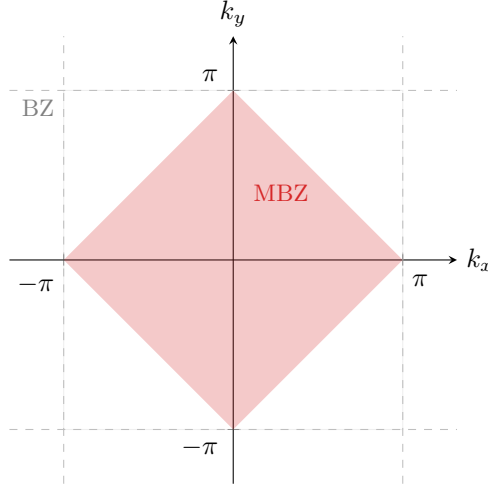


Figure 3.3 | Representation of the MBZ. The dashed lines represent the boundaries of the BZ.

Hence, $\tilde{\epsilon}_{\mathbf{k}} < 0$ inside the MBZ and $\tilde{\epsilon}_{\mathbf{k}} > 0$ outside. Along the boundary ∂MBZ we have $\tilde{\epsilon}_{\mathbf{k}} = 0$.

We conclude this section by proving some analytic properties of the unique HFP of the normal phase, $u^{(s^*)}$. Its self-consistency equation, from Tab. 3.2, reads

$$u^{(s^*)}(\beta, \tilde{\mu}) = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \quad (3.42)$$

We wrote $u^{(s^*)}$ as a function of chemical potential and temperature. Note that the renormalised bands, inside the Fermi-Dirac factor on the right-hand side, depend on $u^{(s^*)}$. We now prove that in the low-temperature limit the above equation implies that $u^{(s^*)}$ is a monotonically descendent function of doping.

Proof 3.7 ($u^{(s^*)}$ decreases with doping at low temperature).

We work at low temperature, $\beta \rightarrow \infty$. The Fermi-Dirac distribution is approximately a step function. Assume the renormalised band is not flat, hence $u^{(s^*)} < 2t/V$. Then Eq. (3.42) becomes

$$\lim_{\beta \rightarrow \infty} u^{(s^*)}(\beta, \tilde{\mu}) = \frac{1}{L_x L_y} \sum_{\epsilon_{\mathbf{k}} \leq \tilde{\mu}} (\cos k_x + \cos k_y)$$

Let $\text{FS}(\delta)$ be the Fermi surface for a given doping level δ . We denote with $\text{Int}(\text{FS})$ be its internal part, the Fermi sea:

$$\begin{aligned} \text{FS} &= \{\mathbf{k} \in \text{BZ} \mid \tilde{\epsilon}_{\mathbf{k}} = \tilde{\mu}\} \\ \text{Int}(\text{FS}) &= \{\mathbf{k} \in \text{BZ} \mid \tilde{\epsilon}_{\mathbf{k}} \leq \tilde{\mu}\} \end{aligned}$$

At half filling, the Fermi surface coincides with the boundary of the MBZ.

$$\delta = 0 \quad \implies \quad \text{FS} = \partial\text{MBZ}$$

This is implied by the fact that renormalised bands are antisymmetric by π shifts,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = -\tilde{\epsilon}_{\mathbf{k}+\pi\sigma}$$

The portion of an s^* -wave band outside the MBZ is mirrored with respect to the portion inside the MBZ. This implies that half the states lie in the MBZ, half out.

At $\delta > 0$ the FS expands in the region $\text{BZ} \setminus \text{MBZ}$ where the form factor $\cos k_x + \cos k_y$ becomes negative. Then Eq. (3.42) becomes

$$\lim_{\beta \rightarrow \infty} u^{(s^*)}(\beta, \tilde{\mu}) \simeq \frac{1}{L_x L_y} \sum_{\epsilon_{\mathbf{k}} \leq 0} |\cos k_x + \cos k_y| - \frac{1}{L_x L_y} \sum_{0 < \epsilon_{\mathbf{k}} \leq \tilde{\mu}} |\cos k_x + \cos k_y|$$

which, being the sum of a positive and a negative term, is maximised whenever $\tilde{\mu} = 0$, so at half filling.

As long as the band is not completely flattened by renormalisation, increasing the doping, the FS expands. Hence

$$\delta_1 > \delta_2 \implies \text{Int}(\text{FS}_1) \subset \text{Int}(\text{FS}_2)$$

where FS_i is the FS associated to the doping level δ_i . Using this property we conclude

$$\tilde{t} > 0 \quad \text{and} \quad \delta_1 > \delta_2 \implies \lim_{\beta \rightarrow \infty} u^{(s^*)}(\beta, \tilde{\mu}) \Big|_{\delta_1} < \lim_{\beta \rightarrow \infty} u^{(s^*)}(\beta, \tilde{\mu}) \Big|_{\delta_2}$$

This proves that $u^{(s^*)}$ decreases with doping at low temperature.

This is the first expected behaviour of the self-consistent functional $u^{(s^*)}$: if there exists a region of parametric space where the shrinking effect is not strong enough to flatten the band, a rise in the doping level must lower $u^{(s^*)}$. Such a region must exist: at least all the segment ($V = 0, \delta$) is included, since on said region of parametric space the HF hamiltonian reduces to the that of the free Fermi gas.

We conclude this section with the analytic computation of $u^{(s^*)}$ at half filling and unitary filling, both at low temperature:

$$\lim_{\beta \rightarrow \infty} u^{(s^*)}[\beta, 0] = \left(\frac{2}{\pi}\right)^2 \quad (3.43)$$

$$\lim_{\beta \rightarrow \infty} u^{(s^*)}[\beta, 2\tilde{t}^{(s^*)}] = 0 \quad (3.44)$$

At unitary filling we set the chemical potential equal to the maximum energy reached by a band of amplitude $4\tilde{t}^{(s^*)}$.

Proof 3.8 (Proof of Eqns. (3.43), (3.44)).

The two proofs are:

1. At half-filling, Eq. (3.42) reduces to

$$\lim_{\beta \rightarrow \infty} u^{(s^*)}(\beta, 0) \simeq \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x + \cos k_y)$$

In thermodynamic limit, performing integration on the MBZ shown in Fig. 3.3:

$$\begin{aligned} \lim_{\beta \rightarrow \infty} u^{(s^*)}(\beta, 0) &\simeq \frac{1}{(2\pi)^2} \int_{-\pi}^{\pi} dk_y \int_{-\pi+|k_y|}^{\pi-|k_y|} dk_x \cos k_x + \frac{1}{(2\pi)^2} \int_{-\pi}^{\pi} dk_x \int_{-\pi+|k_x|}^{\pi-|k_x|} dk_y \cos k_y \\ &= \frac{1}{2\pi^2} \int_{-\pi}^0 dk_y \int_{-\pi-k_y}^{\pi+k_y} dk_x \cos k_x + \frac{1}{2\pi^2} \int_0^{\pi} dk_y \int_{-\pi+k_y}^{\pi-k_y} dk_x \cos k_x \\ &= \frac{1}{\pi^2} \int_{-\pi}^0 dk_y \sin(\pi + k_y) + \frac{1}{\pi^2} \int_0^{\pi} dk_y \sin(\pi - k_y) \\ &= \left(\frac{2}{\pi}\right)^2 \approx 0.405 \end{aligned}$$

which proves Eq. (3.43).

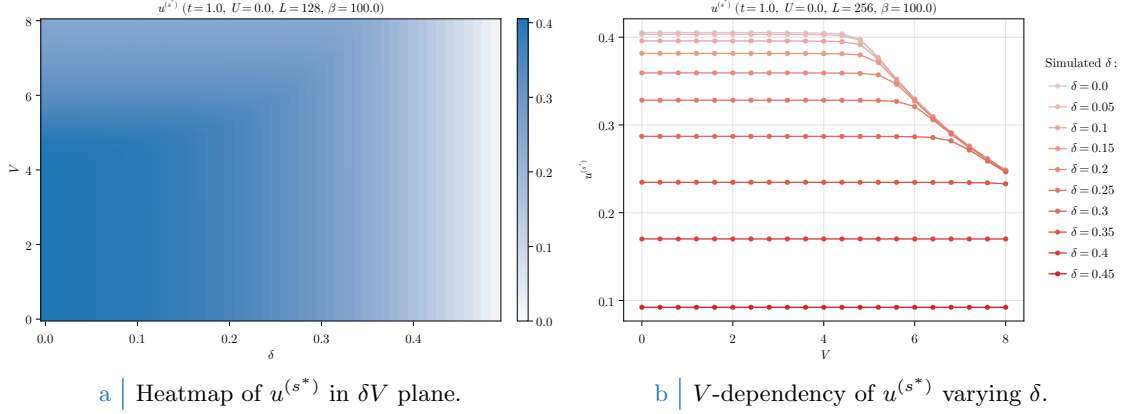


Figure 3.4 | Plot of the self-consistent convergence values for the HFP $u^{(s^*)}$ in the normal phase. To extract the data on the right side coarse-graining precision of the left-hand side data has been exchanged with higher lattice dimensions. As is evident, based on doping level $u^{(s^*)}$ exhibits non-trivial regions of V -independence.

2. At unitary filling we have

$$\lim_{\beta \rightarrow \infty} u^{(s^*)}(\beta, 2\tilde{t}^{(s^*)}) = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x + \cos k_y) = 0$$

which vanishes due to the structure factor periodicity by π . This proves Eq. (3.44).

3.5 Discussion of numeric results

In this section we present and discuss the numeric results, the physical relevance of the found phenomena and the boundaries of the algorithm. For the normal phase, the HFPs “vector” defined in Sec. 1.3.2 has just one component:

$$\mathbf{v} = [u^{(s^*)}]$$

We have two RMPs:

$$\begin{aligned} \tilde{t}_{ij} &\equiv \left(t - \frac{u^{(s^*)} V}{2} \right) \varphi_{ij}^{(s^*)} \\ \tilde{\mu} &\equiv \mu + n(2zV - U) \end{aligned}$$

the second one being only dependent on the model parameters and not on HFPs. Finally, the phase free energy is given by Eq. (3.39). The normal phase is completely independent of U . Because of this we present the results varying the other parameters, keeping everywhere $U = 0$. We ran simulations in the (V, δ) plane, with the setup:

$$U = 0 \quad \Delta n = 10^{-2} \quad \Delta \mathbf{v} = [u^{(s^*)} : 5 \times 10^{-4}]$$

with $\Delta \mathbf{v}$ the tolerance vector defined as in Sec. 1.3.2.

In Fig. 3.4a are reported the results obtained for the HFP $u^{(s^*)}$ on a 128×128 lattice and “low” temperature $\beta = 100$, varying the non-local attraction and the doping level

$$V = 0, 0.1, 0.2, \dots, 4.0 \quad \delta = 0, 0.01, 0.02, \dots, 0.49$$

The expectations of Sec. 3.4 are satisfied: at low V (indicatively $V \lesssim 5$) the trend is monotonically descendent at increasing doping. As V gets bigger, however, $u^{(s^*)}$ starts decreasing in the top-left corner of Fig. 3.4a. This effect is more visible in Fig. 3.4b, where we doubled L and ran simulations with a smaller δ resolution: the convergence value of $u^{(s^*)}$ is approximately independent of V up to some point, where this behaviour is broken. Note that, as was expected from the discussion of Sec. 3.4, the half-filled convergence value of $u^{(s^*)}$ in the V -independent region is approximately given by Eq. (3.43).

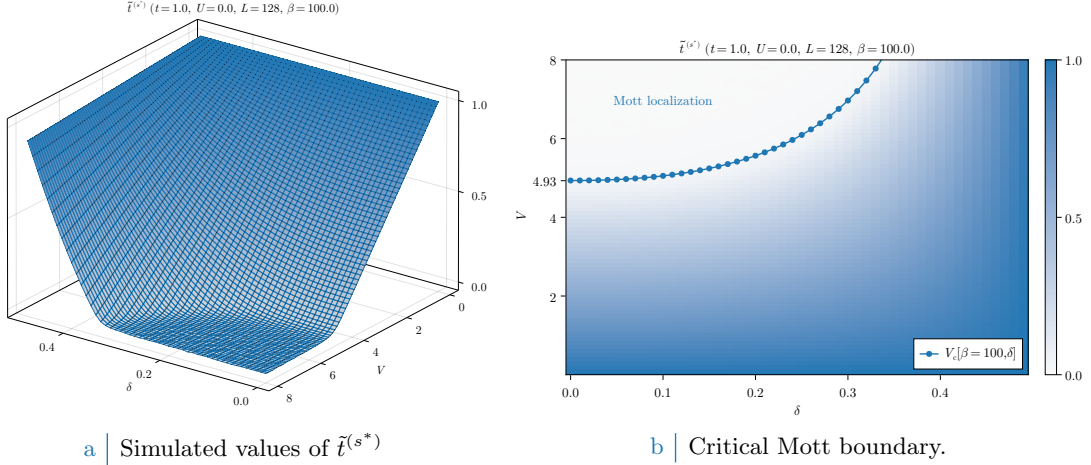


Figure 3.5 | Plot of the self-consistent convergence values for the HFP $u^{(s^*)}$ in the normal phase and the Mott-boundary V_c , computed as is described in text. In the right panel, the tick $V = 4.93$ is specified explicitly as a graphic confirm of the computation of Eq. (3.45). For graphic reasons the points at $V_c > 8$ have been omitted.

From Sec. 3.4 we know that $u^{(s^*)}$ decreases as we increase doping, as long as $\tilde{t}^{(s^*)} > 0$. When V gets large enough, $u^{(s^*)}$ starts decreasing because of its constraints. The net effect is the nullification of $\tilde{t}^{(s^*)}$, which is, a total localisation effect.

3.5.1 Mott localisation effect

The most interesting result we obtained for the normal phase is the possibility of localising electrons by tuning the V and δ . In Fig. 3.4b, at low V the convergence value of $u^{(s^*)}$ is approximately V -independent at sufficiently low temperatures. Let $V_c[\beta, \delta]$ be the critical value such that

$$0 \leq V < V_c[\beta, \delta] \quad : \quad u^{(s^*)} \text{ is } V \text{ independent.}$$

Then within the entire segment $[0, V_c]$ the convergence value is (approximately) the same obtained for $V = 0$. We can define in first instance the critical V as the value such that the hopping correction is maximized,

$$V_c[\beta, \delta] \equiv \frac{2t}{u^{(s^*)}[\beta, \tilde{\mu}(\delta)]|_{V=0}}$$

When V becomes larger than this critical value, the resulting RMP $\tilde{t}^{(s^*)}$ drops to 0 and due to the constraint of Eq. (3.40) the convergence value of $u^{(s^*)}$ is forced to decrease. Note that the constraint of Eq. (3.40) does not prevent $u^{(s^*)}$ from saturating to its upper boundary from below. In Fig. 3.5a the converged values of $\tilde{t}^{(s^*)}$, extracted from the same data of Fig. 3.4a, are plotted in parametric space.

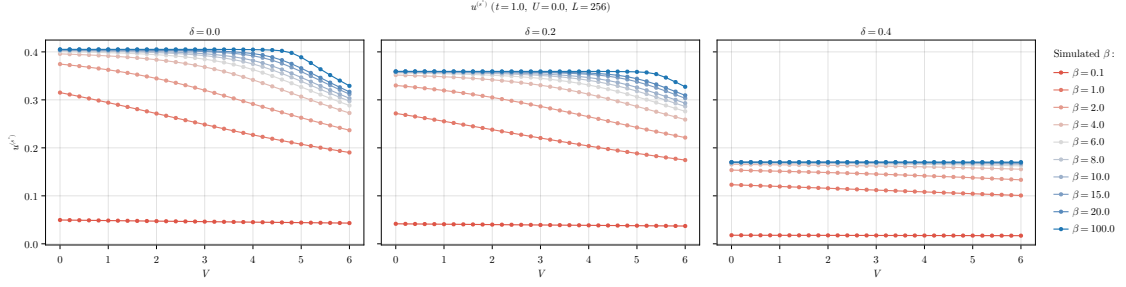
To extract the value of $V_c[\beta, \delta]$ is a non-trivial analytic problem due to the insurgence of elliptic integrals and the smearing effects of temperature. Instead, for the high- β half-filled case we can use Eq. (3.43) and get:

$$V_c[\beta \rightarrow \infty, 0] = \frac{2t}{(2/\pi)^2} \approx 4.93t \quad (3.45)$$

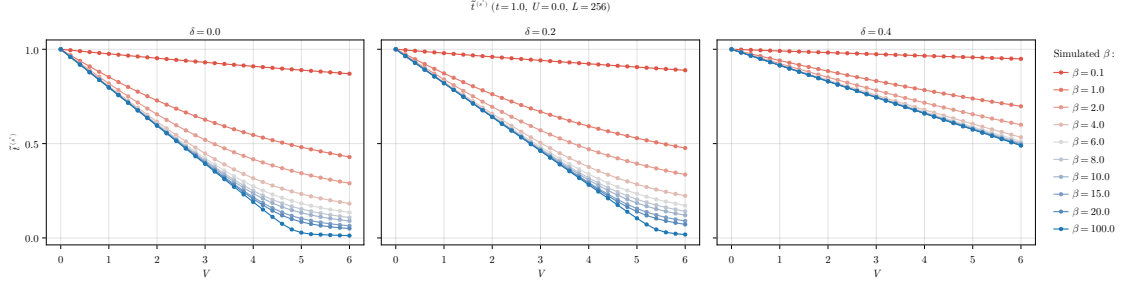
At half-filling, the effective bands amplitude reduction is so strong that small values of V (compared to t) are sufficient to nullify kinetics. In Fig. 3.5b we report, superimposed, the convergence values of the RMP $\tilde{t}^{(s^*)}$ and $V_c[\beta = 100, \delta]$ computed with the simple procedure hereby described. Our prediction of the critical value for V approximates reasonably the localisation region.

Increasing temperature, Mott localisation is suppressed. In Fig. 3.6a we report results for the HFP $u^{(s^*)}$ computed at decreasingly lower temperature and three example doping levels,

$$1.0 \leq \beta \leq 100.0 \quad \delta = 0.0, 0.2, 0.4$$



a | Temperature dependence of $u^{(s*)}$ at various dopings.



b | Temperature dependence of $t^{(s*)}$ at various dopings.

Figure 3.6 | Comparative plots of the HFP $u^{(s*)}$ (top panel) and RMP $t^{(s*)}$ (bottom panel) in null-doped (left plots), “medium-doped” ($\delta = 0.2$, central plots) and “heavily-doped” ($\delta = 0.4$, right plots) contexts. Increasing temperature progressively destroys Mott localisation, reducing the bands shrinking effect.

At high temperature the V -independence region that was visible in Fig. 3.4b completely disappears, making $u^{(s*)}$ a monotonically descendent function. In Fig. 3.6a the same dataset is plotted with focus on the RMP $t^{(s*)}$. Temperature increase disrupts localisation effects, at least at small values of V . However, the bands shrinking effect is still relevant at $\beta = 1.0$, the hopping parameter approximately halving at $V \approx 5 \div 6$.

3.5.2 Solution free energy and entropy densities

In order to compare phases, we need an estimation of free energy of each. The plots of Fig. 3.7 report our findings on the normal phase free energy, computed based on Eq. (3.39). The free energy is intended in unities of t . For a non-flat band, doping increases energy because the band fills up and electrons populate higher energy levels.

Looking the data of Fig. 3.7a, at fixed δ free energy density remains pretty much constant up to the Mott boundary. This is simply because as long as a band with non-zero amplitude is present, the HFP correction of Eq. (3.39) compensates the free energy increase due to bands shrinking,

$$\begin{aligned} -V|u^{(s*)}|^2 &= -\frac{Vu^{(s*)}}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\epsilon_{\mathbf{k}} - \tilde{\epsilon}_{\mathbf{k}}) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \end{aligned}$$

having we used in the second passage

$$\tilde{\epsilon}_{\mathbf{k}} = \epsilon_{\mathbf{k}} + u^{(s*)} V (\cos k_x + \cos k_y)$$

Recalling the first point of the derivation of Sec. 3.3.3, to add the HFP correction to free energy is the same as removing the *tilda* sign from single-state energies, thus using the free model ($V = 0$) free energy density expression.

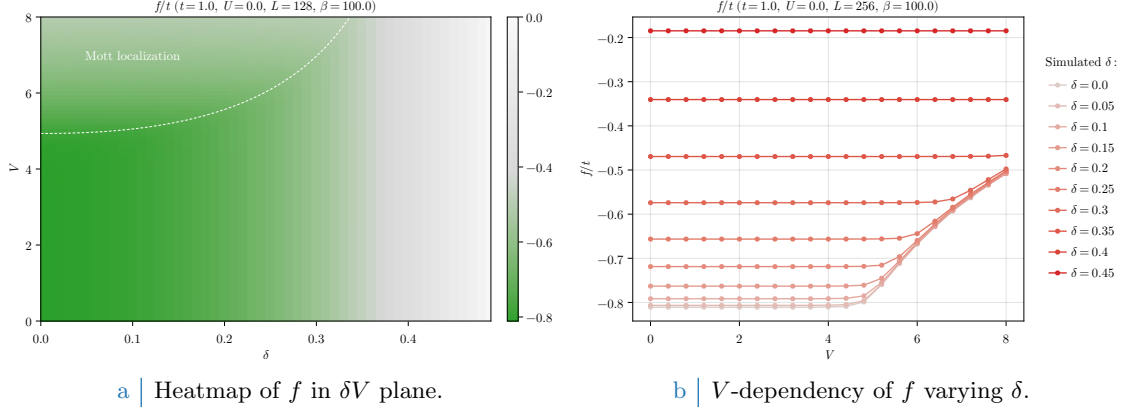


Figure 3.7 | Free energy density of the normal phase. On the left-side panel, the dashed white line has the same meaning as the dotted line of Fig. 3.5b, indicating the Mott transition.

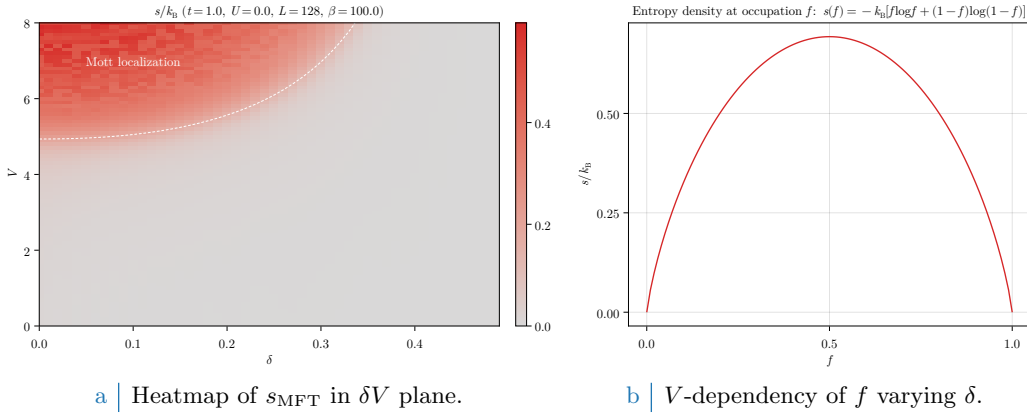


Figure 3.8 | Entropy density of the normal phase (left panel) and theoretical contribution to entropy for a state with occupation f . On the left-side panel, the dashed white line has the same meaning as the dotted line of Fig. 3.5b, indicating the Mott transition.

Mott localisation is energetically expensive. This is visible in both panels of Fig. 3.7. The V -dependent energy increase is an entropic effect: indeed, for a flat band:

$$f(0; \beta, \tilde{\mu}(\delta)) = \frac{1}{2}$$

at any temperature. The expression for the contribution to entropy of a single particle state with occupation f [4] is given by

$$s(f) = -k_{\text{B}} [f \log f + (1 - f) \log(1 - f)]$$

represented in Fig. 3.8b. The maximum is located at $f = 1/2$, then a flat band realises the maximally entropic condition. The Mott insulating region can be seen as a degeneration of the FS interface to the entire BZ, a process that maximises entropy.

All of this holds, as all the rest, at low temperature: by lowering β , finite-temperature effects become more and more relevant and the statements above need to be corrected.

3.5.3 Convergence problems and mixing fine-tuning

The normal phase is simple enough to describe some of the typical convergence problems arising in HF iterative schemes anticipated in Sec. 1.3.2. Bands flattening is a source of algorithmic instability: numerics can lead to values of $u^{(s^*)}$, during iterations, slightly out of bounds with respect to Eq. (??). Even if the numeric effect is small, the consequence can be vast. To heal this

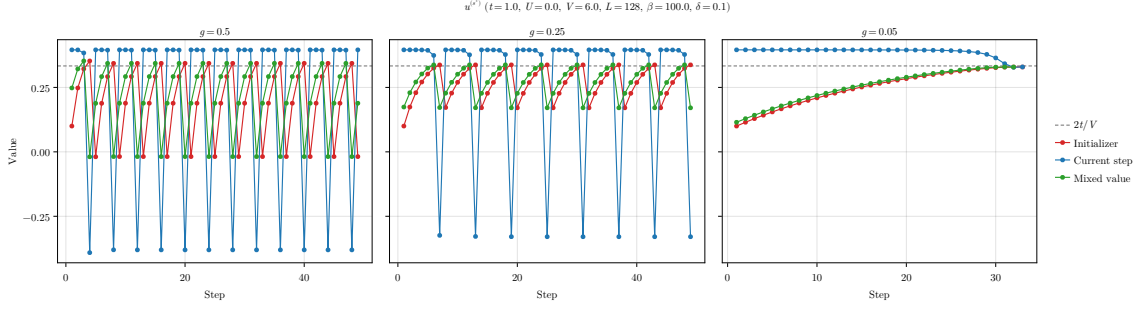


Figure 3.9 | Comparison of three example tracks for the single-valued HFPs vector of the normal phase, simulated at a “Mott-deep” point.

effect and extract the actual convergence values, it is necessary to optimise the mixing parameter $g \in [0, 1]$.

We defined g in Sec. 1.3.2, and operatively in Eq. (1.24). This parameter controls deterministically how, from an initialiser (the HFPs vector we plug in the algorithm) and an output (the HFPs vector we get back from the algorithm) the new initialiser for the following step is generated. We show in Fig. 3.9 the evolutions at different mixing parameters of a simulation at

$$V = 6.0 \quad \delta = 0.1 \quad \beta = 100.0$$

This point is inside the Mott region, as can be checked in Fig. 3.4a. The HFP $u^{(s^*)}$ needs to converge at $2t/V = t/3$ in order to give complete localisation. The initialiser

$$\text{Step 1} \quad : \quad u^{(s^*)} = 0.1$$

was arbitrarily chosen. In each panel of Fig. 3.9, the red track represents the value of $u^{(s^*)}$ we plug in the self-consistency equations, the blue the result of the self-consistent equation, the green one the mixed value of the two based on different g s. The gray dashed line is the theoretical convergence target.

In the first two panels, there is an evident infinite loop occurring whenever the green line crosses the threshold $2t/V$. At those steps, the green point is out of bounds with respect to Eq. (3.40). When this happens, the following red point gives a band with inverted sign, a feature that “breaks” the self-consistency equation and generates a blue point with no physical significance.

To overcome this problem it is necessary to move “slowly”. As is shown in the right panel, the choice of a small enough mixing parameter ensures convergence. In the track, each blue dot is above the target and would produce a band with inverted sign. Mixing blue and red dots with a 95% unbalance on the red ones, we ensure that the green track approaches the target from below. Note, however, that lowering g cannot be reiterated too many times. Indeed, for $g \rightarrow 0$, any point of our map becomes a fixed point.

This kind of problems can turn out to be pretty common for iHF schemes, and to understand how to knob the algorithmic parameters without sacrificing accurateness and precision, as well as avoiding computationally costly and frustrating infinite loops, was an essential part of this algorithm design.

Chapter 4

Antiferromagnetic instability

In this chapter we apply HF method to the AF phase, widely studied for the HM [32]. We focus on one particular realisation of antiferromagnetism, referred to as “commensurate”. It is inherently metallic and only physically sensed at half-filling.

For the commensurate antiferromagnet, the non-local attraction $\hat{H}_V$ adds two main self-consistent effects. First, redefinition of the bands gap: an effect we denote “gap complexification” (Sec. 4.6) and later show is actually suppressed (Sec. ??). Second, bare bands shrinking (Sec. 4.5) similarly to what found for the normal phase.

As for the normal phase, this chapter is equally divided in the same three main parts. First: discussion of AF phase symmetries, based on Chap. 2, and implementation of HF methods. Second: theoretical constraints on the self-consistent solution. Third: numeric results.

4.1 The AF phase and its symmetries

The AF phase is not uniquely defined. Fig. 4.1 presents three (of many) types of antiferromagnets. We represent in the left panel the one we are interested in. The AF structure comes from an uneven and periodic distribution of spin densities with one site periodicity in both directions. The arrows in figure indicate what spin sector is most abundant on a given site. These SDW structures are 2-sites periodic, making the AF phase “commensurate”. The central one presents another simple commensurate AF with higher periodicity. The right panel shows a possible “striped” antiferromagnet, where AF is effectively established along one particular direction; the example we report is 2 sites periodic in y direction, but any other periodicity could work. Striped antiferromagnetism is studied in HM physics [16] as a possible long-range ordering energetically competing

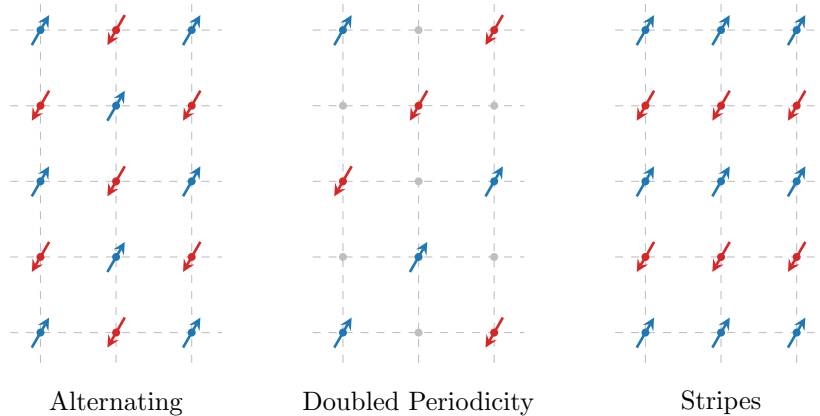


Figure 4.1 Three possible antiferromagnetic structures. Dots and arrows indicate spin abundance on a given site, while empty sites are intended as spin-balanced. Within this text we deal with the lowest-periodicity commensurate phase, the antiferromagnet on left panel.

with anisotropic superconductivity. Nematic antiferromagnets, establishing long-range magnetic ordering along one particular direction, are interesting due to competition and interplay against superconducting ordering at finite doping [25]. Geometrically non-trivial magnetic ordering are at the core of SDW formation in copper oxides at finite doping [26, 27].

In this thesis, the AF phase is described by simultaneous global spin rotation and translational symmetry SSBs:

$$\begin{aligned} \text{SU}(2) &\xrightarrow{\text{SSB}} \text{U}^z(1) \\ \text{T}_1^{(x)} \otimes \text{T}_1^{(y)} &\xrightarrow{\text{SSB}} \text{T}_2^{(x)} \otimes \text{T}_2^{(y)} \end{aligned}$$

We describe a SDW whose period is of two sites, as in the left panel of Fig. 4.1. As done for the normal phase, we allow

$$\text{C}_4 \xrightarrow{\text{SSB}} \text{C}_2$$

in order to capture the same theoretical nematic effects seen in Sec. ???. The theoretical reason to do so for AF phase relies in our previous discussion. As said, planar Hubbard lattices exhibits tendency towards nematic magnetic ordering, thus (even if our Ansatz is the simplest and non evidently nematic) nematic effects are not to be excluded *a priori*.

Commensurate antiferromagnetism is specified by the Ansatz:

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n - (-1)^{x+y+\delta_{\sigma=\uparrow}} m \quad n, m \in [0, 1] \quad (4.1)$$

This Ansatz captures the idea of the staggered distribution of spin populations shown in the left panel of Fig. 4.1. At site $\mathbf{r} = (x, y)$ the two spin populations average to n and differ by $2m$. A detailed justification for this Ansatz in the HM is given in Sec. A.3. Note that $\langle \hat{n}_{\mathbf{r}\sigma} \rangle \leq 1$ due to Pauli exclusion principle. This argument is already sufficient to claim that doping disrupts commensurate antiferromagnetism.

4.1.1 Symmetries of the commensurate AF Ansatz

The Ansatz of Eq. (4.1) breaks translational invariance in each spin sector, but preserves $\text{U}^z(1)$ and $\text{U}^c(1)$ symmetries. Let us prove this. The Ansatz is defined by the following EV being non-zero,

$$\langle \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow} \rangle = 2m \quad (4.2)$$

ignoring the local sign factor. SSB happens establishes when $m \neq 0$. Then:

- Recall the definitions of $\hat{\Psi}_{\mathbf{r}}$ of Eq. (2.7)

$$\hat{\Psi}_{\mathbf{r}} = \begin{bmatrix} \hat{c}_{\mathbf{r}\uparrow} \\ \hat{c}_{\mathbf{r}\downarrow} \end{bmatrix}$$

and the local spin operators of Eq. (2.8),

$$\hat{S}_{\mathbf{r}}^+ = \frac{\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}}{2} \quad \hat{S}_{\mathbf{r}}^- = \frac{\hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\uparrow}}{2} \quad \hat{S}_{\mathbf{r}}^z = \frac{\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\uparrow} - \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}}{2}$$

Consider Eq. (2.19) and its conjugate,

$$[\hat{c}_{\mathbf{r}\sigma}^\dagger, \hat{n}_{\mathbf{r}\sigma}] = -\hat{c}_{\mathbf{r}\sigma}^\dagger \quad [\hat{c}_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}] = \hat{c}_{\mathbf{r}\sigma}$$

The operators $\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}$ and $\hat{S}_{\mathbf{r}}^\pm$ do not commute. Indeed,

$$\begin{aligned} [\hat{S}_{\mathbf{r}}^+, \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}] &= \frac{1}{2} [\hat{c}_{\mathbf{r}\uparrow}^\dagger, \hat{n}_{\mathbf{r}\uparrow}] \hat{c}_{\mathbf{r}\downarrow} - \frac{1}{2} \hat{c}_{\mathbf{r}\uparrow}^\dagger [\hat{c}_{\mathbf{r}\downarrow}, \hat{n}_{\mathbf{r}\downarrow}] \\ &= -\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} \\ [\hat{S}_{\mathbf{r}}^-, \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}] &= \frac{1}{2} \hat{c}_{\mathbf{r}\downarrow}^\dagger [\hat{c}_{\mathbf{r}\uparrow}, \hat{n}_{\mathbf{r}\uparrow}] - \frac{1}{2} [\hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{n}_{\mathbf{r}\downarrow}] \hat{c}_{\mathbf{r}\uparrow} \\ &= \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\uparrow} \end{aligned}$$

Instead, $\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}$ and $\hat{S}_{\mathbf{r}}^z$ commute, because all on-site density operators always do. Then, in order to have $m \neq 0$ in Eq. (4.2), $\text{SU}(2)$ needs to be broken while $\text{U}^z(1)$ needs not.

- From Eq. (2.5) we know that on-site density operators are gauge-invariant separately in both spin sectors. Then no selection rule applies to $\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}$: its EV can be non-zero without breaking $U^z(1)$.

The chosen Ansatz is coherent with the symmetry requirements of Tab. 2.1. Let us rephrase this in the language of HF methods. We need an HFPs vector $\mathbf{v}$, whose components v_i are equipped with self-consistency equations. Commensurate antiferromagnetism is described by those HFPs emerging from Wick's theorem applied to the EHM respecting symmetry requirement. Among these we include explicitly the spin averaged unbalance has a two-sites periodicity,

$$\forall \mathbf{r} = (x, y) \in \mathcal{L} \quad : \quad m \equiv (-1)^{x+y} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \quad (4.3)$$

By allowing for this EV to be non-zero we respect phase symmetry and impose antiferromagnetism to be commensurate. Setting $m \neq 0$ is equivalent to imposing Eq. (4.1).

The on-site averaged density is given by:

$$\forall \mathbf{r} \in \mathcal{L} \quad : \quad n \equiv \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \quad (4.4)$$

Despite of similarity with m , this is not an HFP. First, on-site density respect all EHM symmetries. Second, we are imposing average density as a parametrisation of the model. There is nothing to be searched self-consistently here.

Note that the quantities n, m in Eqns. (4.4), (4.3) are $\mathbf{r}$ -independent. Then it must hold

$$n = \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \quad (4.5)$$

$$m = \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} (-1)^{x+y} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \quad (4.6)$$

From the first equation we gain an operational method to compute density, from the second we get a self-consistency equation for magnetisation. Indeed, the ground-state depends on HFP m in some way to be determined explicitly. While this passage added nothing about the theoretical description of antiferromagnetism (we averaged constants over the lattice), it will be crucial in order to move the discussion to reciprocal space.

4.1.2 The AF Ansatz in reciprocal space

The core parts of the following derivations will be carried out in reciprocal space. It is then useful to comment the reciprocal structure of the AF Ansatz. First, let:

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \quad (4.7)$$

$$\hat{\psi}_{\mathbf{k}\sigma} \equiv \hat{c}_{\mathbf{k}+\pi\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \quad (4.8)$$

Note that $\hat{n}_{\mathbf{k}\sigma}$ is not the Fourier-transform of $\hat{n}_{\mathbf{r}\sigma}$. Recall also the γ -wave form factors of Tab. 2.5. Then:

$$\hat{\psi}_{\mathbf{k}+\pi\sigma}^\dagger = \hat{\psi}_{\mathbf{k}\sigma} \quad (4.9)$$

$$\varphi_{\mathbf{k}+\pi}^{(\gamma)} = -\varphi_{\mathbf{k}}^{(\gamma)} \quad (4.10)$$

which, by the way, implies the property

$$\varphi_{\mathbf{k}+\pi}^{(\gamma)} \psi_{\mathbf{k}+\pi\sigma}^\dagger = -\varphi_{\mathbf{k}}^{(\gamma)} \psi_{\mathbf{k}\sigma} \quad (4.11)$$

that will result useful later.

In the AF phase, we have $\langle \hat{n}_{\mathbf{k}\sigma} \rangle \neq 0$ and $\langle \hat{\psi}_{\mathbf{k}\sigma} \rangle \neq 0$. This is proven by the following chains of equations:

$$\begin{aligned}
\sum_{\mathbf{r} \in \mathcal{L}} \hat{n}_{\mathbf{r}\sigma} &= \sum_{\mathbf{r} \in \mathcal{L}} \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} \\
&\stackrel{\mathcal{F}}{=} \sum_{\mathbf{r} \in \mathcal{L}} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k} \in \text{BZ}} e^{i\mathbf{k} \cdot \mathbf{r}} \hat{c}_{\mathbf{k}\sigma}^\dagger \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}' \in \text{BZ}} e^{-i\mathbf{k}' \cdot \mathbf{r}} \hat{c}_{\mathbf{k}'\sigma} \\
&= \sum_{\mathbf{k} \in \text{BZ}} \sum_{\mathbf{k}' \in \text{BZ}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma} \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} e^{-i(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{r}} \\
&= \sum_{\mathbf{k} \in \text{BZ}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \\
&= \sum_{\mathbf{k} \in \text{BZ}} \hat{n}_{\mathbf{k}\sigma}
\end{aligned} \tag{4.12}$$

where we used Eq. (2.31). Thus:

$$\begin{aligned}
n &= \frac{1}{2} \sum_{\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} n + \overbrace{\frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} (-1)^{x+y} m}^{=0} \right] \\
&= \frac{1}{2L_x L_y} \sum_{\sigma} \sum_{\mathbf{r} \in \mathcal{L}} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \\
&= \frac{1}{2L_x L_y} \sum_{\sigma} \sum_{\mathbf{k} \in \text{BZ}} \langle \hat{n}_{\mathbf{k}\sigma} \rangle
\end{aligned} \tag{4.13}$$

We used the Ansatz of Eq. (4.1). Since $n \neq 0$, in the last sum $\langle \hat{n}_{\mathbf{k}\sigma} \rangle \neq 0$ is impossible for all $\mathbf{k}$ s. Similarly, for the oscillatory part:

$$\begin{aligned}
\sum_{\mathbf{r} \in \mathcal{L}} (-1)^{x+y} \hat{n}_{\mathbf{r}\sigma} &= \sum_{\mathbf{r} \in \mathcal{L}} (-1)^{x+y} \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} \\
&\stackrel{\mathcal{F}}{=} \sum_{\mathbf{r} \in \mathcal{L}} e^{i\boldsymbol{\pi} \cdot \mathbf{r}} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k} \in \text{BZ}} e^{i\mathbf{k} \cdot \mathbf{r}} \hat{c}_{\mathbf{k}\sigma}^\dagger \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}' \in \text{BZ}} e^{-i\mathbf{k}' \cdot \mathbf{r}} \hat{c}_{\mathbf{k}'\sigma} \\
&= \sum_{\mathbf{k} \in \text{BZ}} \sum_{\mathbf{k}' \in \text{BZ}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma} \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} e^{-i[\mathbf{k}' - (\mathbf{k} + \boldsymbol{\pi})] \cdot \mathbf{r}} \\
&= \sum_{\mathbf{k} \in \text{BZ}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k} + \boldsymbol{\pi}\sigma}
\end{aligned} \tag{4.14}$$

$$= \sum_{\mathbf{k} \in \text{BZ}} \hat{\psi}_{\mathbf{k}\sigma} \tag{4.15}$$

where we defined the AF vector $\boldsymbol{\pi} \equiv (\pi, \pi)$. Thus, again using the Ansatz of Eq. (4.1),

$$\begin{aligned}
(-1)^{\delta_{\sigma=\downarrow}} m &= \overbrace{\frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} (-1)^{x+y} n}^{=0} + \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} (-1)^{\delta_{\sigma=\downarrow}} m \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} (-1)^{x+y} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \langle \hat{\psi}_{\mathbf{k}\sigma} \rangle
\end{aligned} \tag{4.16}$$

where we used $1 = ((-1)^{x+y})^2$. If $m \neq 0$, in the last sum we cannot have $\langle \hat{\psi}_{\mathbf{k}\sigma} \rangle \neq 0$ for all $\mathbf{k}$ s. We

Operator	Sector	RWCs	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
	o.s.	Hartree	μ shift by nzV
$\hat{H}_V$		Hartree	μ shift by nzV
	s.s.	Fock	Bands renormalisation and resymmetrization, gap complexification

Table 4.1 | Relevant Wick's contractions and net effect in the AF phase. Note the similarity with Tab. 3.1.

can restrict the sum to the MBZ and write it explicitly for both spins

$$m = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} [\hat{\psi}_{\mathbf{k}\uparrow} + \hat{\psi}_{\mathbf{k}\uparrow}^\dagger] \quad (4.17)$$

$$= -\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} [\hat{\psi}_{\mathbf{k}\downarrow} + \hat{\psi}_{\mathbf{k}\downarrow}^\dagger] \quad (4.18)$$

where we used the property of Eq. (4.9).

Recall now Tabs. 2.2 and 2.3: the relevant Wick's contractions are, for each sector,

$\hat{H}_U$	Hartree contraction
$\hat{H}_V$ (o.s. sector)	Hartree contraction
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions

Their role is anticipated Tab. 4.1. In next sections we break down each one and derive the results by analytically applying HF methods.

4.2 Local Hartree effects: antiferromagnetism in the HM

For the AF phase, the derivation is more elaborated than for the normal phase. It is useful to separate the local and non-local interactions and apply to each one HF methods. Let us start from the local quartic interaction,

$$\hat{H}_U = U \sum_{\mathbf{r} \in \mathcal{L}} \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow}$$

We only need to analyse the Hartree contraction channel, as is reported in Tab. 4.1. The HF decomposition and the Hartree sectors are given in Eq. (1.20).

Before extending to the non-local attraction, we dedicate the end of this section to complete the theoretical description of antiferromagnetism in the conventional, locally repulsive Hubbard model. This will allow us to identify easily what changes due to the presence of $\hat{H}_V$.

4.2.1 Local Hartree channel in the AF phase: analytic derivation

The HF method, applied to a quartic operator, gives the decomposition of Eq. (1.20). Decompose $\hat{H}_U$ in the Hartree channel:

$$\hat{H}_U \simeq U \sum_{\mathbf{r} \in \mathcal{L}} [\hat{n}_{\mathbf{r}\uparrow} \langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \hat{n}_{\mathbf{r}\downarrow} - \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle]$$

Note that in the AF phase, as is reported in Tab. 2.2, the local repulsion decomposes just in the Hartree channel. Then no other term must be included in the above approximation of the quartic operator.

The one-body operators can be rewritten as:

$$\hat{n}_{\mathbf{r}\uparrow} \langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \hat{n}_{\mathbf{r}\downarrow} = \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} (\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}) - \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} (\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow})$$

From Eqns. (4.3) and (4.4), we recognise n and the HFP m . We have the decomposition:

$$\hat{H}_U \simeq nU \sum_{\mathbf{r} \in \mathcal{L}} [\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}] - mU \sum_{\mathbf{r} \in \mathcal{L}} (-1)^{x+y} [\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}] - E_{\text{H}/U}^{(\text{AF})} \quad (4.19)$$

with $E_{\text{H}/U}^{(\text{AF})}$ the HM Hartree contraction energy, discussed in Sec. 4.2.2. The first term of Eq. (4.19) gives a number operator,

$$nU \sum_{\mathbf{r} \in \mathcal{L}} [\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}] = nU \times \hat{N}$$

Then we get the first renormalisation of chemical potential:

$$\mu \rightarrow \mu - nU \quad (4.20)$$

We partially anticipated it in Tab. 4.1. Another shift to μ will come from the non-local attraction. In the second term of Eq. (4.19), apply Eq. (4.14). It follows:

$$mU \sum_{\mathbf{r} \in \mathcal{L}} (-1)^{x+y} [\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}] = mU \sum_{\mathbf{k} \in \text{BZ}} \left[\hat{c}_{\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{k}+\pi\uparrow} - \hat{c}_{\mathbf{k}\downarrow}^\dagger \hat{c}_{\mathbf{k}+\pi\downarrow} \right]$$

Recall the definition of the MBZ given in Eq. (3.41). The full BZ can be seen as the sum of two MBZ shifted by π . Then, using 2π periodicity, the following identity holds:

$$\begin{aligned} \sum_{\mathbf{k} \in \text{BZ}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}+\pi\sigma} &= \sum_{\mathbf{k} \in \text{MBZ}} \left[\hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}+\pi\sigma} + \hat{c}_{\mathbf{k}+\pi\sigma}^\dagger \hat{c}_{\mathbf{k}+2\pi\sigma} \right] \\ &= \sum_{\mathbf{k} \in \text{MBZ}} \left[\hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}+\pi\sigma} + \hat{c}_{\mathbf{k}+\pi\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \right] \end{aligned} \quad (4.21)$$

We write everything in terms of the operators $\hat{n}_{\mathbf{k}\sigma}$ and $\hat{\psi}_{\mathbf{k}\sigma}$, defined in Eqns. (4.7), (4.8). The local interaction is reduced to

$$\hat{H}_U \simeq -mU \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} (-1)^{\delta_{\sigma=\downarrow}} \left(\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^\dagger \right) + nU \hat{N} - E_{\text{H}/U}^{(\text{AF})}$$

We define the gap function:

$$\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$$

This new object allows us to rewrite the local repulsion as:

$$\hat{H}_U \simeq - \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \left(\Delta_{\mathbf{k}\sigma} \hat{\psi}_{\mathbf{k}\sigma} + \Delta_{\mathbf{k}\sigma}^* \hat{\psi}_{\mathbf{k}\sigma}^\dagger \right) + nU \hat{N} - E_{\text{H}/U}^{(\text{AF})} \quad (4.22)$$

The gap function is actually independent of $\mathbf{k}$. It is a constant in reciprocal space. However, we defined it as $\Delta_{\mathbf{k}\sigma}$: this notation will be useful later.

4.2.2 Local Hartree contraction energy

In last section we defined $E_{\text{H}/U}^{(\text{AF})}$ as the local Hartree contraction energy,

$$\begin{aligned} E_{\text{H}/U}^{(\text{AF})} &\equiv U \sum_{\mathbf{r} \in \mathcal{L}} \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle \\ &= U \sum_{\mathbf{r} \in \mathcal{L}} (n + (-1)^{x+y} m) (n - (-1)^{x+y} m) \end{aligned}$$

where we used the Ansatz of Eq. (4.1). Expanding the product above we get:

$$\begin{aligned} (n + (-1)^{x+y} m) (n - (-1)^{x+y} m) &= n^2 + ((-1)^{x+y} - (-1)^{x+y}) nm - ((-1)^{x+y})^2 m^2 \\ &= n^2 - m^2 \end{aligned}$$

The sum over $\mathcal{L}$ gives $L_x L_y$ identical terms, thus we get:

$$E_{\text{H}/U}^{(\text{AF})} = L_x L_y \times U (n^2 - m^2) \quad (4.23)$$

which is the contraction energy in the local Hartree channel.

4.2.3 Antiferromagnetism in the HM

One of the aims of this projects is to identify, theoretically, what changes for each phase due to the presence of the non-local attraction $\hat{H}_V$ with respect to the conventional HM. In this section we give a description of antiferromagnetism in the conventional HM, and later we move to the EHM.

Matrix representation. The HM hamiltonian contains the kinetic operator and the local repulsion,

$$\hat{H}_t + \hat{H}_U$$

Let the reciprocal space spinors:

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\pi\sigma} \end{bmatrix}$$

Despite sharing a similar notation with the spinors of Eq. (2.7), they are inherently different objects. Recall Eq. (2.33): the kinetic sector is represented in reciprocal space by

$$\begin{aligned} \hat{H}_t &= \sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \hat{c}_{\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{k}\sigma} \\ &= \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \left(\hat{c}_{\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{k}\sigma} - \hat{c}_{\mathbf{k}+\pi\sigma}^{\dagger} \hat{c}_{\mathbf{k}+\pi\sigma} \right) \end{aligned}$$

where we used antiperiodicity by π of the bare bands $\epsilon_{\mathbf{k}}$ defined in Eq. (2.34),

$$\epsilon_{\mathbf{k}} = -\epsilon_{\mathbf{k}+\pi}$$

The kinetic operator can be written in terms of the spinors $\hat{\Psi}_{\mathbf{k}\sigma}$,

$$\hat{H}_t = \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \hat{\Psi}_{\mathbf{k}\sigma}^{\dagger} \begin{bmatrix} \epsilon_{\mathbf{k}\sigma} & \\ & -\epsilon_{\mathbf{k}\sigma} \end{bmatrix} \hat{\Psi}_{\mathbf{k}\sigma}$$

The local repulsion $\hat{H}_U$ of Eq. (4.22) can also be written in terms of these spinors:

$$\hat{H}_U = \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \hat{\Psi}_{\mathbf{k}\sigma}^{\dagger} \begin{bmatrix} & -\Delta_{\mathbf{k}\sigma}^* \\ -\Delta_{\mathbf{k}\sigma} & \end{bmatrix} \hat{\Psi}_{\mathbf{k}\sigma} + nU\hat{N} - E_{\text{H/U}}^{(\text{AF})}$$

Collect everything. For the AF phase, the final form of the quadratic hamiltonian is

$$\hat{H}_t + \hat{H}_U \simeq \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \hat{\Psi}_{\mathbf{k}\sigma}^{\dagger} h_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} + nU\hat{N} - E_{\text{H/U}}^{(\text{AF})} \quad (4.24)$$

where we defined the 2×2 matrix:

$$h_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \epsilon_{\mathbf{k}\sigma} & -\Delta_{\mathbf{k}\sigma}^* \\ -\Delta_{\mathbf{k}\sigma} & -\epsilon_{\mathbf{k}\sigma} \end{bmatrix}$$

This is a parametric hamiltonian. We need to identify its ground state for each choice of the parameters $\Delta_{\mathbf{k}\sigma}$. This is done by mapping the problem to the one of a set of spin subject to local magnetic fields: let us show this.

Pseudospin representation. The matrix $h_{\mathbf{k}\sigma}$ can be written in term of the Pauli matrices τ^{α}

$$h_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma} \tau^z - \Delta_{\mathbf{k}\sigma} \tau^x$$

Let us define the pseudospin operator $\hat{\mathbf{s}}_{\mathbf{k}\sigma}$

$$\hat{s}_{\mathbf{k}\sigma}^{\alpha} \equiv \hat{\Psi}_{\mathbf{k}\sigma}^{\dagger} \tau^{\alpha} \hat{\Psi}_{\mathbf{k}\sigma} \quad \text{with} \quad \hat{\mathbf{s}}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{s}_{\mathbf{k}\sigma}^x \\ \hat{s}_{\mathbf{k}\sigma}^y \\ \hat{s}_{\mathbf{k}\sigma}^z \end{bmatrix}$$

EV	Meaning
$\langle \hat{s}_{\mathbf{k}\sigma}^x \rangle$	$\langle \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^\dagger \rangle$
$\langle \hat{s}_{\mathbf{k}\sigma}^y \rangle$	$i \langle \hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}_{\mathbf{k}\sigma}^\dagger \rangle$
$\langle \hat{s}_{\mathbf{k}\sigma}^z \rangle$	$\langle \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\pi\sigma} \rangle$

Table 4.2 | List of the various expectation values (EVs) of each pseudo-spin component and their meaning in terms of fermionic operators.

and the pseudofield $\mathbf{b}_{\mathbf{k}\sigma}$:

$$\mathbf{b}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} -\Delta_{\mathbf{k}\sigma} \\ 0 \\ \epsilon_{\mathbf{k}\sigma} \end{bmatrix}$$

These operators satisfy a $\mathfrak{su}(2)$ algebra up to a constant C ,

$$[\hat{s}_{\mathbf{k}\sigma}^\alpha, \hat{s}_{\mathbf{k}\sigma}^\beta] = C \times i \sum_{\gamma} \epsilon_{\alpha\beta\gamma} \hat{s}_{\mathbf{k}\sigma}^\gamma$$

The proof is conceptually identical to the one presented in Sec. 2.1.2. The actual value of C is irrelevant, as we will see in next paragraphs. **The two set of spin operators are unrelated.** Let us express the pseudospin operator in terms of the operators of Eqns. (4.7), (4.8):

$$\hat{s}_{\mathbf{k}\sigma}^x = \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^\dagger \quad (4.25)$$

$$\hat{s}_{\mathbf{k}\sigma}^y = i \left(\hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}_{\mathbf{k}\sigma}^\dagger \right) \quad (4.26)$$

$$\hat{s}_{\mathbf{k}\sigma}^z = \hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\pi\sigma} \quad (4.27)$$

We report these relations also in Tab. 4.2. Let also:

$$\begin{aligned} \hat{\mathbb{I}}_{\mathbf{k}\sigma} &\equiv \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \mathbb{I}_{2 \times 2} \hat{\Psi}_{\mathbf{k}\sigma} \\ &= \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\pi\sigma} \end{aligned} \quad (4.28)$$

In pseudospin representation we get:

$$\hat{H}_t + \hat{H}_U \simeq \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \mathbf{b}_{\mathbf{k}\sigma} \cdot \hat{\mathbf{s}}_{\mathbf{k}\sigma} + nU \hat{N} - E_{\text{H}/U}^{(\text{AF})} \quad (4.29)$$

The hamiltonian represents a system of spins subject to local magnetic fields. The field $\mathbf{b}_{\mathbf{k}\sigma}$ is orthogonal to the y plane, and is tilted in the z plane by an angle $2\theta_{\mathbf{k}\sigma}$ with:

$$\cos(2\theta_{\mathbf{k}\sigma}) = \frac{\epsilon_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}} \quad \sin(2\theta_{\mathbf{k}\sigma}) = \frac{\Delta_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}} \quad E_{\mathbf{k}\sigma} \equiv \sqrt{\epsilon_{\mathbf{k}\sigma}^2 + |\Delta_{\mathbf{k}\sigma}|^2}$$

A sketch of these angles and the pseudofield is given in Fig. 4.2.

Diagonalisation. To diagonalise the Hubbard hamiltonian $\hat{H}_t + \hat{H}_U$ of Eq. (4.29) means to perform some unitary operation on the $\hat{\Psi}_{\mathbf{k}\sigma}$ spinors which reduces the 2×2 matrix to a diagonal form. This means to diagonalise each $h_{\mathbf{k}\sigma}$. This is obtained through a rotation around the y axis:

$$d_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} W_{\mathbf{k}\sigma}^\dagger \quad \text{with} \quad d_{\mathbf{k}\sigma} \equiv \begin{bmatrix} E_{\mathbf{k}\sigma} & \\ & -E_{\mathbf{k}\sigma} \end{bmatrix} \quad (4.30)$$

The matrix $W_{\mathbf{k}\sigma}$ is unitary, and gives a counter-clockwise rotation by an angle $2\theta_{\mathbf{k}\sigma}$ around the y axis:

$$\begin{aligned} W_{\mathbf{k}\sigma} &\equiv e^{-i\theta_{\mathbf{k}\sigma}\tau^y} \\ &= \cos \theta_{\mathbf{k}\sigma} - i \sin \theta_{\mathbf{k}\sigma} \tau^y \\ &= \begin{bmatrix} \cos \theta_{\mathbf{k}\sigma} & -\sin \theta_{\mathbf{k}\sigma} \\ \sin \theta_{\mathbf{k}\sigma} & \cos \theta_{\mathbf{k}\sigma} \end{bmatrix} \end{aligned} \quad (4.31)$$

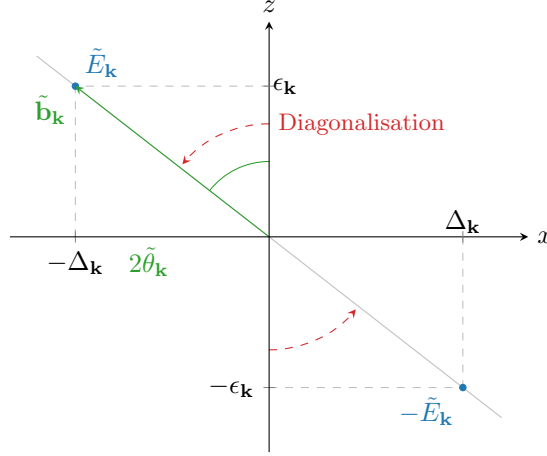


Figure 4.2 | Sketch of the diagonalisation procedure for the HM antiferromagnet. To diagonalise the HF hamiltonian, we align the z axis with the pseudo-field.

Its eigenvalues are $\pm E_{\mathbf{k}\sigma}$, with

$$E_{\mathbf{k}\sigma} = \sqrt{\epsilon_{\mathbf{k}\sigma}^2 + |\Delta_{\mathbf{k}\sigma}|^2} \quad (4.32)$$

In Fig. 4.2 we sketch the pseudo-magnetic field in the xz plane, and the diagonalisation procedure. $W_{\mathbf{k}\sigma}$ is a rotation by an angle $2\theta_{\mathbf{k}\sigma}$ around the y axis, and in the end aligns the original z axis with the pseudofield axis.

Applying the rotation matrix to the spinors we obtain new spinors of $\gamma^{(\pm)}$ fermions,

$$\hat{\Gamma}_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{\gamma}_{\mathbf{k}\sigma}^{(+)} \\ \hat{\gamma}_{\mathbf{k}\sigma}^{(-)} \end{bmatrix} \quad (4.33)$$

In the end we have the non-interacting hamiltonian:

$$\begin{aligned} \hat{H}_t + \hat{H}_U &\simeq \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \hat{\Psi}_{\mathbf{k}\sigma}^{\dagger} h_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} + (\dots) \\ &= \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \hat{\Psi}_{\mathbf{k}\sigma}^{\dagger} W_{\mathbf{k}\sigma}^{\dagger} W_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} W_{\mathbf{k}\sigma}^{\dagger} W_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} + (\dots) \\ &= \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \hat{\Gamma}_{\mathbf{k}\sigma}^{\dagger} d_{\mathbf{k}\sigma} \hat{\Gamma}_{\mathbf{k}\sigma} + (\dots) \\ &= \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} E_{\mathbf{k}\sigma} \left(\hat{\gamma}_{\mathbf{k}\sigma}^{(+)\dagger} \hat{\gamma}_{\mathbf{k}\sigma}^{(+)} - \hat{\gamma}_{\mathbf{k}\sigma}^{(-)\dagger} \hat{\gamma}_{\mathbf{k}\sigma}^{(-)} \right) + (\dots) \end{aligned}$$

where we intend $(\dots) = nU\hat{N} - E_{\text{H/U}}^{(\text{AF})}$. The final non-interacting hamiltonian describes a free Fermi gas of $\gamma^{(\pm)}$ fermions, respectively on the bands $\pm E_{\mathbf{k}\sigma}$. Note that these two bands are separated by a gap $\Delta_{\mathbf{k}\sigma}$.

We could redefine the pseudospin operators dividing them by an appropriate constant, in order to satisfy exactly the $\mathfrak{su}(2)$ algebra, and at the same time multiply the pseudo-magnetic field by the same constant. The product $\mathbf{b}_{\mathbf{k}\sigma} \cdot \hat{\mathbf{s}}_{\mathbf{k}\sigma}$ would be unchanged, and the diagonalisation matrix $W_{\mathbf{k}\sigma}$ would be the same. Because of this, the aforementioned constant C is irrelevant.

4.2.4 Features of the AF solution

This subsection is devoted to discussing briefly some features of the AF solution we found. It is organised as a series of unrelated paragraphs, each to derive a precise property or formula we use in the algorithmic computation. With these observations we conclude the part about the local repulsion and extend to full EHM.

Pseudospin EVs. The expectation value of the original pseudospin operators in the various directions can be expressed in terms of the tilted pseudospins $\hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \tau^\alpha \hat{\Gamma}_{\mathbf{k}\sigma}$,

$$\begin{aligned}\langle \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \tau^x \hat{\Psi}_{\mathbf{k}\sigma} \rangle &= -\sin 2\theta_{\mathbf{k}\sigma} \langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \\ \langle \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \tau^y \hat{\Psi}_{\mathbf{k}\sigma} \rangle &= 0 \\ \langle \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Psi}_{\mathbf{k}\sigma} \rangle &= \cos 2\theta_{\mathbf{k}\sigma} \langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle\end{aligned}$$

where we used the angular relations of Fig. 4.2. Moving to the tilted frame, the z component of the pseudo-spin is just a thermal superposition of “up” and “down” states

$$\langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle = f(E_{\mathbf{k}\sigma}; \beta, \tilde{\mu}) - f(-E_{\mathbf{k}\sigma}; \beta, \tilde{\mu}) \quad (4.34)$$

Let us define the thermal factors:

$$\text{Th}_{\mathbf{k}\sigma}^{(\pm)}[\beta, \mu] \equiv f(-E_{\mathbf{k}\sigma}; \beta, \mu) \pm f(E_{\mathbf{k}\sigma}; \beta, \mu) \quad (4.35)$$

which will often appear in computations. We have:

$$\langle \hat{s}_{\mathbf{k}\sigma}^x \rangle = \frac{\Delta_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}} \text{Th}_{\mathbf{k}\sigma}^{(-)}[\beta, \mu] \quad (4.36)$$

$$\langle \hat{s}_{\mathbf{k}\sigma}^y \rangle = 0 \quad (4.37)$$

$$\langle \hat{s}_{\mathbf{k}\sigma}^z \rangle = -\frac{e_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}} \text{Th}_{\mathbf{k}\sigma}^{(-)}[\beta, \mu] \quad (4.38)$$

Gauge invariance guarantees that we can rotate the pseudo-field on the xy plane, thus obtaining $\langle \hat{s}_{\mathbf{k}\sigma}^y \rangle \neq 0$. From these EVs, we get the expression of the self-consistency equation for m , and the self-consistent formula to compute the density. Note that $E_{\mathbf{k}\sigma}$ is actually spin independent, and so is the thermal factor. We can neglect the spin index from both.

Magnetisation self-consistency equation. Using Eqns. (4.17), (4.18) we get:

$$m = \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \left\langle \left(\hat{\psi}_{\mathbf{k}\uparrow} + \hat{\psi}_{\mathbf{k}\downarrow}^\dagger \right) - \left(\hat{\psi}_{\mathbf{k}\downarrow} + \hat{\psi}_{\mathbf{k}\uparrow}^\dagger \right) \right\rangle$$

From Tab. 4.2, we recognise a pseudospin EV in direction x . From Eq. (4.36)

$$\begin{aligned}m &= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} [\langle \hat{s}_{\mathbf{k}\uparrow}^x \rangle - \langle \hat{s}_{\mathbf{k}\downarrow}^x \rangle] \\ &= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\text{Re}\{\Delta_{\mathbf{k}\uparrow}\} - \text{Re}\{\Delta_{\mathbf{k}\downarrow}\}}{E_{\mathbf{k}}} \text{Th}_{\mathbf{k}}^{(-)}[\beta, \mu] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\Delta_{\mathbf{k}\uparrow}}{E_{\mathbf{k}}} \text{Th}_{\mathbf{k}}^{(-)}[\beta, \mu]\end{aligned} \quad (4.39)$$

In last passage we used $\Delta_{\mathbf{k}\uparrow} = -\Delta_{\mathbf{k}\downarrow}$, which follows from the definition of the gap function. The last passage gives the operational self-consistency equation for the HFP m .

Density. Restricting the sum of Eq. (4.13) to the MBZ and extending to both spins, we have:

$$\begin{aligned}n &= \frac{1}{2L_x L_y} \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \langle \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\pi\sigma} \rangle \\ &= \frac{1}{2L_x L_y} \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \langle \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \hat{\Psi}_{\mathbf{k}\sigma} \rangle \\ &= \frac{1}{2L_x L_y} \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \\ &= \frac{1}{2L_x L_y} \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \left(\langle \hat{\gamma}_{\mathbf{k}\sigma}^{(+)\dagger} \hat{\gamma}_{\mathbf{k}\sigma}^{(+)} \rangle + \langle \hat{\gamma}_{\mathbf{k}\sigma}^{(-)\dagger} \hat{\gamma}_{\mathbf{k}\sigma}^{(-)} \rangle \right) \\ &= \frac{1}{2L_x L_y} \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \text{Th}_{\mathbf{k}\sigma}^{(+)}[\beta, \mu]\end{aligned}$$

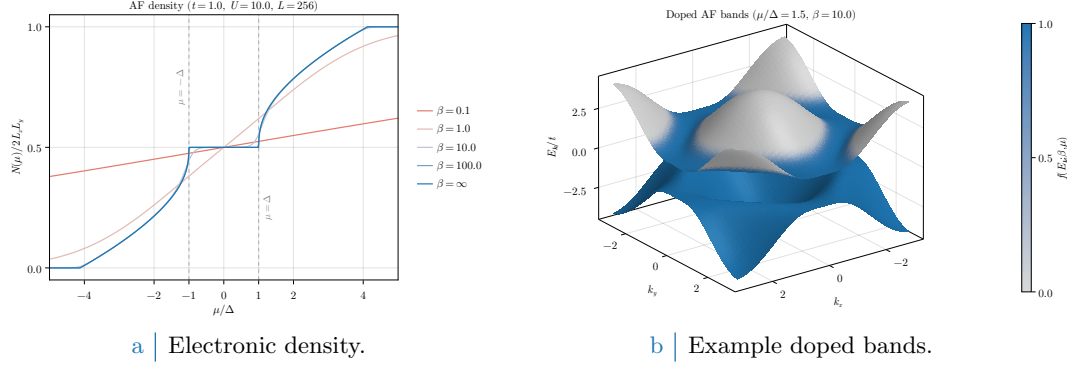


Figure 4.3 Left panel: electronic density in the AF phase as a function of the chemical potential, measured in relation to the gap $\Delta = mU$, at various temperatures. Right panel: example doped bands, with $\mu/\Delta = 1.5$ set arbitrarily and temperature set to $\beta = 10$ to enhance smearing effects around FS. Single-state filling is proportional to color magnitude.

where we used unitarity of the $\hat{\Psi} \rightarrow \hat{\Gamma}$ transformation.

The thermal factor is spin-independent. Moreover, the Bogoliubov bands $E_{\mathbf{k}\sigma}$ are MBZ periodic. Thus we get the two equivalent formulas for density:

$$n = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \text{Th}_{\mathbf{k}\uparrow}^{(+)}[\beta, \mu] \quad (4.40)$$

$$= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \text{Th}_{\mathbf{k}\uparrow}^{(+)}[\beta, \mu] \quad (4.41)$$

The two are perfectly equivalent, but the second one is handier to implement in numeric computations. In Fig. 4.3a we sketch density as a function of the chemical potential. At low temperature, the gap plateau at zero energy is most visible. The lattice is half-filled for any chemical potential inside the plateau range.

Unconventional metallicity. Consider now the gran canonical hamiltonian $\hat{K} \equiv \hat{H} - \mu\hat{N}$ and use Eq. (4.28). The diagonalising matrix remains the same, being

$$-\mu\hat{N} = \sum_{\mathbf{k} \in \text{MBZ}} \begin{bmatrix} -\mu & \\ & -\mu \end{bmatrix} \implies W_{\mathbf{k}\sigma} [h_{\mathbf{k}\sigma} - \mu\mathbb{I}_{2 \times 2}] W_{\mathbf{k}\sigma}^\dagger = d_{\mathbf{k}\sigma} - \mu\mathbb{I}_{2 \times 2} \quad (4.42)$$

thus for non-null chemical potential (i.e. finite doping) the eigenvalues are just shifted, $\pm E_{\mathbf{k}} - \mu$. These two bands have the same number of single-particle states. In Fig. 4.3b we sketch the bands $\pm E_{\mathbf{k}\sigma}$. Intense colour indicates an almost filled single-particle state, lighter gray colour an almost empty single-particle state. The gap is at zero energy, but at any finite doping $\tilde{\mu}$ crosses the upper band.

Instability of the doped antiferromagnet. As we said at the beginning of this chapter, there exist several ways to realise an Hubbard antiferromagnet. For the conventional HM, it is known [14, 15] that the commensurate AF phase is unstable at any finite doping. This is discussed briefly in Sec. A.3. The key takeaway is that insulating antiferromagnets are stable with respect to spin fluctuations, while metallic antiferromagnets are not. The commensurate AF phase is insulating only at half-filling, because the chemical potential lies within the gap.

The mere existence of a self-consistent solution does not guarantee stability: indeed, when in Sec. 1.2.2 we derived the variational solution, we only required it to be a stationary point of the HF energy. In other words, there is no need for the solution to be a minimum. The commensurate AF phase is, in fact, a saddle point of HF energy [14] and any finite doping disrupts commensurate ordering towards uncommensurate phases.

These observations end the description of commensurate antiferromagnetism in the conventional HM. Some more details and observations are provided in App. A.

4.3 Extension to the EHM

We now expand to the EHM, including the non-local interaction $\hat{H}_V$. Our strategy is to generalise the discussion of the section above. A formally identical solution with renormalised quantities,

$$\epsilon_{\mathbf{k}\sigma} \rightarrow \tilde{\epsilon}_{\mathbf{k}\sigma} \quad E_{\mathbf{k}\sigma} \rightarrow \tilde{E}_{\mathbf{k}\sigma} \quad \Delta_{\mathbf{k}\sigma} \rightarrow \tilde{\Delta}_{\mathbf{k}\sigma} \quad \mu \rightarrow \tilde{\mu}$$

is coherent by construction with the symmetry structure of the AF phase. Explicitly, the renormalised bands are $\pm \tilde{E}_{\mathbf{k}\sigma}$, with

$$\tilde{E}_{\mathbf{k}\sigma} \equiv \sqrt{\tilde{\epsilon}_{\mathbf{k}\sigma}^2 + |\tilde{\Delta}_{\mathbf{k}\sigma}|^2}$$

Moreover, we allow $\Delta_{\mathbf{k}\sigma}$ to be a general complex function of $\mathbf{k}$ and σ ,

$$\Delta_{\mathbf{k}\sigma} = \text{Re}\{\Delta_{\mathbf{k}\sigma}\} + i \text{Im}\{\Delta_{\mathbf{k}\sigma}\}$$

We define the renormalised angle $\tilde{\theta}_{\mathbf{k}\sigma}$ through the relations:

$$\sin 2\tilde{\theta}_{\mathbf{k}\sigma} = \frac{|\tilde{\Delta}_{\mathbf{k}\sigma}|}{\tilde{E}_{\mathbf{k}\sigma}} \quad \cos 2\tilde{\theta}_{\mathbf{k}\sigma} = \frac{\tilde{\epsilon}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}} \quad (4.43)$$

and similarly the angle $\tilde{\zeta}_{\mathbf{k}\sigma}$,

$$\sin 2\tilde{\zeta}_{\mathbf{k}\sigma} = \frac{\text{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \quad \cos 2\tilde{\zeta}_{\mathbf{k}\sigma} = \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \quad (4.44)$$

A sketch of these angles is given in Fig. 4.4. For a complex gap function, the new diagonalisation matrix is given by:

$$\begin{aligned} \tilde{W}_{\mathbf{k}\sigma} &\equiv e^{-i\tilde{\theta}_{\mathbf{k}\sigma}\tau^y} e^{-i\tilde{\zeta}_{\mathbf{k}\sigma}\tau^z} \\ &= \left(\cos \tilde{\theta}_{\mathbf{k}\sigma} - i \sin \tilde{\theta}_{\mathbf{k}\sigma} \tau^y \right) \left(\cos \tilde{\zeta}_{\mathbf{k}\sigma} - i \sin \tilde{\zeta}_{\mathbf{k}\sigma} \tau^z \right) \\ &= \begin{bmatrix} \cos \tilde{\theta}_{\mathbf{k}\sigma} & -\sin \tilde{\theta}_{\mathbf{k}\sigma} \\ \sin \tilde{\theta}_{\mathbf{k}\sigma} & \cos \tilde{\theta}_{\mathbf{k}\sigma} \end{bmatrix} \begin{bmatrix} e^{-i\tilde{\zeta}_{\mathbf{k}\sigma}} & \\ & e^{i\tilde{\zeta}_{\mathbf{k}\sigma}} \end{bmatrix} \\ &= \begin{bmatrix} \cos \tilde{\theta}_{\mathbf{k}\sigma} e^{-i\tilde{\zeta}_{\mathbf{k}\sigma}} & -\sin \tilde{\theta}_{\mathbf{k}\sigma} e^{i\tilde{\zeta}_{\mathbf{k}\sigma}} \\ \sin \tilde{\theta}_{\mathbf{k}\sigma} e^{-i\tilde{\zeta}_{\mathbf{k}\sigma}} & \cos \tilde{\theta}_{\mathbf{k}\sigma} e^{i\tilde{\zeta}_{\mathbf{k}\sigma}} \end{bmatrix} \end{aligned}$$

The following relations hold (*tilde* sign indicates the renormalised quantity):

$$\begin{aligned} \langle \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \tau^x \hat{\Psi}_{\mathbf{k}\sigma} \rangle &= -\sin 2\tilde{\theta}_{\mathbf{k}\sigma} \cos 2\tilde{\zeta}_{\mathbf{k}\sigma} \langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \\ \langle \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \tau^y \hat{\Psi}_{\mathbf{k}\sigma} \rangle &= -\sin 2\tilde{\theta}_{\mathbf{k}\sigma} \sin 2\tilde{\zeta}_{\mathbf{k}\sigma} \langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \\ \langle \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Psi}_{\mathbf{k}\sigma} \rangle &= \cos 2\tilde{\theta}_{\mathbf{k}\sigma} \langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \end{aligned}$$

Let us extend Eq. (4.34) and define the renormalised thermal factors,

$$\langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle = f\left(\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) - f\left(-\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \quad (4.45)$$

$$\tilde{\text{Th}}_{\mathbf{k}\sigma}^{(\pm)}[\beta, \tilde{\mu}] \equiv f\left(-\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \pm f\left(\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \quad (4.46)$$

which gives $\langle \hat{s}_{\mathbf{k}\sigma}^z \rangle = -\tilde{\text{Th}}_{\mathbf{k}\sigma}^{(-)}[\beta, \mu]$. Pseudospin expectation values are:

$$\langle \hat{s}_{\mathbf{k}\sigma}^x \rangle = \frac{\text{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{\tilde{E}_{\mathbf{k}\sigma}} \tilde{\text{Th}}_{\mathbf{k}\sigma}^{(-)}[\beta, \mu] \quad (4.47)$$

$$\langle \hat{s}_{\mathbf{k}\sigma}^y \rangle = \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{\tilde{E}_{\mathbf{k}\sigma}} \tilde{\text{Th}}_{\mathbf{k}\sigma}^{(-)}[\beta, \mu] \quad (4.48)$$

$$\langle \hat{s}_{\mathbf{k}\sigma}^z \rangle = -\frac{\tilde{\epsilon}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}} \tilde{\text{Th}}_{\mathbf{k}\sigma}^{(-)}[\beta, \mu] \quad (4.49)$$

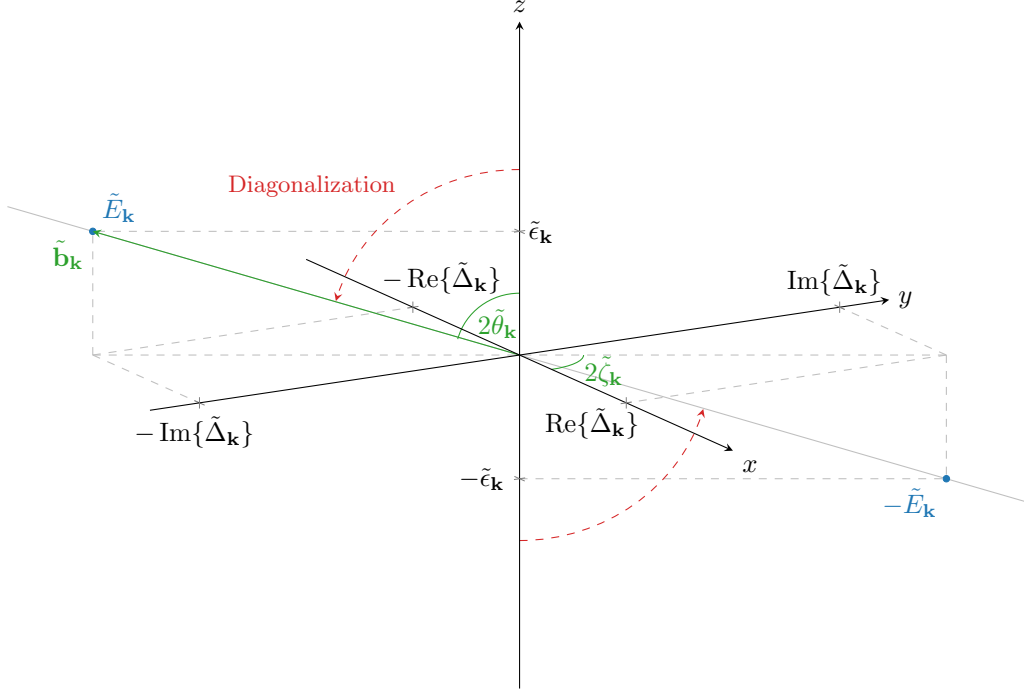


Figure 4.4 | Sketch of the diagonalisation procedure for the EHM antiferromagnet. To diagonalise the HF hamiltonian, we align the z axis with the pseudo-field. This is an extension of Fig. 4.2.

These equations are an extension of Eqns. (4.47), (4.48), (4.49) for the EHM. From now on, in the following sections, we deal with the problem of determining how the various quantities are renormalised.

By renormalising all quantities, the self-consistency equation for magnetisation must stay formally the same of Eq. (4.39)

$$m = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\tilde{\Delta}_{\mathbf{k}\uparrow}}{\tilde{E}_{\mathbf{k}\uparrow}} \tilde{\text{Th}}_{\mathbf{k}\uparrow}^{(-)}[\beta, \mu] \quad (4.50)$$

as well as the formula for density:

$$n = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \text{Th}_{\mathbf{k}\uparrow}^{(+)}[\beta, \mu] \quad (4.51)$$

With this strategy, we get back to the application of HF methods to the full EHM, including the non-local attraction.

4.4 Non-local Hartree effects: chemical potential renormalisation

In last sections we obtained the pseudospin representation of the HM hamiltonian for the AF phase. The main results were Eqns. (4.22), (4.29). Now we must approximate $\hat{H}_V$, reducing to a quadratic operator. We start from the Hartree channel.

The non-local attraction decomposes, in the non-local Hartree channel, in same-spin and

opposite-spin sectors. Using Eq. (1.21) we get:

$$\hat{H}_V \simeq \overbrace{-V \sum_{\langle ij \rangle} \sum_{\sigma} [\langle \hat{n}_{i\sigma} \rangle \hat{n}_{j\sigma} + \hat{n}_{i\sigma} \langle \hat{n}_{j\sigma} \rangle - \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\sigma} \rangle]}^{\text{s.s.}} - \underbrace{V \sum_{\langle ij \rangle} \sum_{\sigma} [\langle \hat{n}_{i\sigma} \rangle \hat{n}_{j\bar{\sigma}} + \hat{n}_{i\sigma} \langle \hat{n}_{j\bar{\sigma}} \rangle - \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\bar{\sigma}} \rangle]}_{\text{o.s.}} + (\text{all the rest})$$

Let $i \rightarrow \mathbf{r} = (x, y)$ and $j \rightarrow \mathbf{r}' = (x', y')$. We use Eq. (4.1), describing oscillatory density for each spin sector. This Ansatz is composed of two parts: the offset n , and the oscillatory part m . The algebraic derivation relative to n part is, of course, the same as for the normal phase. We now treat separately the “operatorial” part of the braced terms of Eq. (??),

$$-V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\langle \hat{n}_{\mathbf{r}\sigma} \rangle \hat{n}_{\mathbf{r}'\sigma} + \hat{n}_{\mathbf{r}\sigma} \langle \hat{n}_{\mathbf{r}'\sigma} \rangle] - V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\langle \hat{n}_{\mathbf{r}\sigma} \rangle \hat{n}_{\mathbf{r}'\bar{\sigma}} + \hat{n}_{\mathbf{r}\sigma} \langle \hat{n}_{\mathbf{r}'\bar{\sigma}} \rangle] \quad (4.52)$$

as well as the its scalar counterpart

$$V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\sigma} \rangle + V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\bar{\sigma}} \rangle \quad (4.53)$$

accounting for energy shift originated as “leftovers” of Wick’s contractions.

4.4.1 Non-local Hartree channel in the AF phase: analytic derivation

Let us start from the operator of Expr. (4.52). By inserting the Ansatz of Eq. (4.1), we get

$$\overbrace{-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}'\sigma} + \hat{n}_{\mathbf{r}\sigma}] + mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} (-1)^{\delta_{\sigma=\uparrow}} \left[(-1)^{x+y} \hat{n}_{\mathbf{r}'\sigma} + (-1)^{x'+y'} \hat{n}_{\mathbf{r}\sigma} \right]}^{\text{s.s.}} - \underbrace{-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}'\bar{\sigma}} + \hat{n}_{\mathbf{r}\sigma}] + mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \left[(-1)^{x+y+\delta_{\sigma=\uparrow}} \hat{n}_{\mathbf{r}'\bar{\sigma}} + (-1)^{x'+y'+\delta_{\bar{\sigma}=\uparrow}} \hat{n}_{\mathbf{r}\sigma} \right]}_{\text{o.s.}}$$

For a square lattice, if $\mathbf{r} = (x, y)$ and $\mathbf{r}' = (x', y')$ are NNs, both the following identities are true

$$\begin{aligned} (-1)^{x'+y'} &= (-1)^{x+y+1} \\ (-1)^{\delta_{\bar{\sigma}=\uparrow}} &= (-1)^{\delta_{\sigma=\uparrow}+1} \end{aligned}$$

We obtain

$$\begin{aligned} &\overbrace{-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\sigma}]}^{\text{s.s. (density)}} - \overbrace{mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} (-1)^{x+y+\delta_{\sigma=\uparrow}} [\hat{n}_{\mathbf{r}\sigma} - \hat{n}_{\mathbf{r}'\sigma}]}^{\text{s.s. (magnetisation)}} \\ &\quad - \underbrace{nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\bar{\sigma}}]}_{\text{o.s. (density)}} + \underbrace{mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} (-1)^{x+y+\delta_{\sigma=\uparrow}} [\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\bar{\sigma}}]}_{\text{o.s. (magnetisation)}} \quad (4.54) \end{aligned}$$

In the above expressions we separated in “density” contributions and “magnetisation” ones. Let us deal with them separately.

Density terms. Consider s.s. and o.s. (density) terms of Expr. (4.54),

$$-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\sigma}] - nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\bar{\sigma}}] \quad (4.55)$$

The discussion is identical to Sec. ?? for the normal phase, with renormalisation of μ being the same of Eq. (3.12).

$$\tilde{\mu} \equiv \mu + n(2zV - U) \quad (4.56)$$

We already included the shift to μ due to the local Hartree channel, anticipated in Eq. (4.20).

Magnetization terms. The magnetisation s.s. and o.s. terms of Expr. (4.54) perfectly cancel out:

$$mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} (-1)^{x+y+\delta_{\sigma=\uparrow}} [(\hat{n}_{\mathbf{r}\sigma} - \hat{n}_{\mathbf{r}'\sigma}) - (\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\bar{\sigma}})] = 0$$

The Hartree channel only affects μ .

4.4.2 Non-local Hartree contraction energy

We now collect double counting terms originated by Wick's contraction in Hartree sector. Consider the scalar Expr. (4.53): it's the sum of two parts, the first coming from s.s. sector, the second from o.s. sector. In computing the contraction energy from Hartree channel, being the operatorial result identical to the one obtained for the normal phase, we expect the same to happen for double counting terms.

s.s. sector double counting energy. For s.s. terms we get

$$\begin{aligned} V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\sigma} \rangle &= V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \overbrace{\left[(n + (-1)^{x+y}m)(n + (-1)^{x'+y'}m) \right]}^{\sigma=\uparrow} \\ &\quad + V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \underbrace{\left[(n - (-1)^{x+y}m)(n - (-1)^{x'+y'}m) \right]}_{\sigma=\downarrow} \end{aligned}$$

The mixed terms (those of order $\mathcal{O}(n) \times \mathcal{O}(m)$) cancel out, and since sign factors necessarily alternate on neighbouring sites, the remainder is just

$$V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\sigma} \rangle = V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} 2(n^2 - m^2)$$

o.s. sector double counting energy. Proceeding identically on the o.s. sector, we get

$$\begin{aligned} V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\bar{\sigma}} \rangle &= V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \overbrace{\left[(n + (-1)^{x+y}m)(n - (-1)^{x'+y'}m) \right]}^{\sigma=\uparrow} \\ &\quad + V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \underbrace{\left[(n - (-1)^{x+y}m)(n + (-1)^{x'+y'}m) \right]}_{\sigma=\downarrow} \end{aligned}$$

which gives

$$V \sum_{\langle ij \rangle} \sum_{\sigma} \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\bar{\sigma}} \rangle = V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} 2(n^2 + m^2)$$

Hence, the total “contraction energy”, given by sum of both contributions, is identical to the normal one of Eq. (??) indeed:

$$E_{H/V}^{(\text{AF})} = -2V \sum_{\langle ij \rangle} \sum_{\sigma} n^2 = 4n^2 V \times \frac{z}{2} L_x L_y \quad (4.57)$$

the minus sign coming from the fact that we are subtracting this energy shift.

4.5 Fock effects: gap and bands renormalisation

We explored the Hartree channel. Let us now move to the effects originated by Fock contractions. We can use the very beginning of Sec. ?? derivation, up to Eq. (3.14),

$$\begin{aligned} V \sum_{\langle ij \rangle} \sum_{\sigma} & \left[\langle \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} \rangle \hat{c}_{j\sigma}^{\dagger} \hat{c}_{i\sigma} + \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} \langle \hat{c}_{j\sigma}^{\dagger} \hat{c}_{i\sigma} \rangle - \langle \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} \rangle \langle \hat{c}_{j\sigma}^{\dagger} \hat{c}_{i\sigma} \rangle \right] \\ & \stackrel{\mathcal{F}}{=} \frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} \sum_{\sigma} [\cos(\delta k_x) + \cos(\delta k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} - E_{F/V}^{(\text{N})} \end{aligned} \quad (4.58)$$

where $E_{F/V}^{(\text{AF})}$ is defined as in Eq. (3.15), with the expectation value being computed on the AF (thermal) ground-state. As before, we discuss the operatorial part and the scalar part separately.

4.5.1 Fock channel in the AF phase: analytic derivation

The commensurate AF phase is described by two EVs being in principle non-zero,

$$|\langle \hat{n}_{\mathbf{k}\sigma} \rangle| \geq 0 \quad |\langle \hat{\psi}_{\mathbf{k}\sigma} \rangle| \geq 0$$

as we specified in Sec. 4.1.2. This implies only two contributions in Eq. (4.58) are non-zero:

$$\mathbf{k} = -\mathbf{k}' \quad \text{or} \quad \mathbf{k} + \boldsymbol{\pi} = -\mathbf{k}'$$

Then we get:

$$\begin{aligned} (\dots) & \stackrel{\mathcal{F}}{=} \frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \\ & \quad \left[\underbrace{\langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}}_{\text{Diagonal terms}} - \underbrace{\langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}}_{\text{Off-diagonal terms}} \right] \end{aligned} \quad (4.59)$$

Now, the above equation presents *diagonal* and *off-diagonal* terms.

Diagonal terms. The diagonal terms of Eq. (4.59) are products of $\hat{n}_{\mathbf{k}\sigma}$ operators. Consider Fig. 3.1a: $\hat{n}_{\mathbf{q}\sigma}$, with $\mathbf{q} \equiv \mathbf{K} + \mathbf{k}$, is coupled to $\langle \hat{n}_{\mathbf{q}'\sigma} \rangle$ with $\mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$. Apply to the diagonal part of Eq. (4.59) this change of variables represented in Fig. 3.1

$$\begin{aligned} & \frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} \\ & = \frac{2V}{L_x L_y} \sum_{\mathbf{q}, \mathbf{q}'} \sum_{\sigma} [\cos(\delta q_x) + \cos(\delta q_y)] \langle \hat{c}_{\mathbf{q}\sigma}^{\dagger} \hat{c}_{\mathbf{q}\sigma} \rangle \hat{c}_{\mathbf{q}'\sigma}^{\dagger} \hat{c}_{\mathbf{q}'\sigma} \\ & = \sum_{\gamma, \sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{n}_{\mathbf{q}\sigma} \rangle \right] \times V \sum_{\mathbf{q}'} \varphi_{\mathbf{q}'}^{(\gamma)*} \hat{n}_{\mathbf{q}'\sigma} \end{aligned} \quad (4.60)$$

where in the last passage we substituted the decomposition of Eq. (2.43). Define as in Eq. (3.2) the EV

$$u_{\mathbf{q}\sigma} \equiv \langle \hat{n}_{\mathbf{q}\sigma} \rangle \quad (4.61)$$

Symmetry	Self-consistency equation	Graph
s^* -wave	$u^{(s^*)} = -\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x + \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. 2.4b
$d_{x^2-y^2}$ -wave	$u^{(d)} = -\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x - \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. 2.4e

Table 4.3 | Symmetry and spin resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$.

The expression in square brackets of Eq. (4.60) is just its γ component, and is expanded in

$$\begin{aligned}
u_{\sigma}^{(\gamma)} &= \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{n}_{\mathbf{q}\sigma} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \left[\varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{n}_{\mathbf{q}\sigma} \rangle + \varphi_{\mathbf{q}+\pi}^{(\gamma)} \langle \hat{n}_{\mathbf{q}+\pi\sigma} \rangle \right] \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} [\langle \hat{n}_{\mathbf{q}\sigma} \rangle - \langle \hat{n}_{\mathbf{q}+\pi\sigma} \rangle] \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\Psi}_{\mathbf{q}\sigma}^{\dagger} \tau^z \hat{\Psi}_{\mathbf{q}\sigma} \rangle \\
&= -\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \frac{\tilde{\epsilon}_{\mathbf{q}\sigma}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\text{Th}}_{\mathbf{q}\sigma}^{(-)}[\beta, \tilde{\mu}]
\end{aligned} \tag{4.62}$$

where, in the last passage, Eq. (4.49) has been used. We are not breaking inversion symmetry (and as a consequence, neither C_2): this implies that, whatever the shape of the renormalised bands, it must be

$$\tilde{\epsilon}_{\mathbf{q}\sigma} = \tilde{\epsilon}_{-\mathbf{q}\sigma} \implies u_{\sigma}^{(p_{\ell})} = 0 \quad \text{with} \quad \ell \in \{x, y\}$$

Renormalised bare bands need to be symmetric functions of the moment. Hence for $\gamma \in \{p_x, p_y\}$ the above sum vanishes. The remaining s^* and d symmetries give legitimate HFPs with relative self-consistency equations, reported in Tab. 4.3 using spin degeneracy.

Recognising harmonics, Eq. (4.60) becomes

$$\begin{aligned}
\frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} \\
= V \sum_{\mathbf{q}, \sigma} \left[u_{\sigma}^{(s^*)} (\cos q_x + \cos q_y) + u_{\sigma}^{(d)} (\cos q_x - \cos q_y) \right] \hat{c}_{\mathbf{q}\sigma}^{\dagger} \hat{c}_{\mathbf{q}\sigma}
\end{aligned}$$

These two factors, together, define the full bands renormalisation

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u_{\sigma}^{(s^*)} V (\cos k_x + \cos k_y) + u_{\sigma}^{(d)} V (\cos k_x - \cos k_y) \tag{4.63}$$

as was for the normal phase in Eq. (3.22). In Sec. 3.3.1 we discussed appearance of anisotropic hopping with d -wave extension of bare bands: here theory predicts something identical. These RMPs are, as before, controlled linearly by V and in principle can lead to relevant changes in the kinetic sector of the hamiltonian.

Off-diagonal terms. Consider off-diagonal terms of Eq. (4.59). These contribute instead to full gap, being out of diagonal in the 2×2 hamiltonian matrix of Eq. (4.24). In pseudo-spin picture, they account for a transverse field. Define $\mathbf{q}, \mathbf{q}'$ as in Fig. 3.1b,

$$\mathbf{q} \equiv \mathbf{K} + \mathbf{k} \quad \mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$$

and rewrite as for Eq. (4.60)

$$\begin{aligned}
& -\frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma} \\
& = -\sum_{\gamma, \sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}\sigma}^\dagger \rangle \right] \times V \sum_{\mathbf{q}' \in \text{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} \hat{\psi}_{\mathbf{q}'\sigma}^\dagger
\end{aligned}$$

Now shift the first sum by an AF vector $\boldsymbol{\pi}$ defining $\mathbf{k} + \boldsymbol{\pi} \equiv \mathbf{q}$, and use Eq. (4.11) as well as reciprocal space periodicity by a BZ,

$$\begin{aligned}
\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}\sigma}^\dagger \rangle &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}+\boldsymbol{\pi}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{k}+\boldsymbol{\pi}\sigma}^\dagger \rangle \\
&= -\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{k}\sigma} \rangle
\end{aligned}$$

which finally gives

$$\begin{aligned}
& -\frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma} \\
& = \sum_{\gamma, \sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \right] \times V \sum_{\mathbf{q}' \in \text{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} \hat{\psi}_{\mathbf{q}'\sigma}^\dagger \quad (4.64)
\end{aligned}$$

Proceed as done previously: define the expectation value

$$v_{\mathbf{q}\sigma} \equiv i \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \quad (4.65)$$

(which is just the AF order parameter with a complex factor) and its γ -wave component, with $\gamma \in \{s^*, d, p_x, p_y\}$:

$$\begin{aligned}
v_{\sigma}^{(\gamma)} &= \frac{i}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \\
&= \frac{i}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \left[\varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle + \varphi_{\mathbf{q}+\boldsymbol{\pi}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}+\boldsymbol{\pi}\sigma} \rangle \right] \\
&= \frac{i}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \left[\langle \hat{\psi}_{\mathbf{q}\sigma} \rangle - \langle \hat{\psi}_{\mathbf{q}\sigma}^\dagger \rangle \right] \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\Psi}_{\mathbf{q}\sigma}^\dagger \tau^y \hat{\Psi}_{\mathbf{q}\sigma} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\text{Th}}_{\mathbf{q}\sigma}^{(-)}[\beta, \tilde{\mu}] \quad (4.66)
\end{aligned}$$

where we used Eq. (4.11), Tab. 4.2 and Eq. (4.48). There is no particular reason to exclude one component or another. In Tab. 4.4 the four derived self-consistency equations are reported, ignoring spin index.

From structure factors definitions of Tab. 2.5, we see that:

$$v_{\sigma}^{(s^*)}, v_{\sigma}^{(d)} \in \mathbb{R} \quad v_{\sigma}^{(p_x)}, v_{\sigma}^{(p_y)} \in i\mathbb{R}$$

Getting back to Eq. (4.64) and inserting these components, we get

$$\begin{aligned}
& -\frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma} \\
& = -iV \sum_{\gamma, \sigma} v_{\sigma}^{(\gamma)} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)*} \hat{\psi}_{\mathbf{q}\sigma}^\dagger \quad (4.67)
\end{aligned}$$

Symmetry	Self-consistency equation	Graph
s^* -wave	$v^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x + \cos k_y) \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. 2.4b
p_x -wave	$v^{(p_x)} = \frac{i\sqrt{2}}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \sin k_x \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. 2.4c
p_y -wave	$v^{(p_y)} = \frac{i\sqrt{2}}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \sin k_y \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. 2.4d
$d_{x^2-y^2}$ -wave	$v^{(d)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x - \cos k_y) \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. 2.4e

Table 4.4 | Symmetry and spin resolved self-consistency equations for the harmonics of $v_{\mathbf{k}}$.

We can restrict to MBZ as

$$\begin{aligned}
& -\frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}+\mathbf{k}+\pi\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}-\pi\sigma} \\
& = -iV \sum_{\gamma, \sigma} \sum_{\mathbf{q} \in \text{MBZ}} \left[v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \hat{\psi}_{\mathbf{q}\sigma}^{\dagger} + v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}+\pi}^{(\gamma)*} \hat{\psi}_{\mathbf{q}+\pi\sigma}^{\dagger} \right]
\end{aligned}$$

and use Eq. (4.11), exchanging terms positions

$$\begin{aligned}
& -\frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}+\mathbf{k}+\pi\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}-\pi\sigma} \\
& = iV \sum_{\gamma, \sigma} \sum_{\mathbf{q} \in \text{MBZ}} \left[v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \hat{\psi}_{\mathbf{q}\sigma} - v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \hat{\psi}_{\mathbf{q}+\pi\sigma} \right] \quad (4.68)
\end{aligned}$$

Now: recalling Tab. 2.5, for the four symmetries we are considering, form factors are real-valued for symmetries s^* and d , and purely imaginary for p_x and p_y . As mentioned before, the same holds for the $v_{\sigma}^{(\gamma)}$ harmonics. Then

$$i v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \in i\mathbb{R} \quad (4.69)$$

and to conjugate a purely imaginary number it suffices to put a minus sign in front of it,

$$-i v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} = \left(i v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \right)^*$$

Therefore, we conclude

$$\begin{aligned}
& -\frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}+\pi\sigma}^{\dagger} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}-\pi\sigma} \\
& = V \sum_{\gamma, \sigma} \sum_{\mathbf{q} \in \text{MBZ}} \left[i v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \hat{\psi}_{\mathbf{q}\sigma} + \text{h.c.} \right] \quad (4.70)
\end{aligned}$$

which is no surprise, since hamiltonian are hermitian operators.

4.5.2 Fock contraction energy

Let us now get back to $E_{\text{F}/V}^{(\text{AF})}$, the double counting energy of Eq. (4.58). As said before, its definition is the same as for the normal phase, Eq. (3.15), with averaging operation being performed on the AF ground state:

$$E_{\text{F}/V}^{(\text{AF})} \equiv \frac{V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} \sum_{\sigma} [\cos(\delta k_x) + \cos(\delta k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \langle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \rangle \quad (4.71)$$

Proceeding exactly as in Eq. (4.59) and making changing variables as in Fig. 3.1, we get

$$E_{F/V}^{(\text{AF})} = \frac{V}{L_x L_y} \sum_{\mathbf{q}, \mathbf{q}'} \sum_{\sigma} [\cos(\delta q_x) + \cos(\delta q_y)] \left[\underbrace{\langle \hat{n}_{\mathbf{q}\sigma} \rangle \langle \hat{n}_{\mathbf{q}'\sigma} \rangle}_{\text{Diagonal terms}} - \underbrace{\langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \langle \hat{\psi}_{\mathbf{q}'\sigma} \rangle}_{\text{Off-diagonal terms}} \right]$$

Apply the decomposition of Eq. (2.43) and use Eqns. (4.61), (4.65),

$$E_{F/V}^{(\text{AF})} = \frac{V}{2L_x L_y} \sum_{\gamma, \sigma} \sum_{\mathbf{q}, \mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} [u_{\mathbf{q}\sigma} u_{\mathbf{q}'\sigma} + v_{\mathbf{q}\sigma} v_{\mathbf{q}'\sigma}] \quad (4.72)$$

Now: it holds

$$u_{\mathbf{q}\sigma} = \langle \hat{n}_{\mathbf{q}\sigma} \rangle \implies u_{\mathbf{q}\sigma} = u_{\mathbf{q}\sigma}^*$$

and then

$$\begin{aligned} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{q}, \mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} u_{\mathbf{q}\sigma} u_{\mathbf{q}'\sigma} &= \left[\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} u_{\mathbf{q}\sigma} \right] \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}' \in \text{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} u_{\mathbf{q}'\sigma}^* \right] \\ &= |u_{\sigma}^{(\gamma)}|^2 \end{aligned}$$

At the same time, being $\hat{\psi}_{\mathbf{q}+\pi\sigma} = \hat{\psi}_{\mathbf{q}\sigma}^\dagger$,

$$v_{\mathbf{q}\sigma} = i \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \implies v_{\mathbf{q}+\pi\sigma} = -v_{\mathbf{q}\sigma}^*$$

Since the form factors satisfy $\varphi_{\mathbf{q}+\pi}^{(\gamma)} = -\varphi_{\mathbf{q}}^{(\gamma)}$, we get:

$$\begin{aligned} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{q}, \mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} v_{\mathbf{q}\sigma} v_{\mathbf{q}'\sigma} &= \left[\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} v_{\mathbf{q}\sigma} \right] \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}' \in \text{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} v_{\mathbf{q}'\sigma} \right] \\ &= v_{\sigma}^{(\gamma)} \left[\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}+\pi}^{(\gamma)*} v_{\mathbf{k}+\pi\sigma} \right] \\ &= v_{\sigma}^{(\gamma)} \left[\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)*} v_{\mathbf{k}\sigma}^* \right] \\ &= |v_{\sigma}^{(\gamma)}|^2 \end{aligned}$$

where in the second passage we employed the change of variables $\mathbf{q}' = \mathbf{k} + \pi$ and used BZ periodicity to map the original sum onto a new sum for $\mathbf{k} \in \text{BZ}$. Inserting above results in Eq. (4.72), the latter becomes:

$$E_{F/V}^{(\text{AF})} = L_x L_y \times \frac{V}{2} \sum_{\gamma, \sigma} \left[|u_{\sigma}^{(\gamma)}|^2 + |v_{\sigma}^{(\gamma)}|^2 \right] \quad (4.73)$$

The new HFPs amplitudes contribute positively to contraction energy. The consequences of this fact will be discussed in Sec. 4.7.1.

4.6 Full MFT hamiltonian and gap complexification

Finally, it is time to collect everything in a single hamiltonian operator. The approximated hamiltonian is:

$$\begin{aligned} \hat{H} - \mu \hat{N} &\simeq \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \tilde{\epsilon}_{\mathbf{k}\sigma} [\hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\pi\sigma}] - \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \left[\tilde{\Delta}_{\mathbf{k}\sigma} \hat{\psi}_{\mathbf{k}\sigma} + \tilde{\Delta}_{\mathbf{k}\sigma}^* \hat{\psi}_{\mathbf{k}\sigma}^\dagger \right] \\ &\quad - \tilde{\mu} \hat{N} - \left[E_{H/U}^{(\text{AF})} + E_{H/V}^{(\text{AF})} + E_{F/V}^{(\text{AF})} \right] \end{aligned} \quad (4.74)$$

where renormalized bands are given by:

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma} + V \sum_{\gamma \in \{s^*, d\}} u_{\sigma}^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*}$$

and full gap function by

$$\tilde{\Delta}_{\mathbf{k}\sigma} = \Delta_{\mathbf{k}\sigma} - iV \sum_{\gamma} v_{\sigma}^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} \quad (4.75)$$

The gap correction is purely imaginary, compared to HM gap $\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$. Since there is no other contribution to the x pseudo-spin projections other than the pure HM ones, spin antisymmetry is preserved in this direction

$$\langle \hat{s}_{\mathbf{k}\sigma}^x \rangle = -\langle \hat{s}_{\mathbf{k}\bar{\sigma}}^x \rangle$$

This property confirms that Eq. (4.50) actually holds. Indeed, we anticipated that the self-consistency equation of the magnetisation should have been identical for the HM and the EHM. The fact that the x component EV of the pseudospin maintains this antisymmetry relation is a confirmation of that, since the proof that lead us to Eq. (4.39) for the HM was based on this property.

One of the net effects of $\hat{H}_V$ on the AF phase, with respect to the conventional HM, is “gap complexification”. This is expressed by:

$$|\operatorname{Im}\{\Delta_{\mathbf{k}\sigma}\}| \geq 0 \quad \implies \quad |\langle \hat{s}_{\mathbf{k}\sigma}^y \rangle| \geq 0$$

while for HM the y pseudo-spin component was identically null.

4.6.1 The relative phase in the zero temperature half-filled ground state

In Sec. A.2, we derive analytically the AF ground state on conventional HM at zero temperature, as a simple Bloch’s transform of a polarized (tilted) pseudo-spins collective pure state. Eq. (A.5) reports the final form of said ground-state. Said quantum state is in general pretty similar to conventional BCS state.

At half-filling and zero temperature we can extend that state to the new complexified gap. The HF approximation of the true ground-state is, in the AF sector,

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k} \in \text{MBZ}} \left[\sin \tilde{\theta}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}^{\dagger} + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}} \cos \tilde{\theta}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\pi\sigma}^{\dagger} \right] |\Omega_{\mathbf{k}\sigma}\rangle \quad (4.76)$$

The proof is identical to the one provided in Sec. A.2 for the HM, apart from renormalisations. Let $|\mathbf{k}\sigma\rangle \equiv \hat{c}_{\mathbf{k}\sigma}^{\dagger} |\Omega_{\mathbf{k}\sigma}\rangle$. Then we can write

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k} \in \text{MBZ}} \left[\sin \tilde{\theta}_{\mathbf{k}\sigma} + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}} \cos \tilde{\theta}_{\mathbf{k}\sigma} \hat{\psi}_{\mathbf{k}\sigma} \right] |\mathbf{k}\sigma\rangle$$

having we used $|\Omega_{\mathbf{k}\sigma}\rangle = \{\hat{c}_{\mathbf{k}\sigma}, \hat{c}_{\mathbf{k}\sigma}^{\dagger}\} |\Omega_{\mathbf{k}\sigma}\rangle$.

The angles appearing in above equations are the same of Eqns. (4.43), (4.44). The relative phase $2\tilde{\zeta}_{\mathbf{k}\sigma}$ is the gap function angle in complex plane, since

$$\begin{aligned} \langle \Psi | \hat{s}_{\mathbf{k}\sigma}^x | \Psi \rangle &= \langle \Psi | \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^{\dagger} | \Psi \rangle \\ &= \sin \tilde{\theta}_{\mathbf{k}\sigma} \cos \tilde{\theta}_{\mathbf{k}\sigma} \left(e^{-2i\tilde{\zeta}_{\mathbf{k}\sigma}} + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}} \right) \\ &= \sin 2\tilde{\theta}_{\mathbf{k}\sigma} \cos 2\tilde{\zeta}_{\mathbf{k}\sigma} \end{aligned}$$

$$\begin{aligned} \langle \Psi | \hat{s}_{\mathbf{k}\sigma}^y | \Psi \rangle &= i \langle \Psi | \hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}_{\mathbf{k}\sigma}^{\dagger} | \Psi \rangle \\ &= i \sin \tilde{\theta}_{\mathbf{k}\sigma} \cos \tilde{\theta}_{\mathbf{k}\sigma} \left(e^{-2i\tilde{\zeta}_{\mathbf{k}\sigma}} - e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}} \right) \\ &= \sin 2\tilde{\theta}_{\mathbf{k}\sigma} \sin 2\tilde{\zeta}_{\mathbf{k}\sigma} \end{aligned}$$

These equations are coherent with Eqns. (4.47), (4.48), thus the involved angles (apart from irrelevant 2π shifts) coincide.

Phase	HFPs
sAF	$\{m, u^{(s*)}, u^{(d)}, v^{(s*)}, v^{(d)}\}$
aAF	$\{m, u^{(s*)}, u^{(d)}, -iv^{(p_x)}, -iv^{(p_y)}\}$

Table 4.5 | Set of HFP for the two possible realization of the AF phase, respectively a symmetric and an antisymmetric imaginary part of the gap.

4.6.2 AF phase flavours and spin degeneracy

In this analysis, antiferromagnetism is realised without breaking inversion symmetry. This translates in the requirement that:

$$\tilde{E}_{\mathbf{k}\sigma} \stackrel{!}{=} \tilde{E}_{-\mathbf{k}\sigma}$$

We know that:

$$\tilde{E}_{\mathbf{k}\sigma} = \sqrt{\tilde{\epsilon}_{\mathbf{k}\sigma}^2 + \text{Re}^2\{\tilde{\Delta}_{\mathbf{k}\sigma}\} + \text{Im}^2\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}$$

Renormalized bare bands as well as the real part of the gap satisfy said inversion symmetry requirement,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \tilde{\epsilon}_{-\mathbf{k}\sigma} \quad \text{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \text{Re}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\}$$

hence inversion symmetry is respected only if

$$\text{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \pm \text{Im}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\}$$

which is: either the imaginary part of the gap is inversion symmetric, or is antisymmetric. Based on this consideration, we are able to discard two of the four self-consistency equations for the v -related HFPs listed in Tab. 4.4: s^* and d -wave symmetries HFPs are nonzero if $\text{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is momentum-even; p_x and p_y -wave symmetries HFPs are nonzero if $\text{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is momentum-odd. The two options cannot coexist, and separate simulations can be implemented in the two symmetry sectors.

We name the two possible symmetries “sAF” (symmetric antiferromagnet) and “aAF” (antisymmetric antiferromagnet). The set of HFPs then restricts to these AF structures, and is reported for both cases in Tab. 4.5. The presence itself of two “AF flavours” is not-so-physical, given that up to now we have no clear physical interpretation for $\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}$. As we will discuss in Sec. ??, it is possible to prove that almost all these new HFPs actually vanish – a fact that is not directly inferrable right now, but can be seen only delving into the discussion of the non-linear map that defines the algorithm. For now, let us proceed integrally with all the HFPs we found and derive the results.

From now on we will omit σ index, working just in the $\uparrow$ sector. As said in Sec. 4.1, the AF phase does not break $\mathbb{Z}_2$ spin inversion symmetry, being the SDW ordering generated by a staggered but globally symmetric spin density distribution among sites. The approximated quadratic hamiltonian of Eq. 4.74 is invariant by a $\sigma \leftrightarrow \bar{\sigma}$ exchange, thus all measurable physical quantities must be.

4.7 AF free energy density

In order to derive the free energy density expression for AF phase, we proceed precisely as in Sec. ??. The only notable difference is in the expression of entropy density: Eq. (??) is written for two opposite bands on MBZ

$$s = s^{(+)} + s^{(-)}$$

with

$$s^{(\pm)} \equiv -\frac{k_B}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \left[\left(1 - f(\pm \tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu})\right) \ln \left(1 - f(\pm \tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu})\right) + f(\pm \tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \ln f(\pm \tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] \quad (4.77)$$

Consider also that we are computing free energy on the gran-canonical hamiltonian

$$\hat{H} - \mu\hat{N} \simeq \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \left[(\tilde{E}_{\mathbf{k}} - \tilde{\mu}) \hat{\gamma}_{\mathbf{k}\sigma}^{(+)\dagger} \hat{\gamma}_{\mathbf{k}\sigma}^{(+)} - (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) \hat{\gamma}_{\mathbf{k}\sigma}^{(-)\dagger} \hat{\gamma}_{\mathbf{k}\sigma}^{(-)} \right] + E_c^{(\text{AF})}$$

being $E_c^{(\text{AF})}$ the total AF contraction energy. Note that on the left-hand side we have μ and on the right-hand side its RMP counterpart $\tilde{\mu}$.

With identical meaning of the notation with respect to Sec. ??, the free energy contributions are:

1. Quasiparticles energy, taking care of spin degeneracy:

$$e_g^{(\text{AF})} = \frac{2}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\tilde{E}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) - \frac{2}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) f(-\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu})$$

2. Contraction energy:

$$e_c^{(\text{AF})} = U(m^2 - n^2) + V \left[2zn^2 - \sum_{\gamma} \left(|u^{(\gamma)}|^2 + |v^{(\gamma)}|^2 \right) \right]$$

3. Entropic energy

$$\begin{aligned} e_s^{(\text{AF})} = & \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \ln \left(1 - f(\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \right) - \frac{2}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\tilde{E}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \\ & + \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \ln \left(1 - f(-\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \right) + \frac{2}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) f(-\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \end{aligned}$$

4. Chemical potential correction

$$e_n^{(\text{AF})} = \mu n$$

Compose everything

$$f^{(\text{AF})} \equiv e_g^{(\text{AF})} + e_c^{(\text{AF})} + e_s^{(\text{AF})} + e_n^{(\text{AF})}$$

and we get

$$\begin{aligned} f^{(\text{AF})} = & \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \left[\ln \left(1 - f(\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \right) + \ln \left(1 - f(-\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \right) \right] \\ & + U m^2 - V \sum_{\gamma} \left[|u^{(\gamma)}|^2 + |v^{(\gamma)}|^2 \right] + n [\mu + n(2zV - U)] \end{aligned}$$

As was for the normal phase, contraction energy and chemical potential energy corrections build up the RMP $\tilde{\mu}$ of Eq. (4.56),

$$\begin{aligned} f^{(\text{AF})} = & \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \left[\ln \left(1 - f(\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \right) + \ln \left(1 - f(-\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \right) \right] \\ & + U m^2 - V \sum_{\gamma} \left[|u^{(\gamma)}|^2 + |v^{(\gamma)}|^2 \right] + \tilde{\mu} n \quad (4.78) \end{aligned}$$

In this expression, we see that $u^{(\gamma)}$ s, $v^{(\gamma)}$ s contribution is negative, while magnetisation contributes positively. At a first glance, this suggests that a system with no magnetisation and non-zero $u^{(\gamma)}$ s, $v^{(\gamma)}$ s would be energetically favoured. In that situation we would have a purely imaginary gap for a non-magnetised phase, which would be not so intuitive to explain physically. In next section we explain how the above expression for the free energy density can be used to give a constraint that proves that said situation is impossible. HFPs are interrelated and the physical solution cannot admit arbitrary values.

4.7.1 Energetic constraints on the HFPs

Let us derive a constraint from the free energy density expression. Collecting the terms directly related to the gap function, the following identity holds:

$$Um^2 - V \sum_{\gamma} |v^{(\gamma)}|^2 = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{|\tilde{\Delta}_{\mathbf{k}}|^2}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] \quad (4.79)$$

The proof is simple and immediate: expanding right-hand side,

$$\begin{aligned} \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{|\tilde{\Delta}_{\mathbf{k}}|^2}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\text{Re}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] \\ &\quad + \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\text{Im}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] \end{aligned}$$

Now:

$$\text{Re}\{\tilde{\Delta}_{\mathbf{k}}\} = Um$$

which implies

$$\begin{aligned} \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\text{Re}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] &= Um \times \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\text{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] \\ &= Um^2 \end{aligned}$$

having we recognised the expression for the magnetisation from Eq. (4.50). Equivalently,

$$\begin{aligned} \text{Im}\{\tilde{\Delta}_{\mathbf{k}}\} &= -V \sum_{\gamma} v^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} \\ &= -V \sum_{\gamma} v^{(\gamma)*} \varphi_{\mathbf{k}}^{(\gamma)} \end{aligned}$$

where we used Eq. (4.69) to exchange the conjugation operation in the last passage. This implies:

$$\begin{aligned} \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\text{Im}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] &= -V \sum_{\gamma} v^{(\gamma)*} \sum_{\mathbf{k} \in \text{MBZ}} \varphi_{\mathbf{k}}^{(\gamma)} \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] \\ &= -V \sum_{\gamma} v^{(\gamma)*} \times v^{(\gamma)} \\ &= -V \sum_{\gamma} |v^{(\gamma)*}|^2 \end{aligned}$$

where in second passage we recognised $v^{(\gamma)}$ given in Eq. (4.66). Then Eq. (4.79) is verified.

A consequence of Eq. (4.79) is that, being the right-hand side positive, for $V > 0$ the total amplitude of the “complexifiers” vector is maximised,

$$\sum_{\gamma} |v^{(\gamma)}|^2 \leq M \quad \text{being} \quad M^{(\text{AF})} \equiv m^2 \frac{U}{V} \quad (4.80)$$

The above constraint predicts that:

- In the strong coupling limit $U \ll V$, gap complexification is suppressed;
- If magnetisation is null, so is the imaginary part of the gap

$$\lim_{m \rightarrow 0} v^{(\gamma)} = 0$$

which is coherent with the entire derivation and takes us back to the normal phase.

Eq. (4.79) is also solved by a null magnetisation and thus null complexifiers. This can be interpreted by the fact that when the constraint is impossible to solve self-consistently, the gap must close.

Eq. (4.80) provides a neat graphic interpretation for the constraint on HFPs in both sAF and aAF phases. At any competition ratio U/V , the product $m\sqrt{U/V}$ is the radius of a sphere. Inside this sphere must lie the vector defined by remaining $v^{(\gamma)}$ HFPs. For our AF flavours, the sphere is actually a circle, being in both cases the total number of HFPs equal to 3, counting also m . Note that the solution:

$$\forall \gamma \quad : \quad v^{(\gamma)} = 0$$

solves the constraint of Eq. (4.80). These considerations do not imply gap complexification is an actual, observed effect.

4.7.2 Free energy at half-filling

Let us discuss the half-filling condition. We start with the identity:

$$\begin{aligned} \ln\left(1 - \frac{1}{e^x + 1}\right) + \ln\left(1 - \frac{1}{e^{-x} + 1}\right) &= \ln\left[\frac{e^x}{(e^x + 1)^2}\right] \\ &= x - 2\ln(1 + e^x) \end{aligned}$$

Let now $x = \beta\tilde{E}_{\mathbf{k}}$. At low temperature for a band with no nodes, we can approximate $x \rightarrow +\infty$ on the entire BZ. This gives

$$x - 2\ln(1 + e^x) \simeq -x$$

Hence we get the half-filling free energy density substituting ,

$$f^{(\text{AF})} \simeq -\frac{2}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \tilde{E}_{\mathbf{k}} + Um^2 - V \sum_{\gamma} \left[|u^{(\gamma)}|^2 + |v^{(\gamma)}|^2 \right]$$

The first term in this sum gives the energy of the fully occupied lower band.

This approximation makes sense when $\tilde{E}_{\mathbf{k}} \neq 0$ everywhere. This is always realised in AF phase: from the constraint of Eq. (4.80) we know magnetisation is necessary for all gap HFPs; for a magnetised system, whatever the symmetries,

$$\tilde{E}_{\mathbf{k}} = \sqrt{\tilde{c}_{\mathbf{k}}^2 + (Um)^2 + |\text{Im}\{\Delta_{\mathbf{k}}\}|^2} \geq Um$$

At half filling and low temperature, the system is well approximated by a completely filled electronic band $-\tilde{E}_{\mathbf{k}}$. This gives the aforementioned approximation of the free energy $f^{(\text{AF})}$.

4.8 Discussion of numeric results

In this section we present numeric results. For the AF phase, the HFPs vector, as we thoroughly showed, has two components

$$\mathbf{v} = \begin{bmatrix} m \\ u^{(s^*)} \end{bmatrix} \quad \mathbf{v} \in \mathbb{R}^2$$

The resulting model RMPs are, as before:

$$\begin{aligned} \tilde{t}_{ij} &\equiv \left(t - \frac{u^{(s^*)}V}{2} \right) \varphi_{ij}^{(s^*)} \\ \tilde{\mu} &\equiv \mu + n(2zV - U) \end{aligned}$$

In the end, the computational scenario is not so different from the normal phase. The only real difference is our choice of parametric boundaries for simulations:

$$U = 0, 0.05, 0.1, \dots, 8.0 \quad V = 0, 0.05, 0.1, \dots, 8.0 \quad \delta = 0.0$$

We set $\delta = 0.0$. As we anticipated in Sec. 4.2.4 (and discuss in Sec. A.3), any finite doping disrupts commensurate antiferromagnetism, a property that makes any simulation at $\delta > 0$ of low physical significance. Following this consideration, we omit data at finite $\delta \neq 0$.

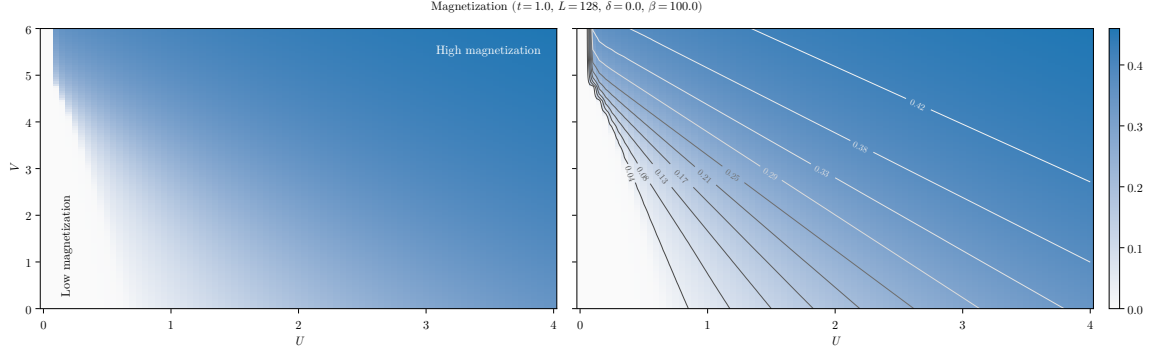
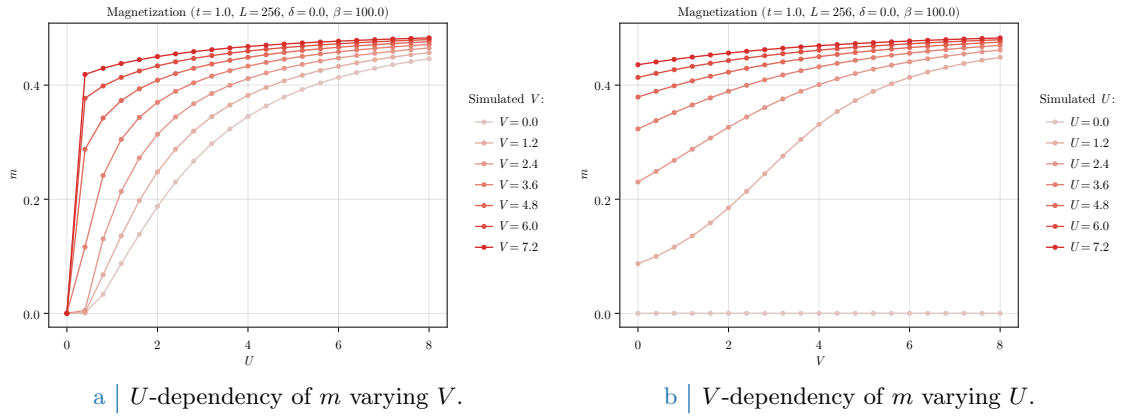


Figure 4.5 | Plot of the self-consistent convergence values for m (left panel) and isomagnetic curves computed on the same data (right panel), along which magnetisation is constant. For the sake of graphic clarity, only the points at $0 \leq U \leq 4$ and $0 \leq V \leq 6$ are plotted. [Something broke at $U = 0.05$, $V \geq 5$, change with new data.]



a | U -dependency of m varying V .

b | V -dependency of m varying U .

Figure 4.6 | Plot of the self-consistent convergence values for the HFPs m and $u^{(s*)}$ in the AF phase.

4.8.1 Magnetization boost

The convergence values for m computed in (U, V) plane are shown in the panels of Fig. 4.5, along with a level curves plot to identify isomagnetic curves (right panel). Note that, to increase graphic clarity in heatmaps bottom left sector, from the plots are omitted the points at $U \geq 4$ and $V \geq 6$. Tolerances on density and HFPs have been set to

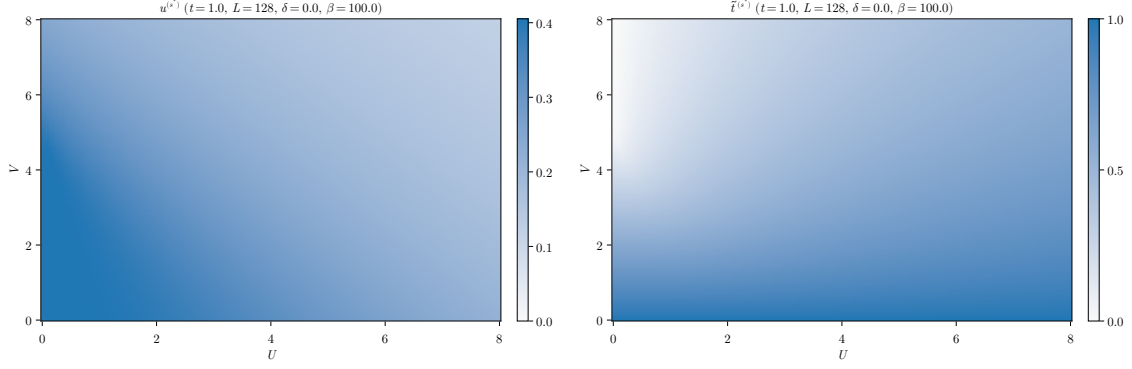
$$\Delta n = 10^{-2} \quad \Delta \mathbf{v} = \begin{bmatrix} m: 5 \times 10^{-4} \\ u^{(s*)}: 5 \times 10^{-4} \end{bmatrix}$$

From heatmaps we see that $\hat{H}_V$ actually boosts magnetisation. Since the only observable net effect of the non-local interaction is to lower t by shrinking bands, at any fixed point (t, U, V) the model can be effectively mapped to

$$\text{EHM: } (t, U, V) \rightarrow \text{HM: } (\tilde{t}, U, 0)$$

being $\tilde{t}$ dependent on V . For the pure HM, magnetisation is controlled by the ratio U/t . To lower t is physically the same as to increase U , and from this comes the magnetisation precession visible in Fig. 4.5. Isomagnetic lines on the right panel show, for U significantly greater than 0, linear trends whose angular coefficient depend on m itself. The low- U limit of isomagnetic lines presents some oscillations that are imputable to numeric discretisation of parametric space. [Why isomagnetic curves are lines? Something about $U/(V - V_c)$ ratio being constant?]

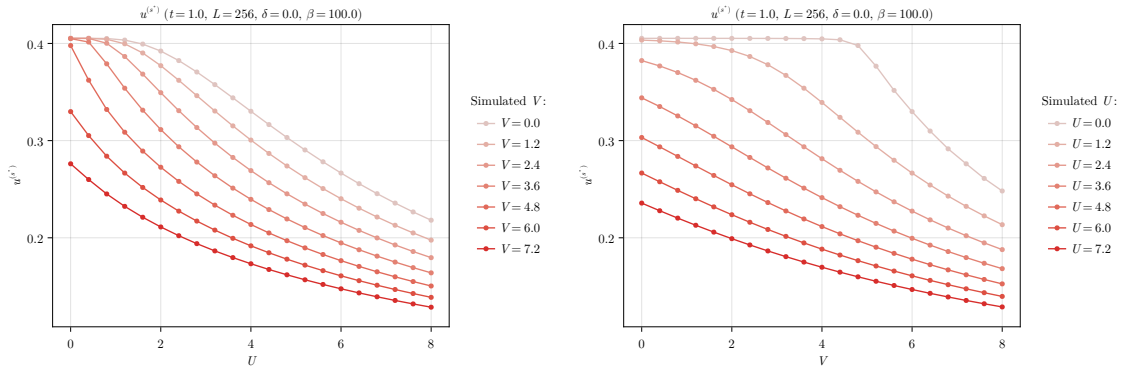
Data shown in Fig. 4.6 confirm clearly this V -controlled magnetisation “boost”. The effect is more enhanced at low values of U . As is visible from Fig. 4.6b, magnetisation does not always



a | Heatmap of $u^{(s*)}$ in UV plane.

b | Heatmap of $t^{(s*)}$ in UV plane.

Figure 4.7 | Data for the HFP $u^{(s*)}$ and $\tilde{t}^{(s*)}$ varying both U and V .



a | U -dependency of $u^{(s*)}$ varying V .

b | V -dependency of $u^{(s*)}$ varying U .

Figure 4.8 | Plot of the self-consistent convergence values for the HFPs m and $u^{(s*)}$ in the AF phase.

saturate the same way. For example, considering the curve at $U = 1.2$, magnetisation varies following a somewhat tilted sigmoid shape, making boost most effective around $V \approx 2 \div 4$. Because of m saturation, this behaviour is lost at higher U s.

4.8.2 Bands shrinking

The bands shrinking effect was discussed in Sec. ?? within C_4 symmetry, and then confirmed in Sec. ?? when trying to extend to nematic structures. As was for the normal phase, we have proven that $\hat{H}_V$ net effect on the hamiltonian kinetic part is to shrink bands,

$$t \rightarrow \tilde{t} < t$$

For the normal phase, in Sec. 3.5.1 we presented numeric results showing a Mott-localized region appears in (δ, V) plane, whose $\delta \rightarrow 0$ limit was approximately $V = 4.93t$, as shown in Fig. 3.5b.

The convergence values for the HFP $u^{(s*)}$ are shown in Fig. 4.7a, and more in detail in the two panels of Fig. 4.8. In general, $u^{(s*)}$ decreases monotonically both in U and in V . As we already discussed in normal phase, the $U = 0$ data of Fig. 4.8b present a plateau up to a Mott-critical value. This effect is absent at positive U s. Both interactions disrupt the HFP; however, the two limits

$$U \gg 1 \quad \text{and} \quad V \gg 1$$

are much different. By computing the value of $\tilde{t}^{(s*)}$,

$$\tilde{t}^{(s*)} = 2t - u^{(s*)}V$$

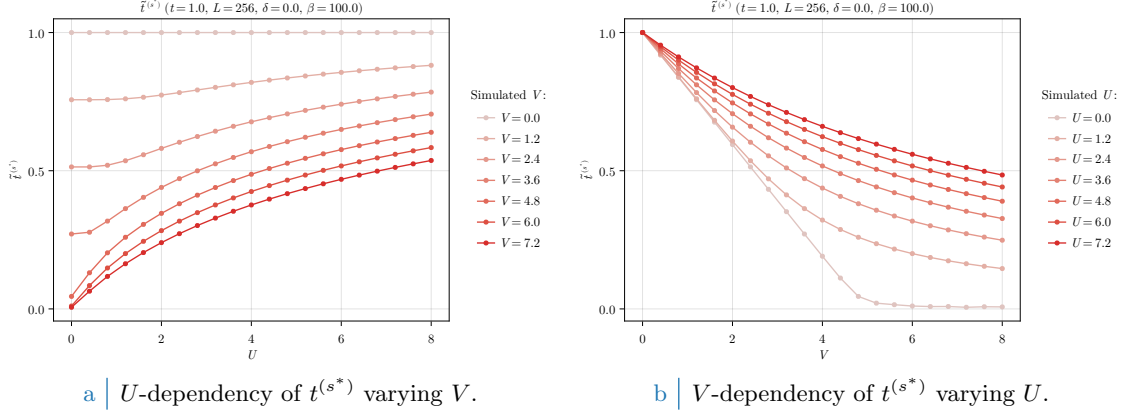


Figure 4.9 | Plot of the self-consistent convergence values for the HFPs m and $u^{(s^*)}$ in the AF phase.

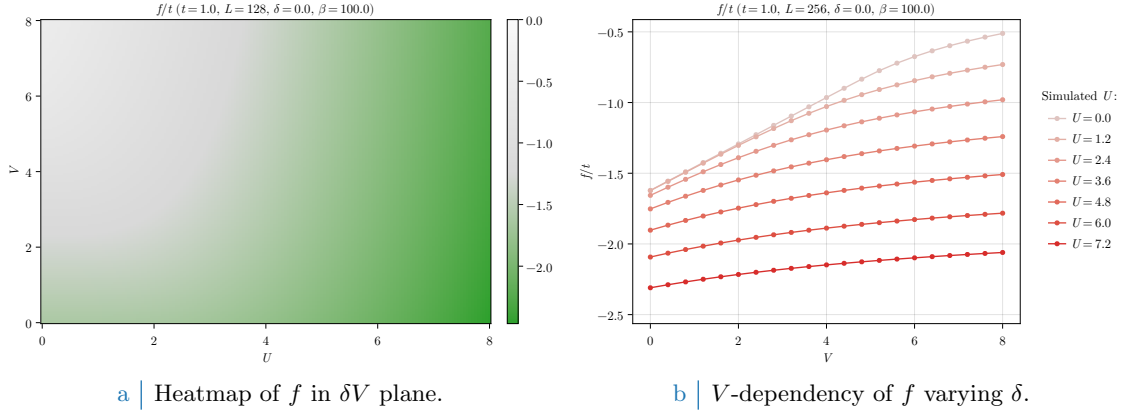


Figure 4.10 | Free energy density of the AF phase.

we obtain the data of Fig. 4.7b. The limit $U \gg 1$ taken at fixed V , also shown in Fig. 4.9a, provides “ U -induced antiferromagnetism”, with bands shrinking effect being almost irrelevant as V lowers; on the contrary, the opposite limit $V \gg 1$ taken at fixed U , also shown in Fig. 4.9b, shrinks bands strongly providing a “Mott-induced antiferromagnetism”.

In (U, V) space there is not Mott-localised region of the kind we had in (δ, V) space for the normal phase (see Fig. 3.5b). At half filling, complete localisation only occurs at $U = 0$, as can be seen in the bottom left sector of Fig. 4.9a. Any finite value of U disrupts bands flattening, thus – being the local repulsion an essential ingredient in Hubbard antiferromagnetism – we can claim, based on numeric evidence, that bands flattening is incompatible with a commensurate AF phase on the half-filled EHM.

4.8.3 Solution free energy and entropy densities

In Fig. 4.10 the solution free energy is plotted. In the right panel, Fig. 4.10b, we see that antiferromagnetism becomes more stable at low V , high U . The opposite limit of weak local repulsion, strong non-local attraction (top-left corner of Fig. 4.10a) instead increases f , reaching zero energy at $U = 0$, $V > 4.93t$. The latter is the critical value discussed in Sec. 3.5.1 and computed in Eq. (3.45).

It is then evident how, even if the macroscopic magnetic properties are the same along one isomagnetic line of those plotted in Fig. 4.5, right panel, and thus the realised state exhibits an identical value of m , much different is the energetic content of the state itself. While in the $U \gg 1$, fixed V limit free energy gain comes from gap amplification – which moves down the Bogoliubov bands $\tilde{E}_{\mathbf{k}}$ –, in the $V \gg 1$, fixed U limit energies of the lower Bogoliubov band (full at low temperature) are shifted upwards via the Mott localisation effect.

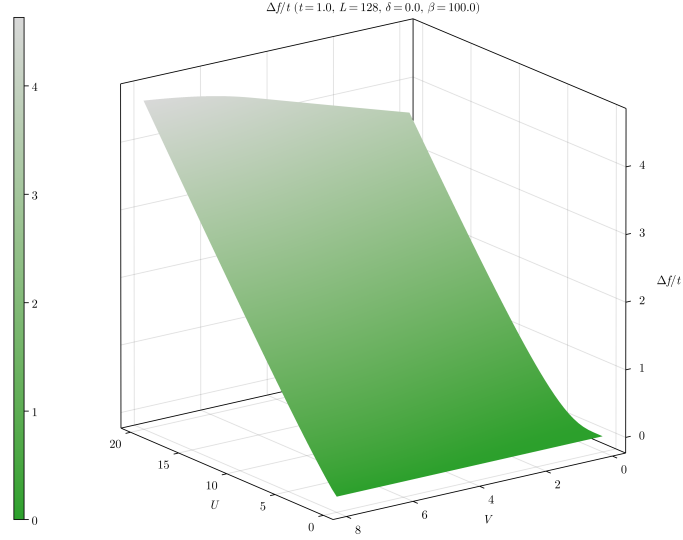


Figure 4.11 | Difference of free energies for the normal phase minus the AF phase.

The AF phase is always energetically favoured at half-filling, with respect to the normal phase. Let us define the energy difference,

$$\Delta f \equiv f^{(N)} - f^{(AF)}$$

shown for a slightly more coarse-grained dataset in Fig. 4.11, we see immediately how $\Delta f \geq 0$ in all (U, V) plane. The normal phase data of Sec. 3.5 having been simulated at $U = 0$ due to their intrinsic U independence. The AF phase is simulated in the same range $0 \leq V \leq 8$, but for $0 \leq U \leq 10$. The comparable normal phase dataset was just a repetition of the $U = 0$ data at all U s. As we expect, in the $V \gg 1$ and low U limit the energy difference is null: in this limit the two phases tend to coincide, since at $U = 0.0$ no magnetisation is physically possible. Moreover, the $U \gg 1$ limit makes the AF phase energetically favoured. Overall, at half filling the interesting free fermions Mott-localisation effect is, at least when comparing these phases, existent but surpassed by the high energetic gain of establishing an AF structure at infinitesimal U . The situation could change at finite doping, a regime we do not investigate limitedly to AF Physics.

At low temperature and half filling, it is not worth plotting entropy density. Half-filled commensurate antiferromagnets are insulators, with two gapped Bogoliubov bands. At low temperature, electronic distribution fills completely the lower band and empties the upper one. Because of this, insulating gapped systems inherently are low entropic.

4.8.4 Fixed-point interpretation of HF solution

As a final remark discussing the mathematical structure of the algorithm, we anticipated in Sec. 1.3.2 how the reject-mix-iterate scheme upon which our HF algorithm is built can be seen as an effective search for the singular fixed point of a nonlinear map. A physical solution is represented as a HFP vector $\mathbf{v}$ such that the algorithm is a local identity,

$$A(\mathbf{v}) = \mathbf{v}$$

To predict for a given initializer $\mathbf{v}_0$ the path it will follow through HFPs space is, in general, highly non-trivial due to the intrinsic nonlinearity of the algorithmic map defined by self-consistency equations for the various HFPs. An example of such nonlinearity is given in the derivation of Secs. ??, ??.

In Fig. 4.12 we sketch this concept in a practical situation. In figure are plotted various algorithmic paths through AF HFPs space, with operational setup at

$$U = 3 \quad V = 6 \quad L = 128 \quad \beta = 100 \quad \delta = 0$$

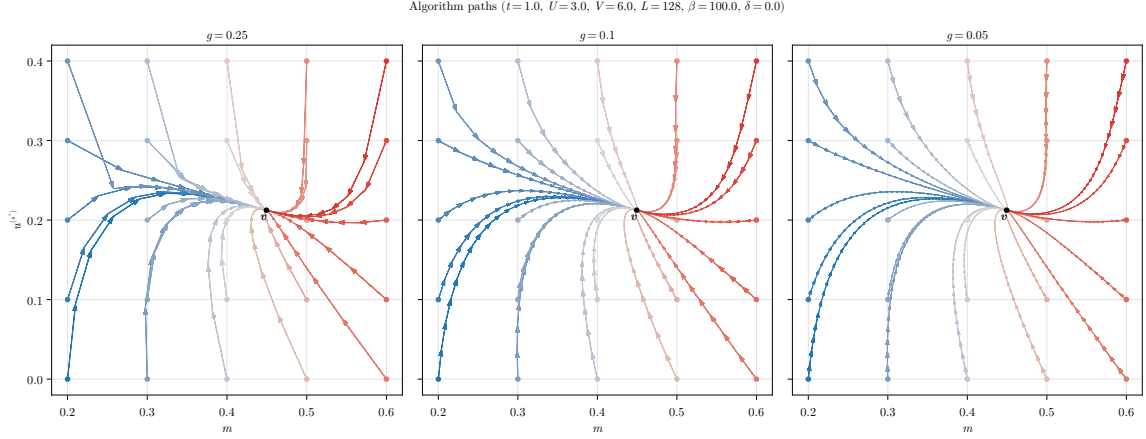


Figure 4.12 | Quiver plot of algorithmic evolution in HFPs space at fixed model parametrization. The black dot $\mathbf{v}$ represents the nonlinear map fixed point, while the colored paths the sequences of HFPs coordinates simulated via iterative HF procedure, starting from different HFPs initializers (colored dots).

At this point, the self-consistent HF “fixed-point” solution is given by

$$m \approx 0.449 \quad u^{(s^*)} \approx 0.212$$

indicated in panels as the black dot. The three panels show different map evolutions varying the mixing parameter within three choices, $g \in \{0.25, 0.1, 0.05\}$. We chose various initializers in the set:

$$\mathbf{v}_0 = \begin{bmatrix} m_0 \\ u_0^{(s^*)} \end{bmatrix} : m_0 \in \{0.2, 0.3, \dots, 0.6\} \quad \text{and} \quad u_0^{(s^*)} \in \{0.0, 0.1, \dots, 0.4\}$$

In figures, different colours indicate different evolutions, while the tip size indicates the magnitude of the algorithmic step in passing from one iteration to the following.

All paths are able to converge to $\mathbf{v}$, and the quantity of steps necessary to reach the fixed point, a sort of attractor, increases as we lower g . To select the correct g turns out to be the real algorithm fine-tuning knob: as we saw for the normal phase, some model parametrisation could lead to unavoidable algorithmic instabilities. The problem is that we do not know beforehand where the instabilities will occur, and how. The correct g is the compromise between keeping low the number of steps (larger mixing) and moving slow enough to not diverge when encountering unstable points (smaller mixing). By recursively fine-tuning g repeating non-converged simulation with more care, we are essentially making the nonlinear map “softer” and curvier, as is in the right panel of Fig. 4.12. It’s not always resource convenient to do so, so for most simulations we stick to a rougher map (like the left panel one) by setting $g = 0.5$.

Appendix A

Antiferromagnetism in the HM

In this appendix we give some details about the antiferromagnetic phase in the conventional Hubbard model,

$$\hat{H} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \left[\hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} + \text{h.c.} \right] + U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \quad t, U > 0 \quad (\text{A.1})$$

approximated by HF methods. We use the results discussed in Sec. 4.2.3.

A.1 The AF ground-state for the conventional HM

Let the Nambu spinors:

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\pi\sigma} \end{bmatrix}$$

Let also the gap function:

$$\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$$

We know from Eq. (4.29) that the approximated hamiltonian is:

$$\hat{H}_t + \hat{H}_U \simeq \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \mathbf{b}_{\mathbf{k}\sigma} \cdot \hat{\mathbf{s}}_{\mathbf{k}\sigma} + nU \hat{N} - E_{\text{H}/U}^{(\text{AF})}$$

where:

$$\mathbf{b}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} -\Delta_{\mathbf{k}\sigma} \\ 0 \\ \epsilon_{\mathbf{k}\sigma} \end{bmatrix} \quad \hat{s}_{\mathbf{k}\sigma}^{\alpha} \equiv \hat{\Psi}_{\mathbf{k}\sigma}^{\dagger} \tau^{\alpha} \hat{\Psi}_{\mathbf{k}\sigma} \quad \hat{\mathbf{s}}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{s}_{\mathbf{k}\sigma}^x \\ \hat{s}_{\mathbf{k}\sigma}^y \\ \hat{s}_{\mathbf{k}\sigma}^z \end{bmatrix}$$

Through the unitary matrix $W_{\mathbf{k}\sigma}$ given in Eq. (4.31), the hamiltonian is mapped to a gas of non-interacting fermions collected in the $\hat{\Gamma}$ spinors of Eq. (4.33),

$$\hat{\Gamma}_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{\gamma}_{\mathbf{k}\sigma}^{(+)} \\ \hat{\gamma}_{\mathbf{k}\sigma}^{(-)} \end{bmatrix}$$

The final non-interacting hamiltonian is:

$$\hat{H}_t + \hat{H}_U \simeq \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} E_{\mathbf{k}\sigma} \left(\hat{\gamma}_{\mathbf{k}\sigma}^{(+)\dagger} \hat{\gamma}_{\mathbf{k}\sigma}^{(+)} - \hat{\gamma}_{\mathbf{k}\sigma}^{(-)\dagger} \hat{\gamma}_{\mathbf{k}\sigma}^{(-)} \right) + nU \hat{N} - E_{\text{H}/U}^{(\text{AF})}$$

with $E_{\mathbf{k}\sigma}$ the Bogoliubov bands of Eq. (4.32). In next section we give the explicit form of the many-body ground-state at zero temperature and half-filling. From now on, limitedly to this appendix, we use the simplified notation:

$$\Delta_{\mathbf{k}\uparrow} \rightarrow \Delta$$

with $\Delta_{\mathbf{k}\uparrow} = -\Delta_{\mathbf{k}\downarrow}$, and $\Delta = mU$.

A.2 Explicit AF ground state at half-filling and $\beta \rightarrow \infty$

Let $|\uparrow_{\mathbf{k}\sigma}\rangle$ and $|\downarrow_{\mathbf{k}\sigma}\rangle$ be respectively the “up” and “down” $\mathbf{k}$ -th pseudo-spin states. Let $|\overline{\downarrow}_{\mathbf{k}\sigma}\rangle$ be the “down” state in the tilted frame. The ground state is given by

$$|\Psi\rangle \equiv \bigotimes_{\sigma} \bigotimes_{\mathbf{k} \in \text{MBZ}} |\overline{\downarrow}_{\mathbf{k}\sigma}\rangle \quad (\text{A.2})$$

We can link the “down” states via an inverse rotation,

$$\begin{aligned} |\overline{\downarrow}_{\mathbf{k}\sigma}\rangle &= e^{-i\theta_{\mathbf{k}\sigma} \hat{s}_{\mathbf{k}\sigma}^y} |\downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos \theta_{\mathbf{k}\sigma} |\downarrow_{\mathbf{k}\sigma}\rangle - i \sin \theta_{\mathbf{k}\sigma} \hat{s}_{\mathbf{k}\sigma}^y |\downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos \theta_{\mathbf{k}\sigma} |\downarrow_{\mathbf{k}\sigma}\rangle - \sin \theta_{\mathbf{k}\sigma} (\hat{s}_{\mathbf{k}\sigma}^+ - \hat{s}_{\mathbf{k}\sigma}^-) |\downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos \theta_{\mathbf{k}\sigma} |\downarrow_{\mathbf{k}\sigma}\rangle - \sin \theta_{\mathbf{k}\sigma} |\uparrow_{\mathbf{k}\sigma}\rangle \end{aligned} \quad (\text{A.3})$$

having we used

$$\hat{s}_{\mathbf{k}\sigma}^{\pm} \equiv \frac{\hat{s}_{\mathbf{k}\sigma}^x \pm i \hat{s}_{\mathbf{k}\sigma}^y}{2}$$

Use now Eqns. (4.28) and (4.27),

$$\frac{\hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}_{\mathbf{k}\sigma}^z}{2} = \hat{n}_{\mathbf{k}\sigma} \quad \frac{\hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}_{\mathbf{k}\sigma}^z}{2} = \hat{n}_{\mathbf{k}+\pi\sigma}$$

Evaluating the above operators on the “up” and “down” states, we get

$$\begin{aligned} \frac{1}{2} \langle \uparrow_{\mathbf{k}\sigma} | \hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}_{\mathbf{k}\sigma}^z | \uparrow_{\mathbf{k}\sigma} \rangle &= 1 & \frac{1}{2} \langle \uparrow_{\mathbf{k}\sigma} | \hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}_{\mathbf{k}\sigma}^z | \uparrow_{\mathbf{k}\sigma} \rangle &= 0 \\ \frac{1}{2} \langle \downarrow_{\mathbf{k}\sigma} | \hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}_{\mathbf{k}\sigma}^z | \downarrow_{\mathbf{k}\sigma} \rangle &= 0 & \frac{1}{2} \langle \downarrow_{\mathbf{k}\sigma} | \hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}_{\mathbf{k}\sigma}^z | \downarrow_{\mathbf{k}\sigma} \rangle &= 1 \end{aligned}$$

Thus, up to a phase

$$|\uparrow_{\mathbf{k}\sigma}\rangle \sim \hat{c}_{\mathbf{k}\sigma}^{\dagger} |\Omega_{\mathbf{k}\sigma}\rangle \quad |\downarrow_{\mathbf{k}\sigma}\rangle \sim \hat{c}_{\mathbf{k}+\pi\sigma}^{\dagger} |\Omega_{\mathbf{k}\sigma}\rangle$$

being $|\Omega_{\mathbf{k}\sigma}\rangle$ the $(\mathbf{k}, \sigma)$ Hilbert subspace vacuum.

It follows that composing Eqns. (A.2) and (A.3) the ground state $|\Psi\rangle$ can be written as:

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k} \in \text{MBZ}} \left[\sin \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}^{\dagger} + e^{2i\zeta_{\mathbf{k}\sigma}} \cos \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\pi\sigma}^{\dagger} \right] |\Omega_{\mathbf{k}\sigma}\rangle \quad (\text{A.4})$$

having we absorbed in the relative phase $e^{2i\zeta_{\mathbf{k}\sigma}}$ the two states phases differences as well as the original sign difference of Eq. (A.3). Now, inserting the explicit form of the ground state from equation above:

$$\begin{aligned} \langle \overline{\downarrow}_{\mathbf{k}\sigma} | \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^{\dagger} | \overline{\downarrow}_{\mathbf{k}\sigma} \rangle &= (e^{2i\zeta_{\mathbf{k}\sigma}} + e^{-2i\zeta_{\mathbf{k}\sigma}}) \sin \theta_{\mathbf{k}\sigma} \cos \theta_{\mathbf{k}\sigma} \\ &= \cos 2\zeta_{\mathbf{k}\sigma} \sin 2\theta_{\mathbf{k}\sigma} \end{aligned}$$

but from Eq. (4.25)

$$\begin{aligned} \langle \overline{\downarrow}_{\mathbf{k}\sigma} | \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^{\dagger} | \overline{\downarrow}_{\mathbf{k}\sigma} \rangle &= \langle \overline{\downarrow}_{\mathbf{k}\sigma} | \hat{s}_{\mathbf{k}\sigma}^x | \overline{\downarrow}_{\mathbf{k}\sigma} \rangle \\ &= \sin 2\theta_{\mathbf{k}\sigma} \end{aligned}$$

since the pseudofield has no y component. This implies necessarily

$$\cos 2\zeta_{\mathbf{k}\sigma} \implies \zeta_{\mathbf{k}\sigma} = k\pi \quad \text{with } k \in \mathbb{Z}$$

and lets us conclude from Eq. (A.4)

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k} \in \text{MBZ}} \left[\sin \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}^{\dagger} + \cos \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\pi\sigma}^{\dagger} \right] |\Omega_{\mathbf{k}\sigma}\rangle \quad (\text{A.5})$$

For low-temperature physics, we can expect the ground-state density matrix to be highly dominated by this BCS-like zero-temperature ground-state, which highlights the essential features of the system.

Finally, using Eq. (4.33), we check that correctly the ground-state describes a perfect degenerate $\gamma^{(-)}$ quasiparticles Fermi gas,

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k} \in \text{MBZ}} \gamma_{\mathbf{k}\sigma}^{(-)\dagger} |\Omega_{\mathbf{k}\sigma}\rangle \quad (\text{A.6})$$

The AF phase can be thought of as a long-range coordination between fermionic states related by a π shift in reciprocal space, producing a new free gas where all interactions have been “condensed” in these AF pairs.

A.3 Instability of the commensurate AF phase

The commensurate AF phase is unstable at any finite doping. This result is discussed in [14, 15], and here we recall just the main passages. Consider the local spin operator defined in Eq. (2.8),

$$\hat{S}_j^\alpha = \frac{1}{2} \hat{\Psi}_j^\dagger \tau^\alpha \hat{\Psi}_j \quad \hat{S}_j^+ \equiv \frac{\hat{S}_j^x + i\hat{S}_j^y}{2} \quad \hat{S}_j^- \equiv \frac{\hat{S}_j^x - i\hat{S}_j^y}{2}$$

Let us take a continuum limit, $L_x \rightarrow \infty$ and $L_y \rightarrow \infty$, while the lattice spacing goes to zero.

Continuous SSBs are connected to the presence of a stable gapless Goldstone mode. Such a Goldstone mode is essentially a sharp peak at zero frequency for some dressed response function. For the commensurate AF phase, the low-energy collective excitations are transverse spin fluctuations about the antiferromagnetic ground-state, describing SDWs. In order to see if the Goldstone mode is stable for a given doping δ , we need to investigate whether the respective response function diverges at zero frequency. If it does, we have a gapless Goldstone mode and the phase is stable; if not, the phase is unstable. We work at $\omega = 0$ in order to capture gapless modes.

Consider the interaction representation of spin operators. Our new local spin operators are given by

$$\hat{\mathbf{S}}(\mathbf{r}, t)$$

The retarded transverse spin-spin response is given by:

$$\chi^{-+}(\mathbf{r}, t; \mathbf{r}', t') \equiv iH(t - t') \langle \Psi | [\hat{S}^-(\mathbf{r}, t), \hat{S}^+(\mathbf{r}', t')] | \Psi \rangle$$

being $H(x)$ the Heaviside function. The argument carried by Singh and Tešanović relies on random phase approximation (RPA). Let χ_0^{-+} be the non-interacting transverse response function. Then

$$\chi^{-+} \simeq \frac{\chi_0^{-+}}{1 - U\chi_0^{-+}} \quad (\text{RPA})$$

We move to momentum-frequency space. Within RPA approximation, low-lying spin waves collective excitations at wavevector $\mathbf{q}$ give a stable gapless Goldstone mode if:

$$1 - U\chi_0^{-+}(\mathbf{q}, 0) = 0 \quad (\text{A.7})$$

Commensurate antiferromagnetism is stable if the above equation is satisfied at $\mathbf{q} \rightarrow \pi$. Indeed, the wavevector π expresses two-sites periodicity in both directions.

Now, as we specified in Sec. 4.1, commensurate antiferromagnetism is described by the simultaneous SSB:

$$\text{SU}(2) \xrightarrow{\text{SSB}} \text{U}^z(1) \\ \text{T}_1^{(x)} \otimes \text{T}_1^{(y)} \xrightarrow{\text{SSB}} \text{T}_2^{(x)} \otimes \text{T}_2^{(y)}$$

Thus translational invariance is halved in both directions. The lattice $\mathcal{L}$ is separated in the two sublattices $\mathcal{S}_1, \mathcal{S}_2$ of Fig. 2.1. As a consequence, the non-interacting response function χ_0^{+-} does not depend strictly on $\mathbf{r}, \mathbf{r}'$ themselves. Instead, it depends only on which sublattice $\mathbf{r}$ and $\mathbf{r}'$ belong to. Because of this χ_0^{+-} is representable as a 2×2 matrix χ_0^{+-} . Then Eq. (A.7) is actually an eigenvalue equation,

$$\mathbb{I} - U\chi_0^{+-}(\mathbf{q}, 0) = 0$$

with $\mathbf{0}$ the 2×2 zero matrix. Diagonalising $\chi_0^{+-}(\mathbf{q}, 0)$ (which is symmetric due to lattice exchange symmetry) we can express the above equation in terms of the diagonal matrix containing its eigenvalues.

Consider the largest eigenvalue $\lambda_\delta(\mathbf{k}, \omega)$ of the matrix $[\chi_0^{+-}]$ at doping δ . We work in momentum-frequency space. The phase is unstable if

$$1 - U\lambda_\delta \neq 0 \quad (\text{A.8})$$

because, as we said, in that case we do not have a pole for the transverse spin-spin response function. Singh and Tešanović [14] found

$$\lambda_0(\mathbf{k} \rightarrow \boldsymbol{\pi}, 0) = \frac{1}{U} \quad (\text{half filling}) \quad (\text{A.9})$$

$$\lambda_\delta(\mathbf{k} \rightarrow \boldsymbol{\pi}, 0) = \frac{1}{U} + \frac{2\epsilon_F}{\Delta} N(\epsilon_F) \quad (\text{finite doping}) \quad (\text{A.10})$$

The reason for the distinction above is the intrinsic metallic nature of the system for infinitesimal doping. The half-filled AF phase is insulating, because the chemical potential lies within the gap. For an insulator Fermi energy is not defined. Instead, at finite filling the phase is metallic because the chemical potential crosses the upper band.

From Eq. (A.8) and (A.9), we see that the half filled antiferromagnet is stable towards transverse spin fluctuations, because χ^{+-} develops a divergence at zero frequency. At any finite doping, since we are describing a metal, a FS exists and thus $N(\epsilon_F)$ is finite. This makes the phase unstable, and this is the reason we do not discuss the AF phase outside of half-filling.

As we discussed at the beginning of Chap. 4, antiferromagnetism can establish in many ways. For instance, striped antiferromagnets are of particular interest [16]. In order to investigate the most stable physical realisation of an antiferromagnet on the conventional HM, on a pure HF level, we should:

1. Choose an initialiser wavevector, say $\mathbf{k} = \boldsymbol{\pi}$;
2. Solve the self-consistency equations for the HFPs vector for an antiferromagnet oscillating at wavevector $\mathbf{k}$;
3. Compute the free energy F as a functional of the wavevector around $\mathbf{k}$;
4. Look for gradient descent of the free energy, $\nabla F < 0$;
5. If the gradient descent is found, update $\mathbf{k} \rightarrow \mathbf{k} + \delta\mathbf{k}$ and repeat from step 2, otherwise halt.

The sketched procedure can lead to a lot of complications. For instance, the fact that at incommensurate wavevector $\mathbf{k} \neq \boldsymbol{\pi}$ the unit cell is not as simple as a pair of sites, but can become much larger requiring more refined computational solutions than our pseudospin representation of the HM hamiltonian in the commensurate AF channel.

[Extend to EHM: is it true that:

$$\chi^{-+} \simeq \frac{\chi_0^{-+}}{1 - [U - 2V(\cos q_x + \cos q_y)]\chi_0^{-+}} \quad (\text{RPA})$$

?)

A.4 iHF results

For the conventional HM, antiferromagnetism is described by a single HFP, the magnetisation m . Its self-consistency Eq. (4.39) reads:

$$m_{i+1} = \frac{U}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{m_i}{2E_{\mathbf{k}}[\Delta_i]} [f(-E_{\mathbf{k}}[m_i]; \beta, \mu) - f(E_{\mathbf{k}}[m_i]; \beta, \mu)] \quad (\text{A.11})$$

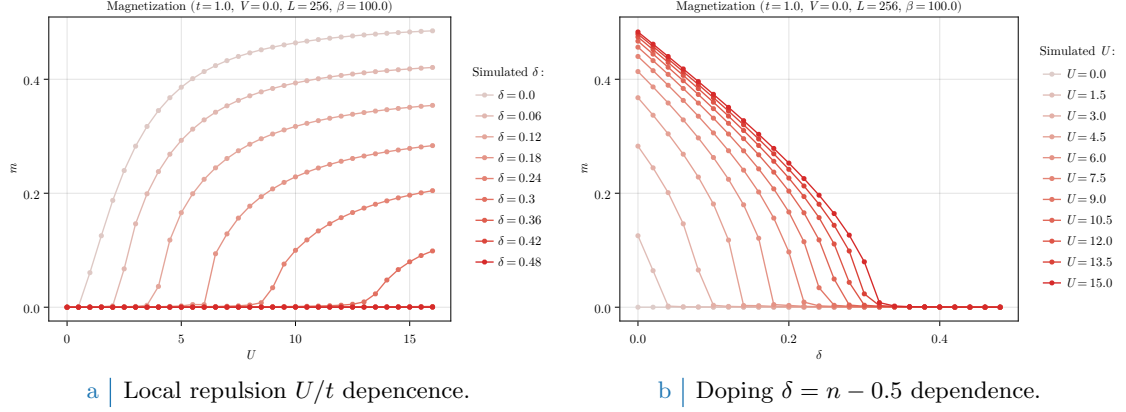


Figure A.1 | Plots of the zero-temperature AF instability order parameter m as a function of both the local repulsion U/t (Fig. A.1a) and the doping $\delta = n - 1/2$ (Fig. A.1b).

where we indicated as m_i the value of magnetisation at step i . We ran iHF simulations in the ranges:

$$0 \leq U \leq 16 \quad 0 \leq \delta \leq 0.48$$

We considered also $\delta > 0$, although – as we specified in previous section – a doped commensurate antiferromagnet is actually unstable.

In Fig. A.1 magnetisation m at $\beta = 100$ is plotted. As is visible in Fig. A.1a, ignoring instability issues, doping disrupts AF ordering; the horizontal asymptote lowers as δ gets bigger. Also, for $\delta > 0$, a finite magnetisation m exists only for $U \geq U_c[\delta]$ (a critical value parametrically dependent on doping).

Fig. A.1b shows the dependence of the magnetisation on doping varying the local repulsion U . Identical analysis have been performed for different temperatures, leading to analogous results as long as $\beta \gtrsim 10$ and to $m \simeq 0$ for higher temperatures.

A.5 Analytic expression of self-consistent magnetisation

In this subsection we provide an analytic expression to compute m self-consistently at half-filling. Multiply Eq. (4.39) on both sides by U . Since due to symmetry $\mu = 0$, we have the finite-temperature self-consistent equation

$$\Delta = \frac{U}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\Delta}{\sqrt{\epsilon_{\mathbf{k}}^2 + \Delta^2}} \tanh\left(\frac{\beta}{2} \sqrt{\epsilon_{\mathbf{k}}^2 + \Delta^2}\right)$$

where we used the relation

$$\text{Th}_{\mathbf{k}}^{(-)}[\beta, 0] = \tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right)$$

In thermodynamic limit $L \rightarrow \infty$

$$\frac{1}{U} = \frac{1}{2} \int_{\text{BZ}} \frac{d\mathbf{k}}{(2\pi)^2} \frac{1}{\sqrt{\epsilon_{\mathbf{k}}^2 + \Delta^2}} \tanh\left(\frac{\beta}{2} \sqrt{\epsilon_{\mathbf{k}}^2 + \Delta^2}\right)$$

Substitute the momentum integral with an integral over energies:

$$\frac{2}{U} = \int_{-4t}^{4t} \frac{d\epsilon \rho(\epsilon)}{\sqrt{\epsilon^2 + \Delta^2}} \tanh\left(\frac{\beta}{2} \sqrt{\epsilon^2 + \Delta^2}\right)$$

being $\rho(\epsilon)$ the 2D square lattice DoS. Its explicit form is given by [28]

$$\rho(\epsilon) = \frac{1}{2\pi^2} K\left(\sqrt{1 - \left(\frac{\epsilon}{4t}\right)^2}\right) \quad K(x) \equiv \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - x^2 \sin^2 \theta}}$$

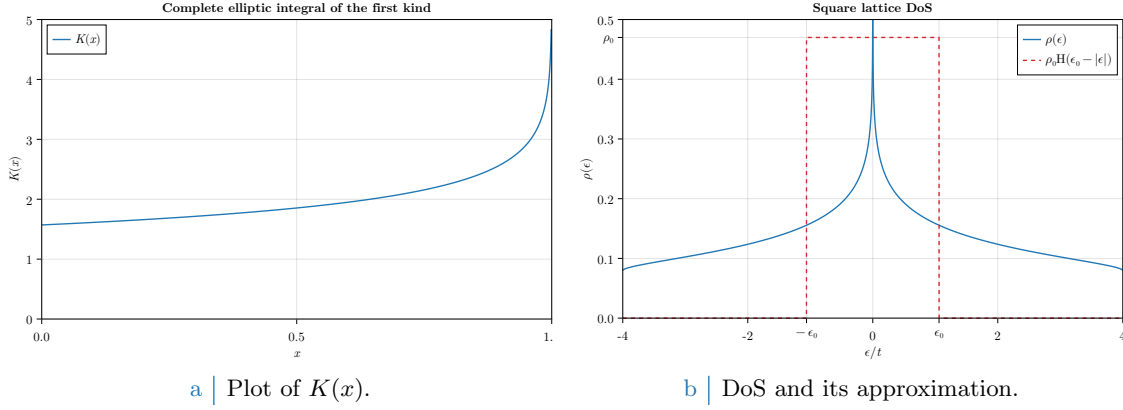


Figure A.2 | Plots of the complete elliptic integral of the first kind $K(x)$ and the density of states for the planar square lattice.

$K(x)$ is the complete elliptic integral of the first kind, shown in Fig. A.2a.

As is plotted in Fig. A.2b, the resulting DoS is logarithmically peaked at $\epsilon = 0$, on the MBZ. An extremely rough but common approximation is to substitute the actual DoS with a step function,

$$\rho(\epsilon) \simeq \rho_0 H(\epsilon_0 - |\epsilon|) \quad \text{with} \quad 2\rho_0\epsilon_0 = 1$$

Here, the peak parameters ϵ_0 and ρ_0 are chosen in order to maintain a correct DoS integral, considering also the 2 pre-factor of spin degeneracy. Within this extremely rough approximation, we have

$$\frac{2}{U} \simeq \rho_0 \int_{-\epsilon_0}^{\epsilon_0} \frac{d\epsilon}{\sqrt{\epsilon^2 + \Delta^2}} \tanh\left(\frac{\beta}{2} \sqrt{\epsilon^2 + \Delta^2}\right)$$

Suppose the system is magnetised, $\Delta > 0$. At low temperature, $\beta \gg 1$, as a first approximation we can substitute the hyperbolic tangent with unity, obtaining

$$\begin{aligned} \frac{2}{\rho_0 U} &\simeq \int_{-\epsilon_0}^{\epsilon_0} \frac{d\epsilon}{\sqrt{\epsilon^2 + \Delta^2}} \\ &= \int_{-\epsilon_0/\Delta}^{\epsilon_0/\Delta} \frac{dx}{\sqrt{1+x^2}} \\ &= \sinh^{-1}\left(\frac{\epsilon_0}{\Delta}\right) - \sinh^{-1}\left(-\frac{\epsilon_0}{\Delta}\right) \\ &= 2 \sinh^{-1}\left(\frac{\epsilon_0}{\Delta}\right) \end{aligned}$$

thus

$$m \simeq \frac{\epsilon_0}{U \sinh(1/\rho_0 U)} \quad (\text{A.12})$$

having we used $\Delta = mU$. In the left panel of Fig. A.3 we sketch this function of U varying ρ_0 , ϵ_0 . The same data of Fig. A.1a, $\delta = 0$ are plotted as gray dots. The shown curves are not fits.

Now: at large U ,

$$\lim_{U \rightarrow +\infty} m \simeq \epsilon_0 \rho_0 = \frac{1}{2}$$

which gives the correct saturation limit at high local repulsion. In the opposite limit,

$$\begin{aligned} \lim_{U \rightarrow 0} m &\simeq \lim_{U \rightarrow 0} \frac{2\epsilon_0/U}{\exp(1/\rho_0 U) - \exp(-1/\rho_0 U)} \\ &\simeq \lim_{U \rightarrow 0} \frac{2\epsilon_0}{U} \exp\left(-\frac{1}{\rho_0 U}\right) \end{aligned} \quad (\text{A.13})$$

giving a near-0 exponential suppression of magnetisation, typical of MFT self-consistent solutions. In the right panel of Fig. A.3 we sketch this approximation.

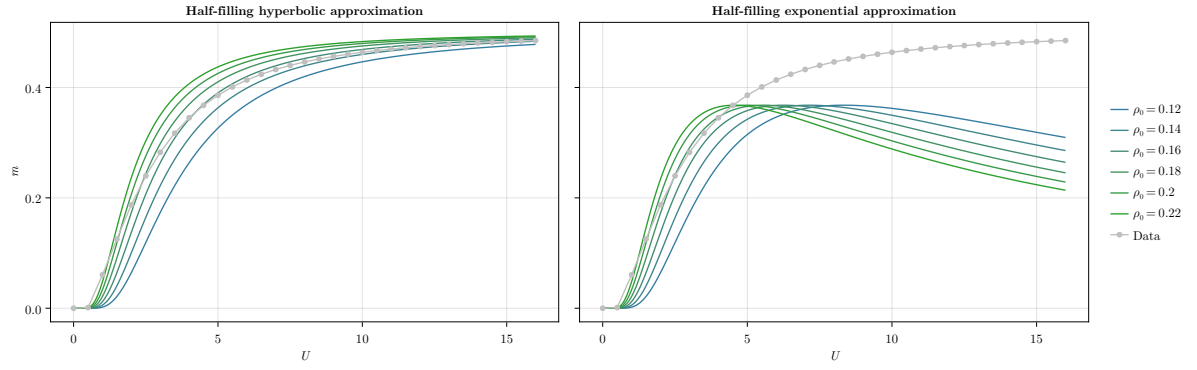


Figure A.3 | Sketch of the analytic expression for m . The left panel shows a family of coloured curves described by Eq. (A.12). The right panel shows a family of coloured curves described by Eq. (A.13), only reasonable at $U \rightarrow 0$.

Bibliography

- [1] J. G. Bednorz and K. A. Muller. “Possible high-Tc superconductivity in the Ba-La-Cu-O system”. en. In: *Zeitschrift für Physik B Condensed Matter* 64.2 (June 1986), pp. 189–193. ISSN: 1431-584X. DOI: [10.1007/BF01303701](https://doi.org/10.1007/BF01303701). URL: <https://doi.org/10.1007/BF01303701> (visited on 04/03/2026).
- [2] Liangzi Deng et al. “Ambient-pressure 151 K superconductivity in HgBa₂Ca₂Cu₃O₈ via pressure quench”. In: *Proceedings of the National Academy of Sciences* 123.11 (Mar. 2026), e2536178123. DOI: [10.1073/pnas.2536178123](https://www.pnas.org/doi/10.1073/pnas.2536178123). URL: <https://www.pnas.org/doi/10.1073/pnas.2536178123> (visited on 04/03/2026).
- [3] B. Keimer et al. “From quantum matter to high-temperature superconductivity in copper oxides”. en. In: *Nature* 518.7538 (Feb. 2015), pp. 179–186. ISSN: 0028-0836, 1476-4687. DOI: [10.1038/nature14165](https://www.nature.com/articles/nature14165). URL: <https://www.nature.com/articles/nature14165> (visited on 03/15/2025).
- [4] Giuseppe Grosso and Giuseppe Pastori Parravicini. *Solid State Physics*. Second Edition. Academic Press, 2014.
- [5] D. J. Scalapino. “A common thread: The pairing interaction for unconventional superconductors”. en. In: *Reviews of Modern Physics* 84.4 (Oct. 2012), pp. 1383–1417. ISSN: 0034-6861, 1539-0756. DOI: [10.1103/RevModPhys.84.1383](https://link.aps.org/doi/10.1103/RevModPhys.84.1383). URL: <https://link.aps.org/doi/10.1103/RevModPhys.84.1383> (visited on 04/03/2026).
- [6] Daniel P. Arovas et al. “The Hubbard Model”. In: *Annual Review of Condensed Matter Physics* 13. Volume 13, 2022 (2022), pp. 239–274. ISSN: 1947-5462. DOI: <https://doi.org/10.1146/annurev-conmatphys-031620-102024>. URL: <https://www.annualreviews.org/content/journals/10.1146/annurev-conmatphys-031620-102024>.
- [7] J. E. Hirsch. “Two-dimensional Hubbard model: Numerical simulation study”. In: *Phys. Rev. B* 31 (7 Apr. 1985), pp. 4403–4419. DOI: [10.1103/PhysRevB.31.4403](https://link.aps.org/doi/10.1103/PhysRevB.31.4403). URL: <https://link.aps.org/doi/10.1103/PhysRevB.31.4403>.
- [8] Piers Coleman. *Introduction to Many-Body Physics*. Cambridge University Press, 2015.
- [9] Hari Padma et al. “Beyond-Hubbard pairing in a cuprate ladder”. en. In: *Physical Review X* 15.2 (May 2025), p. 021049. ISSN: 2160-3308. DOI: [10.1103/PhysRevX.15.021049](https://link.aps.org/doi/10.1103/PhysRevX.15.021049). URL: <https://link.aps.org/doi/10.1103/PhysRevX.15.021049> (visited on 04/03/2026).
- [10] R. Micnas, J. Ranninger, and S. Robaszkiewicz. “Superconductivity in narrow-band systems with local nonretarded attractive interactions”. In: *Reviews of Modern Physics* 62.1 (Jan. 1990). Publisher: American Physical Society, pp. 113–171. DOI: [10.1103/RevModPhys.62.113](https://link.aps.org/doi/10.1103/RevModPhys.62.113). URL: <https://link.aps.org/doi/10.1103/RevModPhys.62.113> (visited on 01/06/2026).
- [11] Antoine Georges et al. “Dynamical mean-field theory of strongly correlated fermion systems and the limit of infinite dimensions”. en. In: *Reviews of Modern Physics* 68.1 (Jan. 1996), pp. 13–125. ISSN: 0034-6861, 1539-0756. DOI: [10.1103/RevModPhys.68.13](https://link.aps.org/doi/10.1103/RevModPhys.68.13). URL: <https://link.aps.org/doi/10.1103/RevModPhys.68.13> (visited on 03/15/2025).
- [12] Robin Scholle et al. “Comprehensive mean-field analysis of magnetic and charge orders in the two-dimensional Hubbard model”. In: *Phys. Rev. B* 108 (3 July 2023), p. 035139. DOI: [10.1103/PhysRevB.108.035139](https://link.aps.org/doi/10.1103/PhysRevB.108.035139). URL: <https://link.aps.org/doi/10.1103/PhysRevB.108.035139>.

- [13] Zhangkai Cao et al. “Dominant p -wave pairing induced by nearest-neighbor attraction in the square-lattice extended Hubbard model”. In: *Phys. Rev. B* 111 (2 Jan. 2025), p. 024509. DOI: [10.1103/PhysRevB.111.024509](https://doi.org/10.1103/PhysRevB.111.024509). URL: <https://link.aps.org/doi/10.1103/PhysRevB.111.024509>.
- [14] Avinash Singh and Zlatko Tešanović. “Collective excitations in a doped antiferromagnet”. en. In: *Physical Review B* 41.1 (Jan. 1990), pp. 614–631. ISSN: 0163-1829, 1095-3795. DOI: [10.1103/PhysRevB.41.614](https://doi.org/10.1103/PhysRevB.41.614). URL: <https://link.aps.org/doi/10.1103/PhysRevB.41.614> (visited on 12/12/2025).
- [15] Avinash Singh, Zlatko Tešanović, and Ju H. Kim. “Doped Hubbard antiferromagnet: Instability and effective interactions”. en. In: *Physical Review B* 44.14 (Oct. 1991), pp. 7757–7759. ISSN: 0163-1829, 1095-3795. DOI: [10.1103/PhysRevB.44.7757](https://doi.org/10.1103/PhysRevB.44.7757). URL: <https://link.aps.org/doi/10.1103/PhysRevB.44.7757> (visited on 12/12/2025).
- [16] Alexander Wietek et al. “Stripes, Antiferromagnetism, and the Pseudogap in the Doped Hubbard Model at Finite Temperature”. en. In: *Physical Review X* 11.3 (July 2021), p. 031007. ISSN: 2160-3308. DOI: [10.1103/PhysRevX.11.031007](https://doi.org/10.1103/PhysRevX.11.031007). URL: <https://link.aps.org/doi/10.1103/PhysRevX.11.031007> (visited on 12/12/2025).
- [17] Kazuyoshi Yoshimi, Takeo Kato, and Hideaki Maebashi. “Enhanced Spin Susceptibility toward the Charge-Ordering Transition in a Two-Dimensional Extended Hubbard Model”. In: *Journal of the Physical Society of Japan* 78.10 (Oct. 2009), p. 104002. ISSN: 0031-9015. DOI: [10.1143/JPSJ.78.104002](https://doi.org/10.1143/JPSJ.78.104002). URL: <https://journals.jps.jp/doi/10.1143/JPSJ.78.104002> (visited on 04/17/2026).
- [18] Ashvin Vishwanath. “Dirac Nodes and Quantized Thermal Hall Effect in the Mixed State of d-wave Superconductors”. In: *Physical Review B* 66.6 (Aug. 2002). arXiv:cond-mat/0104213, p. 064504. ISSN: 0163-1829, 1095-3795. DOI: [10.1103/PhysRevB.66.064504](https://doi.org/10.1103/PhysRevB.66.064504). URL: <http://arxiv.org/abs/cond-mat/0104213> (visited on 01/05/2026).
- [19] P. Senarath Yapa et al. “Mixed-symmetry superconductivity and the energy gap”. en. In: *Physica C: Superconductivity and its Applications* 633 (June 2025), p. 1354719. ISSN: 09214534. DOI: [10.1016/j.physc.2025.1354719](https://doi.org/10.1016/j.physc.2025.1354719). URL: <https://linkinghub.elsevier.com/retrieve/pii/S0921453425000723> (visited on 01/06/2026).
- [20] Joel Hutchinson and Frank Marsiglio. “Mixed temperature-dependent order parameters in the extended Hubbard model”. en. In: *Journal of Physics: Condensed Matter* 33.6 (Feb. 2021), p. 065603. ISSN: 0953-8984, 1361-648X. DOI: [10.1088/1361-648X/abc801](https://doi.org/10.1088/1361-648X/abc801). URL: <https://iopscience.iop.org/article/10.1088/1361-648X/abc801> (visited on 01/06/2026).
- [21] C. Zhou and H. J. Schulz. “Density of states and tunneling spectra in two-dimensional d-wave superconductors”. In: *Physical Review B* 45.13 (Apr. 1992). Publisher: American Physical Society, pp. 7397–7401. DOI: [10.1103/PhysRevB.45.7397](https://doi.org/10.1103/PhysRevB.45.7397). URL: <https://link.aps.org/doi/10.1103/PhysRevB.45.7397> (visited on 01/06/2026).
- [22] James F. Annett. “Symmetry of the order parameter for high-temperature superconductivity”. In: *Advances in Physics* 39.2 (Apr. 1990). Publisher: Taylor & Francis eprint: <https://doi.org/10.1080/00018739000101481>, pp. 83–126. ISSN: 0001-8732. DOI: [10.1080/00018739000101481](https://doi.org/10.1080/00018739000101481). URL: <https://doi.org/10.1080/00018739000101481> (visited on 01/06/2026).
- [23] Manfred Sgrist and Kazuo Ueda. “Phenomenological theory of unconventional superconductivity”. In: *Reviews of Modern Physics* 63.2 (Apr. 1991). Publisher: American Physical Society, pp. 239–311. DOI: [10.1103/RevModPhys.63.239](https://doi.org/10.1103/RevModPhys.63.239). URL: <https://link.aps.org/doi/10.1103/RevModPhys.63.239> (visited on 01/07/2026).
- [24] Anna E. Böhrer et al. “Nematicity and nematic fluctuations in iron-based superconductors”. en. In: *Nature Physics* 18.12 (Dec. 2022), pp. 1412–1419. ISSN: 1745-2481. DOI: [10.1038/s41567-022-01833-3](https://doi.org/10.1038/s41567-022-01833-3). URL: <https://www.nature.com/articles/s41567-022-01833-3>.

- [25] Mingpu Qin et al. “Absence of Superconductivity in the Pure Two-Dimensional Hubbard Model”. In: *Physical Review X* 10.3 (July 2020), p. 031016. DOI: [10.1103/PhysRevX.10.031016](https://doi.org/10.1103/PhysRevX.10.031016). URL: <https://link.aps.org/doi/10.1103/PhysRevX.10.031016> (visited on 03/21/2026).
- [26] Jan Zaanen and Olle Gunnarsson. “Charged magnetic domain lines and the magnetism of high- T_c oxides”. In: *Physical Review B* 40.10 (Oct. 1989), pp. 7391–7394. DOI: [10.1103/PhysRevB.40.7391](https://doi.org/10.1103/PhysRevB.40.7391). URL: <https://link.aps.org/doi/10.1103/PhysRevB.40.7391> (visited on 03/21/2026).
- [27] Masaru Kato et al. “Soliton Lattice Modulation of Incommensurate Spin Density Wave in Two Dimensional Hubbard Model -A Mean Field Study-”. In: *Journal of the Physical Society of Japan* 59.3 (Mar. 1990), pp. 1047–1058. ISSN: 0031-9015. DOI: [10.1143/JPSJ.59.1047](https://doi.org/10.1143/JPSJ.59.1047). URL: <https://journals.jps.jp/doi/10.1143/JPSJ.59.1047> (visited on 03/21/2026).
- [28] Eugene Kogan and Godfrey Gumbs. “Green’s Functions and DOS for Some 2D Lattices”. en. In: *Graphene* 10.01 (2021), pp. 1–12. ISSN: 2169-3439, 2169-3471. DOI: [10.4236/graphene.2021.101001](https://doi.org/10.4236/graphene.2021.101001). URL: <https://www.scirp.org/journal/paperinformation?paperid=104591> (visited on 03/26/2026).
- [29] Charles L. Cleveland and Rodrigo Medina A. “Obtaining a Heisenberg Hamiltonian from the Hubbard model”. en. In: *American Journal of Physics* 44.1 (Jan. 1976), pp. 44–46. ISSN: 0002-9505, 1943-2909. DOI: [10.1119/1.10537](https://doi.org/10.1119/1.10537). URL: <https://pubs.aip.org/ajp/article/44/1/44/1050180/Obtaining-a-Heisenberg-Hamiltonian-from-the> (visited on 05/08/2026).
- [30] Eva Pavarini et al., eds. *Many-body physics: from Kondo to Hubbard: lecture notes of the Autumn School on Correlated Electrons 2015: at Forschungszentrum Jülich, 21-25 September 2015*. en. Band 5. Meeting Name: Autumn School on Correlated Electrons. Forschungszentrum Jülich, 2015. ISBN: 978-3-95806-074-6.
- [31] Gabriele Giuliani and Giovanni Vignale. *Quantum Theory of the Electron Liquid*. Cambridge University Press, 2005.
- [32] Michele Fabrizio. *A Course in Quantum Many-Body Theory*. Springer, 2022.
- [33] Eva Pavarini et al., eds. *Correlated electrons: from models to materials: lecture notes of the Autumn School Correlated Electrons 2012: at Forschungszentrum Jülich, 3-7 September 2012*. eng. Schriften des Forschungszentrums Jülich. Reihe Modeling and Simulation Band 2. Meeting Name: Autumn School Correlated Electrons. Jülich: Forschungszentrum Jülich, Zentralbibliothek, Verl, 2012. ISBN: 978-3-89336-796-2.